

LightAPI Interfaces

User Guide

Manual Version: V4.00

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1 Preface

1.1 LightAPI Overview

LightAPI is the HTTP message part of "Multiple Controls in One" and adopts HTTP non-persistent connection mechanism. During the process of an interaction, the client notifies the server of the requested resource through the HTTP URL, and the server returns information about the URL-specified resources in Json format.

1.2 About This Guide

This guide describes interfaces of the system, PTZ, media, storage, network, image, alarm, serial port, smart services, etc, and uses the following conventions for parameter requirements.

Format	Meaning
M	Parameter is mandatory.
C	Parameter is required according to conditions.
O	Parameter is optional.

1.3 Important Notice

All the LAPI interfaces in this manual adopt Digest authentication. For more information, please refer to <https://tools.ietf.org/html/rfc2617>

The calculation rules of digest authentication are as follows:

If qop=Undefined

Digest=MD5(MD5(A1):<nonce>:MD5(A2))

If qop="auth:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))

If qop="auth-int:"

Digest=MD5(MD5(A1):<nonce>:<nc>:<cnonce>:<qop>:MD5(A2))

A1 and A2 bits calculate the 2 pieces of data content of the digest

For MD5

A1=<user>:<realm>:<password>;

For MD5-sess

A1=MD5(<user>:<realm>:<password>:<nonce>:<cnonce>.

If qop is undefined or if qop="auth:"

A2=<request-method>:<uri-directive-value>;

If qop="auth-int:"

A2=<request-method>:<uri-directive-value>:MD5(<request-entity-body>).

1.4 Four HTTP Methods Supported by LightAPI

Method	GET	PUT	POST	DELETE
Usage	Get	Update	Create	Delete
URL resource type	Array object name/array member ID	Array object name/array member ID	Array object name	Array object name/array member ID



NOTE!

PUT and POST differ in:

POST is only used to increase the number of array member objects that can be dynamically changed, and the URL only specifies an array name; PUT is used to update the resource object specified by an URL, and the URL specifies an object.

The object ID in the PUT method is determined by the client. The object ID in the POST method is assigned by the server as the index of the newly added array object, and the ID is returned to the client via the "CreatedID" field; if the object contains any index variable, then the server will update the index variable in the object of the POST request. An interface that supports POST also supports DELETE.

1.5 URL Rules

The following example illustrates the URL composition rules.

/LAPI/V1.0[/Channels/<ID>]/<service-name>/<resource-name>[/<child-resource-name>][/<ID>]][?<key>=<value>]

Composition	Description
[]:	Indicates parameter (character string or unsigned long).
<>	Indicates video input channel resource of a device.
<ID>	Indicates video input channel ID (unsigned long).
<service-name>	Services are divided into different types, for example, system, media, network, etc.
<resource-name>	Includes physical resources and virtual resources.

The following example illustrates rules for locating multi-channel resources.

/LAPI/V1.0[/Channels/<ID>]/<service-name>/<resource-name>[?ChannelID=<ID>]

Composition	Description
[/Channels/<ID>]	Locates resources in the sub domain of a specified channel of the device.
[?ChannelID=<ID>]	Locates resources of a specified channel of the device.

1. Example code 1

```
PUT /LAPI/V1.0/System/Time/NTP HTTP/1.1
{
  "Enabled":1,
  "AddressType":0,
  "Address":"24.56.178.140",
  "Port":123,
  "SynchronizeInterval":60
}
```

When using PUT/POST method, the HTTP message body usually needs to contain JSON data. The request must be sent to the HTTP/HTTPS port of the device.

2. Example code 2

HTTP/1.1 200 Ok

```
{
  "Response":
  {
    "ResponseURL": "URL",
    "ResponseCode": 0,
    "SubResponseCode": 101,
    "ResponseString": "Succeed",
    "StatusCode": 0,
    "Data": "null" or {...}
  }
}
```

ResponseURL: Indicates the URL carried in the HTTP request. When the request uses the GET/PUT/DELETE method, the URL is the same as the URI in the request. For the POST request, the URL is the URL of a newly created object. The client can use the URL to directly obtain the new object.

ResponseCode: Indicates the system processing result. 0 means success.

ResponseString: Indicates the analysis of system processing results.

Data: Indicates the response to the request (GET), which in most cases is in JSON format. If the URLs are the same, the data format of the PUT request is the same as the data format of the GET response. The Data item in the following tables lists the data of each API. If the server failed to handle the request, the data is "null". If the HTTP message uses the PUT/POST method, the data in the response is "null".

1.6 Get structure Info

Please visit www.json.org for more information about JSON. JSON string values can include any Unicode characters (except "\" and control characters).

2 Functions

This section describes the procedure of calling multiple interfaces. For descriptions about calling interfaces of a single service, refer to Chapter 3.

2.1 Alarm Notification

2.1.1 Function Introduction

The upper platform can call the corresponding interfaces to obtain alarm information from devices.

2.1.2 Interface Overview

Interface	Description
/LAPI/V1.0/System/Event/Subscription	Subscribe to/unsubscribe from alarm
/LAPI/V1.0/System/Event/Subscription/<ID>	Update alarm
/LAPI/V1.0/System/Event/Notification/Alarm	Alarm notification

2.1.3 Interface Calling Procedure

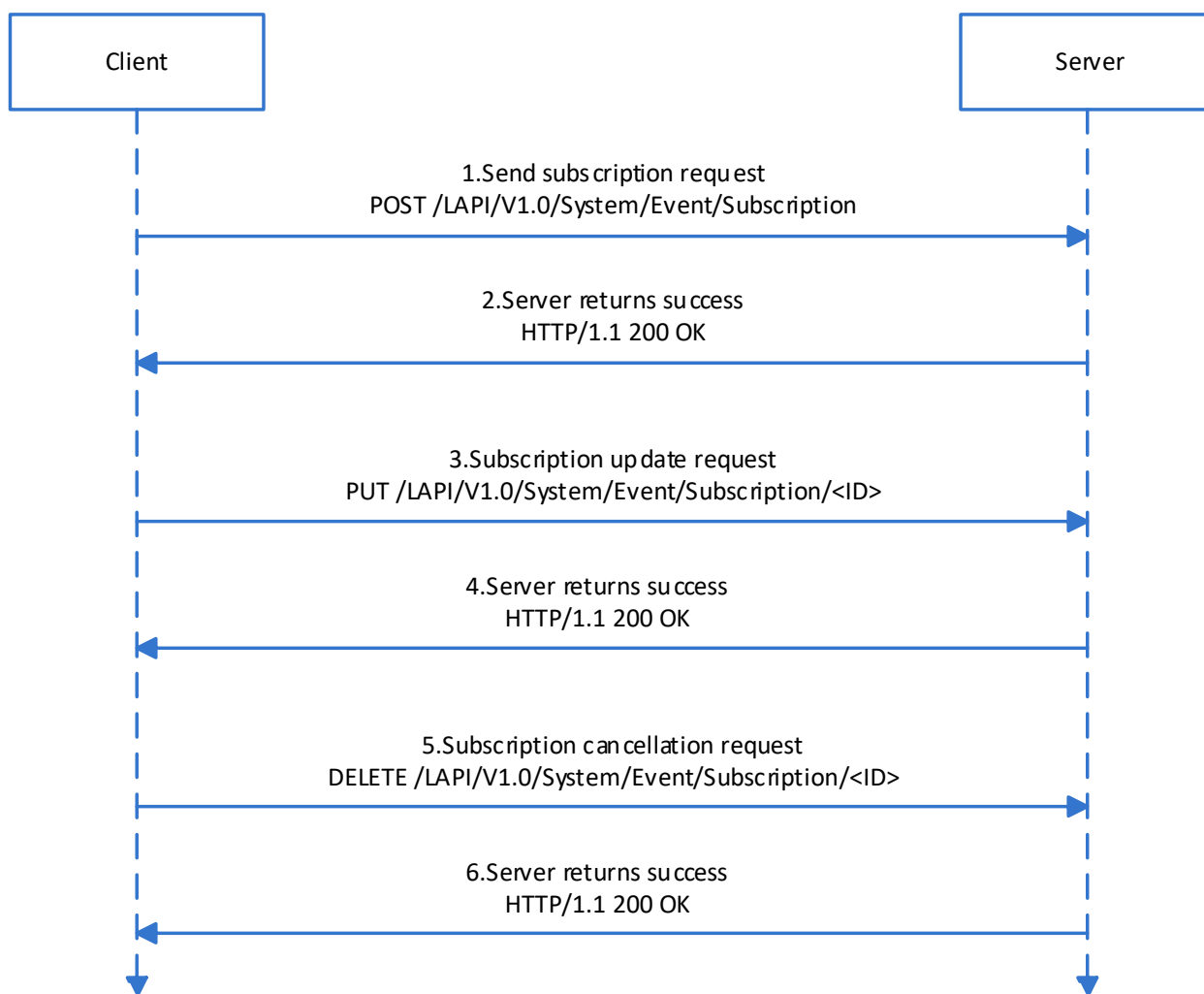


图 2-1 Alarm subscription

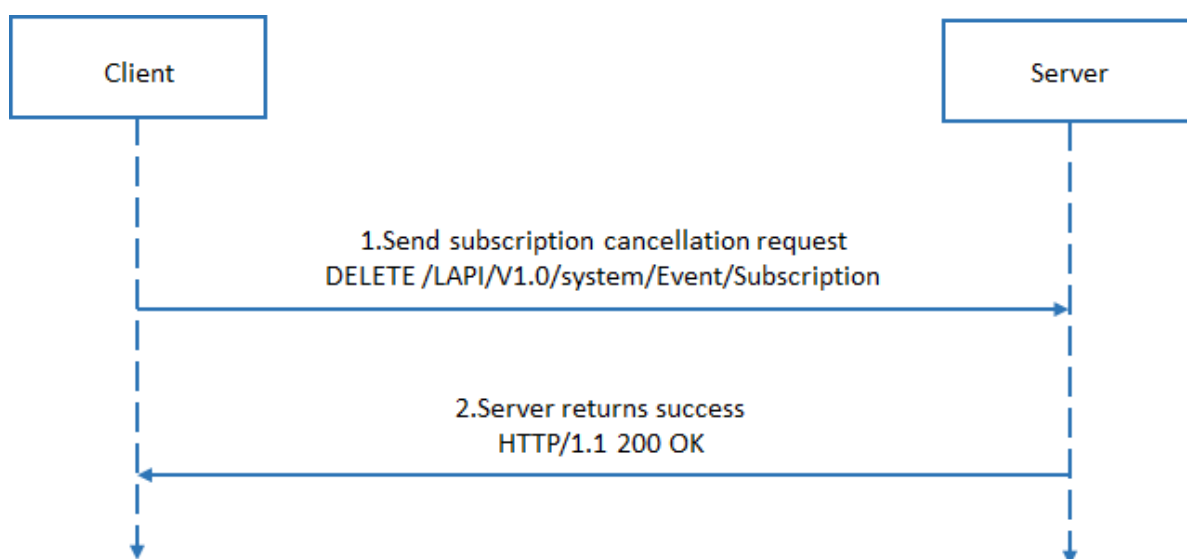


图 2-2 Subscription cancellation

The procedure is described as follows:

- Alarm notification

Step 1: The client subscribes to alarms from the device (server). See </LAPI/V1.0/System/Event/Subscription>.



NOTE!

The parameters carried by this interface represent the client's IP and port to which the device (server) pushes alarms and the duration of subscription, respectively.

Step 2: The parameters returned by the device (server) represent the subscription identifier (ID), subscribed info (Reference), subscription start time (CurrentTime) and end time (TerminationTime), respectively. The subscription identifier (ID) is used for subsequent alarm update or subscription cancellation.

Step F3: The client periodically sends the update interface to refresh subscription till the subscription end time (TerminationTime).

Step 4: The device (server) returns the updated subscription time.

Step 5: After an alarm occurred, the device pushes the alarm to the client. See </LAPI/V1.0/System/Event/Notification/Alarm>

- Cancel alarm subscription

Step 1: The client cancels the alarm subscription. See </LAPI/V1.0/System/Event/Subscription/<ID>>.

Step 2: The server cancels the subscription and stops pushing alarms.

2.2 Search by Image

2.2.1 Function Introduction

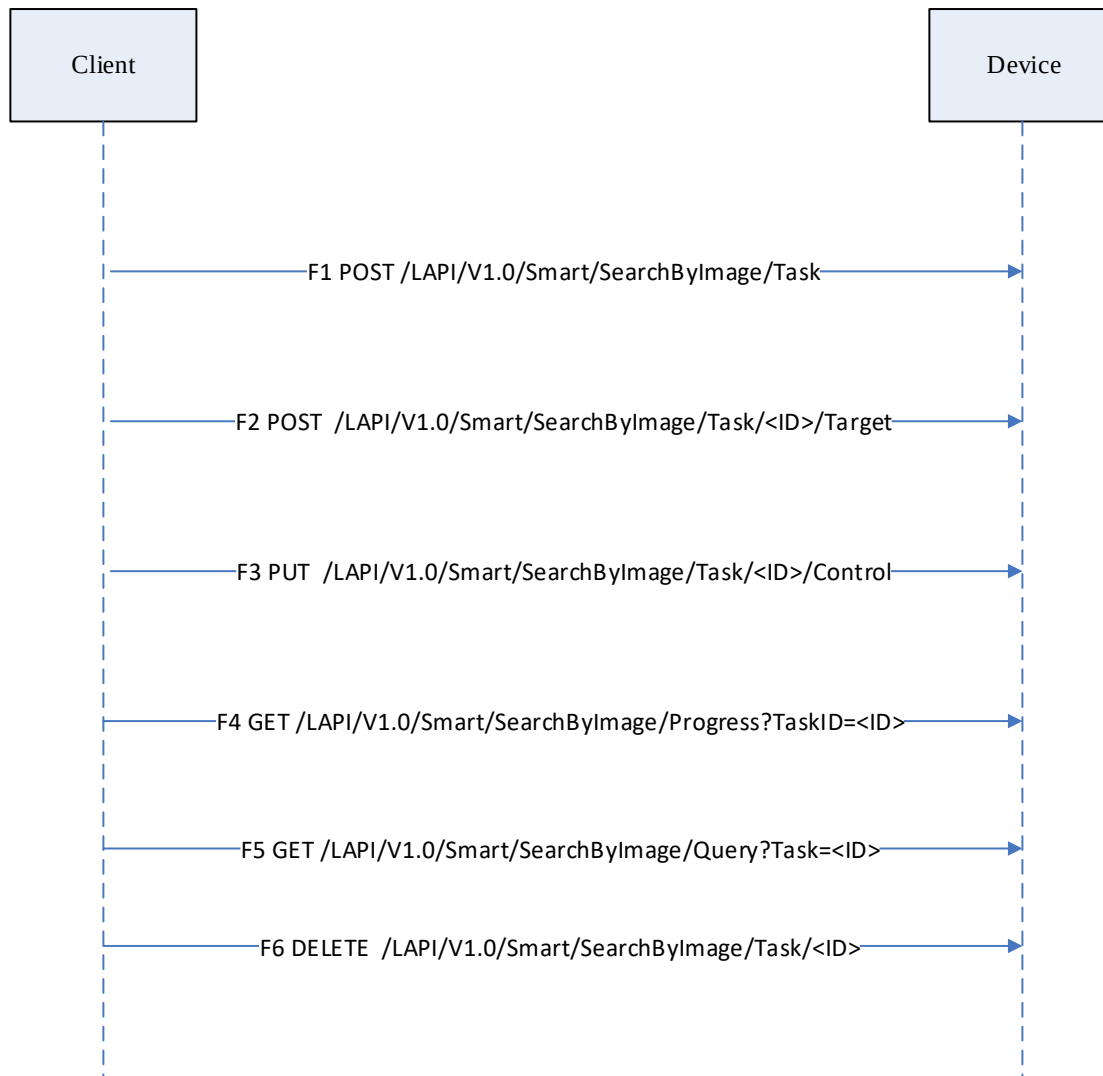
The upper platform sends an image with the search range to the device capable of image search. The device returns search results similar to the target in the received image.

2.2.2 Interface Overview

Interface	Description
POST /LAPI/V1.0/Smart/SearchByImage/Task	Create a search-by-image task
PUT /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/Target	Specify search-by-image parameters
PUT /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/Control	Start a search-by-image task
GET /LAPI/V1.0/Smart/SearchByImage/Progress?TaskID=<ID>	Get the progress of a search-by-image task
GET /LAPI/V1.0/Smart/SearchByImage/Query?Task=<ID>	Get the result of a search-by-image task
DELETE /LAPI/V1.0/Smart/SearchByImage/Task/<ID>	Delete a search-by-image task

2.2.3 Interface Calling Procedure

1. Search Pass-through Library



F1: Create a task.

F2: Set search criteria (image, library).

F3: Start analysis task

F4: Get task progress. Proceed to F5 when task execution is completed.

F5: Get paginated search results.

F6: Delete the task.

2. Search Face Library



F1: Create a task.

F2: Set search criteria (image, library).

F3: Start analysis task

F4: Get task progress. Proceed to F5 when task execution is completed.

F5: Get paginated search results.

F6: Delete the task.

Optional. You can continue using an existing task by updating search criteria (see blue steps above) if the task is executed but not deleted.

3 Interfaces

3.1 System

3.1.1 Time

1. /LAPI/V1.0/System/Time

URL	/LAPI/V1.0/System/Time
Description	Get device time.
Method	GET
Input Data	None
Success Return Data	TimeCfgInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/System/Time
Description	Set device time.
Method	PUT
Input Data	TimeCfgInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/System/Time/SyncMode

URL	/LAPI/V1.0/System/Time/SyncMode
Description	Get device's time sync mode.
Method	GET
Input Data	None
Success Return Data	SyncModeInfo
Note	Applicable to IPC.
Status	

URL	/LAPI/V1.0/System/Time/SyncMode
Description	Set device's time sync mode.
Method	PUT
Input Data	SyncModeInfo
Success Return Data	None
Note	Applicable to IPC.
Status	

3. /LAPI/V1.0/System/Time/NTP

URL	/LAPI/V1.0/System/Time/NTP
Description	Get NTP server info.
Method	GET
Input Data	None
Success Return Data	{ "Num": , "NTPServerInfos":[< NTPServerInfo >...] }
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/System/Time/NTP
Description	Set NTP server info.
Method	PUT
Input Data	{ "Num": , "NTPServerInfos":[< NTPServerInfo >...] }
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of NTP servers. Currently only one is allowed, so the maximum value of Num is 1.	1
NTPServerInfos	M	array	NTP server list.	
NTPServerInfo	M	unsigned long	NTP server info. See NTPServerInfo Json Block .	

4. /LAPI/V1.0/System/Time/DST

URL	/LAPI/V1.0/System/Time/DST
Description	Get DST info.
Method	GET
Input Data	None
Success Return Data	DSTCfgInfo
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/System/Time/DST
Description	Set DST info.
Method	PUT
Input Data	DSTCfgInfo
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.
Status	

3.1.2 Reboot

1. /LAPI/V1.0/System/Reboot

URL	/LAPI/V1.0/System/Reboot
Description	Restart device.
Method	PUT
Input Data	None
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.
Status	

3.1.3 Logs

1. /LAPI/V1.0/System/Logs

URL	/LAPI/V1.0/System/Logs
Description	Get logs.
Method	POST
Input Data	{ "Num": "QueryInfos": [< QueryInfo >, < QueryInfo >,...], "Limit": , "Offset": }
Success Return Data	{ "Total": , "Offset": "Num":, "Logs":[< Log >, < Log >, ...] }
Note	Applicable to NVR and VMS.
Status	

Param	Requirement	Type	Description	Example
-------	-------------	------	-------------	---------

Num(Input Data)	M	unsigned long	Number of search criteria. Must not exceed 16 types.	2
QueryInfos	C	array	Search criteria info list. Optional when Num is 0.	-
QueryInfo	M	Json Block	Search criteria info.	See QueryInfo Json Block.
Limit	M	unsigned long	Number of logs to search each time. Max 200.	100
Offset	M	unsigned long	Sequence number to start search from. The sequence number starts from 0.	0
Num(Return Data)	M	unsigned long	Number of logs returned for the page	100
Total	M	unsigned long	Total number of logs	2850
Logs	C	array	Log info list. Optional when Num is 0.	-
Log	M	Json Block	Log info. See Log Json Block.	-

2. /LAPI/V1.0/System/AlarmLogs

URL	/LAPI/V1.0/System/AlarmLogs
Description	Get alarm logs.
Method	POST
Input Data	<pre>{ "Num": "QueryInfos": [<QueryInfo>, <QueryInfo>,...], "Limit": , "Offset": }</pre>
Success Return Data	<pre>{ "Total":, "Offset":, "Num":, "AlarmLogs":[<AlarmLog>, <AlarmLog>...] }</pre>
Note	Applicable to VMS.
Status	

Param	Requirement	Type	Description	Example
Num(Input Data)	M	unsigned long	Number of search criteria. Must not exceed 16 types.	2
QueryInfos	C	array	Search criteria info list. Optional when Num is 0.	-

QueryInfo	M	Json Block	Search criteria info.	See QueryInfo Json Block.
Limit	M	unsigned long	Number of logs to search each time. Max 200.	100
Offset	M	unsigned long	Sequence number to start search from. The sequence number starts from 0.	0
Num(Return Data)	M	unsigned long	Number of alarm logs returned for the page.	100
Total	M	unsigned long	Total number of alarm logs.	2850
AlarmLogs	C	array	Alarm log info list. Optional when Num is 0.	-
AlarmLog	M	Json Block	Alarm log info. See AlarmLog Json Block.	-

3.1.4 DeviceInfo

1. /LAPI/V1.0/System/DeviceInfo

URL	/LAPI/V1.0/System/DeviceInfo
Description	Get device info.
Method	GET
Input Data	None
Success Return Data	DeviceInfo
Notes	It is meaningless if the ID in the return value of DeviceInfo is 0. Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/System/DeviceInfo

URL	/LAPI/V1.0/Channels/<ID>/System/DeviceInfo
Description	Get info about the device connected to a channel.
Method	GET
Input Data	None
Success Return Data	DeviceInfo
Notes	Applicable to NVR.
Status	

3. /LAPI/V1.0/Channels/System/DeviceInfos

URL	/LAPI/V1.0/Channels/System/DeviceInfos
------------	--

Description	Get device info of all the channels.
Method	GET
Input Data	None
Success Return Data	DeviceInfoList
Notes	Applicable to NVR.
Status	

3.1.5 ChannelInfos

1. /LAPI/V1.0/Channels/System/ChannelDetailInfos

URL	/LAPI/V1.0/Channels/System/ChannelDetailInfos
Description	Get detailed info of all channels.
Method	GET
Input Data	None
Success Return Data	<pre>{ "Nums": , "DetailInfos": [<DetailInfo>, <DetailInfo>, <DetailInfo>, ...] }</pre>
Note	Applicable to NVR.
Status	

Param	Requirement	Type	Description	Example
Nums	M	unsigned long	Total number of video channels of the device.	3
DetailInfos	C	Array	Detailed video channel info list. Optional when Num is 0.	-
DetailInfo	M	Json Block	Detailed video channel info. Optional when Nums is 0.	See DetailInfo Json Block.

3.1.6 Upgrade

1. /LAPI/V1.0/System/Upgrade

URL	/LAPI/V1.0/System/Upgrade
Description	Upgrade the firmware of a device.
Method	PUT
Input Data	<pre>{ "UpgradeType": }</pre>
Success Return Data	<pre>{ "UpgradeTaskID": }</pre>
Note	To perform a local upgrade, use /LAPI/V1.0/System/UploadFirmware to upload the

	firmware version. Use the /LAPI/V1.0/System/UpgradeStatus interface to get upgrade status. Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
UpgradeType	M	unsigned long	Upgrade type: 1: Cloud upgrade 2: By USB drive 3: Local upgrade	3
UpgradeTaskID	C	string	Upgrade task ID. Used to verify the upgrade task. Required when upgrading NVR locally.	"2b188bd214ae1e2b0d96d5b7636d3bd0@1504663113"

2. /LAPI/V1.0/System/UploadFirmware?UpgradeTaskID=<UpgradeTaskID>

URL	/LAPI/V1.0/System/UploadFirmware?UpgradeTaskID=<UpgradeTaskID>
Description	Upload firmware version for device upgrade.
Method	POST
Message body	Accept-Encoding: gzip, deflate Content-Length: 25665652 Content-Type: multipart/form-data; boundary=-----5851473516029 -----5851473516029 Content-Disposition: form-data; name="UpgradeFile"; filename="Program.bin" Content-Type: application/octet-stream File content -----5851473516029—
Note	Note: The lines started with Content-Dispostion and Content-Type and the blank line are contents of the standard HTTP message body. Use the /LAPI/V1.0/System/UpgradeStatus interface to get upgrade status. Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
UpgradeTaskID	C	string	Upgrade task ID. Used to verify the upgrade task. Required when upgrading NVR locally. Returned by the /LAPI/V1.0/System/Upgrade interface.	"2b188bd214ae1e2b0d96d5b7636d3bd0@1504663113"

3. /LAPI/V1.0/System/UpgradeStatus

URL	/LAPI/V1.0/System/UpgradeStatus
Description	Get device upgrade status.
Method	GET
Input Data	None
Success Return Data	UpgradeStatusInfo
Note	Applicable to IPC and NVR.
Status	

4. /LAPI/V1.0/Channels/System/Upgrade

URL	/LAPI/V1.0/Channels/System/Upgrade
Description	Upgrade firmware of the device connected to a video channel.
Method	PUT
Input Data	{ "Channels":[<ID>,<ID>,...], "UpgradeType": }
Success Return Data	{ "UpgradeTaskID": }
Note	To perform a local upgrade, use /LAPI/V1.0/Channels/System/UploadFirmware?UpgradeTaskID=<UpgradeTaskID> to upgrade the firmware version. Use the /LAPI/V1.0/Channels/System/UpgradeStatuses interface to get upgrade status. Applicable to NVR.
Status	

Param	Requirement	Type	Description	Example
Channels	M	array	Channel ID list. Used only to locally upgrade a device connected to a channel.	
ID	M	unsigned long	Video channel ID.	1
UpgradeType	M	unsigned long	Upgrade type: 1: Cloud upgrade 2: By USB drive 3. Local upgrade	1
UpgradeTaskID	C	string	Upgrade task ID. Required when locally upgrading IPC on NVR.	"956087319"

5. /LAPI/V1.0/Channels/System/UploadFirmware?UpgradeTaskID=<UpgradeTaskID>

URL	/LAPI/V1.0/Channels/System/UploadFirmware?UpgradeTaskID=<UpgradeTaskID>
Description	Upload the firmware version for the device connected to the corresponding channel.
Method	POST
Message body	Accept-Encoding: gzip, deflate Content-Length: 14231621 Content-Type: multipart/form-data; boundary=-----178201537128716 -----178201537128716 Content-Disposition: form-data; name="UpgradeFile"; filename="IPC_D1201-B5022P20D1702_JPEG.zip" Content-Type: application/octet-stream File content -----178201537128716—
Success Return Data	None
Note	Note: The lines started with Content-Disposition and Content-Type and the blank line are contents of the standard HTTP message body. Use the /LAPI/V1.0/Channels/System/UpgradeStatus interface to get upgrade status. Applicable to NVR.
Status	

Param	Requirement	Type	Description	Example
UpgradeTaskID	C	string	Upgrade task ID. Used to verify the upgrade task. Required when locally upgrading IPC on NVR. Returned by the /LAPI/V1.0/Channels/System/Upgrade interface.	"956087319"

6. /LAPI/V1.0/Channels/System/UpgradeStatuses

URL	/LAPI/V1.0/Channels/System/UpgradeStatuses
Description	Get upgrade status of the device connected to a video channel.
Method	GET
Input Data	None
Success Return Data	UpgradeStatusInfoList
Note	Applicable to NVR.
Status	

3.1.7 Json Block

1. DeviceInfo

DeviceInfo Json Block	{ "ID";, "DeviceName";, "DeviceType";, "DeviceCode";, "DeviceModel";, "SerialNumber";, "FirmwareVersion";, "HardwareID";, "UbootVersion"; }
----------------------------------	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Video channel ID.	1
DeviceName	M	string	Device name.	"NVR-B200-E4"
DeviceType	C	unsigned long	Device type. 0: IPC 1: NVR 2: DVR 3: VMS Applicable to NVR.	0
DeviceCode	M	unsigned long	Device ID	12
DeviceModel	M	string	Device model. Length: 1-63.	"NVR-B200-E4"
SerialNumber	M	string	Device serial number. Length: 0-25.	"210235C1MPF1580 00503"
FirmwareVersion	M	string	Firmware version, program version. Length: 0-255.	"R2318P05"
HardwareID	M	string	Hardware ID. Length: 0-255.	"A"
UbootVersion	M	string	UBOOT version. Length: 0- 255.	"V1.4"

2. TimeCfgInfo

TimeCfgInfo Json Block	{ "TimeZone": , "DeviceTime": , "DateFormat": , "HourFormat": }
-----------------------------------	--

Param	Requirement	Type	Description	Example
-------	-------------	------	-------------	---------

TimeZone	M	string	Time zone string. Range: [1,255] "GMT-12:00" "GMT-11:00" "GMT-10:00" "GMT-09:00" "GMT-08:00" "GMT-07:00" "GMT-06:00" "GMT-05:00" "GMT-04:30" "GMT-04:00" "GMT-03:30" "GMT-03:00" "GMT-02:00" "GMT-01:00" "GMT-00:00" "GMT+01:00" "GMT+02:00" "GMT+03:00" "GMT+03:30" "GMT+04:00" "GMT+04:30" "GMT+05:00" "GMT+05:30" "GMT+05:45" "GMT+06:00" "GMT+06:30" "GMT+07:00" "GMT+08:00" "GMT+09:00" "GMT+09:30" "GMT+10:00" "GMT+11:00" "GMT+12:00" "GMT+13:00"	"GMT+08:00"
DeviceTime	M	unsigned long	UTC time (unit: second)	1477104949
DateFormat	C	unsigned long	Date format: 0: YYYY-MM-DD 1: MM-DD-YYYY 2: DD-MM-YYYY Required for NVR.	0
HourFormat	C	unsigned long	Time format: 0: 12 hour format 1: 24 hour format Required for NVR.	1

3. DeviceInfoList

DeviceInfoList Json Block	{ "Nums": , "DeviceInfos": [< DeviceInfo >] }
--------------------------------------	--

Param	Requirement	Type	Description	Example
Nums	M	unsigned long	Number of channel IDs with connected devices.	1
DeviceInfos	C	array	Device info list. Optional when Nums is 0.	
DeviceInfo	C	Json Block	Device name. Optional. See DeviceInfo Json Block.	

4. DetailInfo

DetailInfo Json Block	{ "ID": , "Name": , "Status": , "IsPoEPort": , "PoEStatus": , "StreamNums": , "DeviceType": , "AddressInfo": AddressInfo , "AccessProtocol": , "OffReason": , "RemoteIndex": , "Manufacturer": , "DeviceModel": , "GBID": , "AddType": , }
----------------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Channel ID.	1
Name	M	string	Channel name. Range: [0, 63].	"IP Camera 01"
Status	M	unsigned long	Channel status: 0: Offline 1: Online 2: Idle	0
IsPoEPort	M	unsigned long	Whether is PoE port: 0: No 1: Yes Required by NVR.	1

PoEStatus	C	unsigned long	PoE port status: 1: Normal 2: Insufficient power 4: The load exceeds the rated power. 8: No need to supply power. Absent when IsPoEPort is 0.	1
StreamNums	C	unsigned long	Number of media streams supported. Absent when Status is 2.	3
DeviceType	C	unsigned long	Type of device connected to the channel. 0: Fixed camera 1: PTZ camera 2: Fisheye camera (fisheye lens with dewarping function) 3: Wide-angle lens camera (including fisheye or wide-angle lens but without dewarping function) Required by NVR. Absent when Status is 2.	0
AddressInfo	C	Json Block	Device address info. Absent when Status is 2. See Addressinfo Json Block	
AccessProtocol	C	unsigned long	Access protocol: 1: ONVIF 2: Private 3: GB/T28181 100: Custom 1 101: Custom 2 102: Custom 3 103: Custom 4 104: Custom 5 105: Custom 6 106: Custom 7 107: Custom 8 108: Custom 9 109: Custom 10 110: Custom 11 111: Custom 12 112: Custom 13 113: Custom 14 114: Custom 15 115: Custom 16 116: Custom 17 117: Custom 18 118: Custom 19	1

			119: Custom 20 120: Custom 21 121: Custom 22 122: Custom 23 123: Custom 24 124: Custom 25 125: Custom 26 126: Custom 27 127: Custom 28 128: Custom 29 129: Custom 30 130: Custom 31 131: Custom 32 Required by NVR. Absent when Status is 2.	
OffReason	C	unsigned long	Cause of offline device: 0: Offline (connecting) 1: Online 2: Offline (incorrect username or password) 3: Offline (network disconnection) 4: Offline (failed to set live stream) 5: Offline (failed to start live stream) 6: Offline (media stream interruption) 7: Offline (timeout) 8: Offline (IPC connected via GB/T28181 is waiting for registration packets) 9: Offline (IPC connected via GB/T28181 changed stream transport protocol) 10: Offline (PoE power error) 11: Offline (bandwidth not enough) 12: Offline (access denied for weak password) Required by NVR. Absent when Status is 2.	1
RemoteIndex	C	unsigned long	Remote channel ID (channel ID of multi-channel device) Required by NVR. Absent when Status is 2.	1
Manufacturer	C	string	Manufacturer. Range: [0, 31] Required by NVR.	Uniview

			Absent when Status is 2.	
DeviceModel	C	String	Device model. Range: [0, 31] Required by NVR. Absent when Status is 2.	IPC342E-DLVIR
GBID	C	String	GB/T28181 resource ID. Range: [0, 31]	
AddType	C	unsigned long	Adding type: 0: Add manually 1: Plug and Play Required by NVR. Absent when Status is 2.	0

5. AddressInfo

AddressInfo Json Block	{ "AddressType": , "Address": , "Port": , "AccessAddress": }
-----------------------------------	---

Param	Requirement	Type	Description	Example
AddressType	M	unsigned long	IP address type. 0: IPv4 1: IPv6	0
Address	M	string	Device IP address. Range: [0,128]. Required by NVR.	206.5.99.17
Port	M	unsigned long	Device port. Required by NVR.	80
AccessAddress	M	string	Device access address. Range: [0, 63]. Required for NVR.	http://206.5.99.17

6. SyncModelInfo

SyncModelInfo Json Block	{ "Mode": }
-------------------------------------	-------------------

Param	Requirement	Type	Description	Example
Mode	M	unsigned long	Time sync mode 0: Sync time with any server 1: Sync time with computer's system time 2: Sync time with photo server	0

			3: Sync time with NTP server 4: Sync time with management server (non-Onvif) 5: Sync time with management server (Onvif)	
--	--	--	--	--

7. NTPServerInfo

NTPServerInfo Json Block	{ "Enabled": , "AddressType": , "IPAddress": , "DomainName": , "Port": , "SynchronizeInterval": }
-------------------------------------	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable NTP server 0: Disable 1: Enable	0
AddressType	M	unsigned long	Address type 0: IPv4 1: IPv6 (currently unavailable) 2: Domain name (applicable to NVR and VMS).	0
IPAddress	C	string	NTP server IP address. Length: [0, 64]. Required when AddressType is 0.	0.0.0.0
DomainName	C	string	NTP server domain name. Length: [0, 64]. Required when AddressType is 2.	pool.ntp.org
Port	C	unsigned long	NTP port. Range: [1-65535] Configuration is not supported by IPC.	123
SynchronizeInterval	M	unsigned long	Sync interval. Range for NVR and VMS: 5/10/15/30 minutes, 1/2/3/6/12h, 1 day, 1 week Range for IPC: 30-3600s	10

			All the time needs to be converted to seconds.	
--	--	--	--	--

8. DSTCfgInfo

DSTCfgInfo Json Block	<pre> { "Enabled": , "BeginTime": { "Month": , "WeekNum": , "WeekDay": , "Hour": }, "EndTime": { "Month": , "WeekNum": , "WeekDay": , "Hour": }, "DSTBias": } </pre>
----------------------------------	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable DST. 0: Disable 1: Enable	1
BeginTime	M	Json Block	DST start time	-
Month	M	unsigned long	Start month (compliant with ISO8601 standards) 1: January 2: February 3: March 4: April 5: May 6: June 7: July 8: August 9: September 10: October 11: November 12: December	2
WeekNum	M	unsigned long	Which week of the start month 1: First week 2: Second week	1

			3: Third week 4: Fourth week 5: Last week	
WeekDay	M	unsigned long	Which day of the xx week of the start month (compliant with ISO8601 standards) 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday 7: Sunday	6
Hour	M	unsigned long	Start time (unit: hour, 24 hour format), compliant with ISO8601 standards: 0: 00:00 1: 01:00 2: 02:00 3: 03:00 4: 04:00 5: 05:00 6: 06:00 7: 07:00 8: 08:00 9: 09:00 10: 10:00 11: 11:00 12: 12:00 13: 13:00 14: 14:00 15: 15:00 16: 16:00 17: 17:00 18: 18:00 19: 19:00 20: 20:00 21: 21:00 22: 22:00 23: 23:00	0
EndTime	M	Json Block	DST end time	-
DSTBias	M	unsigned long	Time offset (unit: minute). Options: 30, 60, 90, 120	60

9. QueryInfo

QueryInfo Json Block	<pre>{ "QryType": , "QryCondition": , "QryData": }</pre>
---------------------------------	--

Param	Requirement	Type	Description	Example
QryType	M	unsigned long	Search criteria: 0: User name 1: Organization name 2: Device name 3: Channel name 4: Time (unit: second, UTC format) 5: Service type (equivalent to main type) 6: Operation type (equivalent to sub type) 7: Operation object 8: Alarm type 9: Alarm source name 10: Alarm level 11: Alarm status (acknowledged or not) 12: User who acknowledged the alarm 13: Time when alarm was acknowledged 14: Alarm device ID 15: Alarm channel ID 16: Alarm sub type 17: Server Only Time, Service Type, and Operation Type are applicable to NVR. On VMS, operation logs support search by username, time, service type, operation type, and operation object; alarm logs support search by time, alarm type, alarm source name, alarm level, alarm acknowledgement status/user/time, alarm device ID, alarm channel ID, alarm sub type, and server.	4

QryCondition	M	unsigned long	Logical type of search criteria. 0: Equal to 1: Greater than 2: Less than 3: Greater than or equal to 4: Less than or equal to 5: Not equal to 6: Fuzzy matching 7: IN GROUP 8: Ascending order 9: Descending order For NVR, only “equal to”, “greater than or equal to”, and “less than or equal to” are applicable. For VMS, all are applicable.	0
QryData	C	string	Rvalue of search criteria. Alarm types are represented using character strings. When searching by multiple alarm types (up to 16 allowed), separate alarm types by comma. Optional when QryCondition is 6 or 7. NVR only supports search by one type.	"DiskOffline"

10. Log

Log Json Block	<pre>{ "Time": , "MainType": , "SubType": , "ID": , "LoginName": , "IP": , "DetailInfo": , "OperObject": , "Result": }</pre>
----------------	--

Param	Requirement	Type	Description	Example
Time	M	unsigned long	Log occurrence time (unit: second, UTC format)	1476088399
MainType	M	string	Log main type, See Alarm&Operate Log Type .	"alarm"
SubType	M	string	Log sub type, See Alarm&Operate Log Type .	"motionDetectionStart"

ID	C	unsigned long	Required when the logs involve a channel ID. 0 means the device itself; from 1 on it indicates a specific channel ID. When involving input/output alarm configuration and alarms, it means input/output alarm ID. For NVR, the ID is channel ID*100+alarm input ID. For example, the channel ID of the device itself is 0. 101 means the first input/output alarm ID of the first channel. For VMS, the ID is configured on the VMS.	1
LoginName	M	string	Operating user. Length range: [0, 64]	Admin
IP	M	string	Operating user's IP. Length range: [0, 64]	206.5.3.13
DetailInfo	C	string	Detailed info. Currently only PoE port status is displayed. Length range: [0, 64] Required by NVR.	
OperObject	C	string	Operation object. Length range: [0, 64] Required by VMS.	bsdkabik
Result	C	unsigned long	Operation result. Required by VMS.	1234

11. AlarmLog

AlarmLog Json Block	<pre>{ "AlarmID": , "AlarmType":, "AlarmSubType":, "AlarmLevel":, "ServerID":, "DevID":, "ChlID":, "AlarmSrc":, "AlarmTime":, "AlarmChecked":, "AlarmCheckUser":, "AlarmCheckTime":, "AlarmCheckDesc":</pre>
----------------------------	--

	}
--	---

Param	Requirement	Type	Description	Example
AlarmID	M	unsigned long	Alarm log ID.	
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"alarm"
AlarmSubType	O	string	Alarm sub type. See Alarm&Operate Log Type .	"motionDetectionStart"
AlarmLevel	M	unsigned long	Alarm level. 0: Critical 1: Major 2: Minor 3: Warning 4: Prompt	0
ServerID	M	unsigned long	Server ID.	3
DevID	M	unsigned long	Device ID.	1
ChlID	M	unsigned long	Channel ID.	831
AlarmSrc	M	string	Alarm source info. Length range: [1, 128].	"Camera1"
AlarmTime	M	unsigned long	Alarm occurrence time (unit: second, UTC format)	
AlarmChecked	M	boolean	Whether alarm is acknowledged. 0: No 1: Yes	1
AlarmCheckUser	M	string	User who acknowledged alarm. Length range: [0, 64].	"admin"
AlarmCheckTime	M	unsigned long	Alarm acknowledgement time (unit: second, UTC format)	1476088399
AlarmCheckDesc	M	string	Remarks given when acknowledging an alarm. Length range: [0, 516]	

12. Alarm&Operate Log Type

To be added the attachment



Alarm&Operate
Log Type.xlsx

13. AlarmInfo

AlarmInfo Json Block	<pre>{ "AlarmType": , "TimeStamp": , "AlarmSeq": , "AlarmSrcID": , "AlarmSrcType":, "AlarmSrcName": }</pre>
---------------------------------	---

Param	Requirement	Type	Description	Example
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"MotionAlarmOn"
TimeStamp	M	unsigned long	Alarm time.	1489040894
AlarmSeq	C	unsigned long	Alarm sequence number. Applicable to VMS and IPC.	327
AlarmSrcID	C	unsigned long	Resource ID. Applicable to NVR and VMS. For NVR, the following rules are applicable: 1. AlarmSrcType is 0-7, AlarmSrcID indicates HDD index. 2. AlarmSrcType is 8, AlarmSrcID is video channel ID. 3. AlarmSrcType is 9, AlarmSrcID is alarm input channel ID. 4. AlarmSrcType is 10, AlarmSrcID is 0 by default and indicates the system itself.	860
AlarmSrcType	C	unsigned long	Type of alarm source: 0: Local HDD 1: Disk enclosure-1 HDD 2: Disk enclosure-2 HDD 3: RAID HDD 4: NAS HDD 5: SAN HDD 6: eSATA HDD	2

			7: SD card 8: Video channel 9: Alarm input channel 10: System Applicable to NVR.	
AlarmSrcName	C	string	Alarm source name. Length range: [0, 128]. Applicable to VMS.	"206.9.252.7_V_6"

14. ResInfo

ResInfo Json Block	<pre>{ "Reference": , "Action":, "Num":, "ResInfos": [{ "ResType": , "ResID": },] }</pre>
-------------------------------	---

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"http://192.168.0.13:80/LAPI/V1.0/System/ALarmEvent/CommonAlarm/<SubscribersID>"
Action	M	unsigned long	Sub event type of action to change resource: 0: Add 1: Delete 2: Edit 3: Online 4: Offline 5: Alarm notification by E-map	0
Num	M	unsigned long	Number of changed resources.	5
ResInfos	M	array	Info about resource data changes.	
ResType	M	unsigned long	Resource type. 0: User resource 1: Device resource 2: Channel resource 3: User permission resource	2

			4: Sequence resource 5: Hotspot resource 6: Hot zone resource 7: Alarm resource 8: Time template 9: System permission resource 10: Device permission resource 11: Organization resource 12: Alarm task resource 13: Primary/replica resource 0xff: Invalid resource type	
ResID	M	unsigned long	Resource ID.	1206

15. VMSCCommon

VMSCCommon Json Block	<pre> { "OrgID": , "DevID": , "DevName": , "ChlIndex": , "Permission": } </pre>
----------------------------------	---

Param	Requirement	Type	Description	Example
OrgID	M	unsigned long	Organization ID.	1
DevID	M	unsigned long	Parent device ID.	1
DevName	M	string	Parent device name. Length range: [1, 256]	"123"
ChlIndex	M	unsigned long	Channel index.	1
Permission	M	unsigned long	Channel permission.	214

16. UpgradeStatusInfo

UpgradeStatusInfo Json Block	<pre> { "ID": "Status" ;, "Percent": } </pre>
---	---

Param	Requirement	Type	Description	Example
ID	O	unsigned long	The value is meaningless in /LAPI/V1.0/System/UpgradeStatus interface. The default is 0.	1

			In /LAPI/V1.0/Channels/System/UpgradeStatuses, the value is the video channel ID.	
Status	M	unsigned long	Upgrade status. Currently these statuses are supported: 0: Initializing 1: Searching 2: Downloading 3: Transferring 4: Upgrading 5: Upgrade error 6: Already the latest 7: Higher version is available 8: Upgrade completed 9: Failed to get upgrade progress 10: Version does not match 11: Cannot load multiple upgrade processes simultaneously 12: Timeout occurred while getting upgrade progress	2
Percent	C	unsigned long	Progress of the statuses. Range: [0,100]. Required when Status is 2, 3 or 4.	50

17. UpgradeStatusInfoList

UpgradeStatusInfoList	Json Block	{ "Nums":, "UpgradeStatuses":[< UpgradeStatusInfo >,< UpgradeStatusInfo >,.....] }
------------------------------	-------------------	---

Param	Requirement	Type	Description	Example
Nums	M	unsigned long	Number of IPC channels.	1
UpgradeStatuses	C	array	Status info list. Required when Nums is greater than 0.	-
UpgradeStatusInfo	C	Json Block	Status info. Required when Nums is greater than 0.	See UpgradeStatusInfo Json Block.

3.2 PTZ

3.2.1 PTZCtrl

1. /LAPI/V1.0/Channels/<ID>/PTZ/PTZCtrl

URL	/LAPI/V1.0/Channels/<ID>/PTZ/PTZCtrl
Description	Operate PTZ of a channel.
Method	PUT
Input Data	PTZCommandInfo
Success Return Data	None
Note	For IPC, the ID is always 0. Applicable to IPC, NVR, and VMS.
Status	

3.2.2 Presets

1. /LAPI/V1.0/Channels/<ID>/PTZ/Presets

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Presets
Description	Get PTZ presets of a video channel.
Method	GET
Input Data	None
Success Return Data	PresetInfoList
Note	For IPC, the ID is always 0. Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Presets
Description	Add a PTZ preset for a video channel.
Method	POST
Input Data	PresetInfo
Success Return Data	None
Note	For IPC, the ID is always 0. Applicable to IPC and VMS.
Status	

2. /LAPI/V1.0/Channels/<ID>/PTZ/Presets/<ID>

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Presets/<ID>
Description	Edit a PTZ preset of a video channel.
Method	PUT

Input Data	PresetInfo
Success Return Data	None
Note	For IPC, the ID of the channels is always 0. Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Presets/<ID>
Description	Delete a PTZ preset of a video channel.
Method	DELETE
Input Data	None
Success Return Data	None
Note	For IPC, the ID of the channels is always 0. Applicable to IPC, NVR, and VMS.
Status	

3. /LAPI/V1.0/Channels/<ID>/PTZ/Presets/<ID>/Goto

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Presets/<ID>/Goto
Description	Rotate PTZ of a channel to a preset.
Method	PUT
Input Data	None
Success Return Data	None
Note	For IPC, the ID of the channels is always 0. Applicable to IPC, NVR, and VMS.
Status	

3.2.3 Capabilities

1. /LAPI/V1.0/Channels/<ID>/PTZ/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Capabilities
Description	Get PTZ capability of a channel.
Method	GET
Input Data	None
Success Return Data	PTZCapabilityInfo
Note	Applicable to IPC and NVR.
Status	

3.2.4 AreaZoom

1. /LAPI/V1.0/Channels/<ID>/PTZ/AreaZoomIn

URL	/LAPI/V1.0/Channels/<ID>/PTZ/AreaZoomIn
Description	Perform drag-to-zoom-in to a channel.
Method	PUT
Input Data	PTZZoomAreaInfo
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.
Status	

2. /LAPI/V1.0/Channels/<ID>/PTZ/AreaZoomOut

URL	/LAPI/V1.0/Channels/<ID>/PTZ/AreaZoomOut
Description	Perform drag-to-zoom-out to a channel.
Method	PUT
Input Data	PTZZoomAreaInfo
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.
Status	

3.2.5 Patrols

1. /LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>/Start

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>/Start
Description	Start a patrol route of a channel.
Method	PUT
Input Data	None
Success Return Data	None
Note	The corresponding ID of Patrols is the patrol route ID. Applicable to IPC, NVR, and VMS.
Status	

2. /LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>/Stop

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>/Stop
Description	Stop a patrol route of a channel.
Method	PUT

Input Data	None
Success Return Data	None
Note	The corresponding ID of Patrols is the patrol route ID. Applicable to IPC, NVR, and VMS.
Status	

3. /LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>
Description	Get a patrol route of a channel.
Method	GET
Input Data	None
Success Return Data	PatrolInfo
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>
Description	Edit a patrol route of a channel.
Method	PUT
Input Data	PatrolInfo
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols/<ID>
Description	Delete a patrol route of a channel.
Method	DELETE
Input Data	None
Success Return Data	None
Note	The corresponding ID of Patrols is the patrol route ID. Applicable to IPC and VMS.
Status	

4. /LAPI/V1.0/Channels/<ID>/PTZ/Patrols

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols
Description	Get all preset patrol routes of a channel.
Method	GET
Input Data	None
Success Return Data	PatrolInfoList
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols
Description	Edit preset patrol routes of a channel in batches.
Method	PUT
Input Data	PatrollInfoList
Success Return Data	None
Note	Applicable to NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/PTZ/Patrols
Description	Add a preset patrol route for a channel.
Method	POST
Input Data	PatrollInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

5. /LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/Start

URL	/LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/Start
Description	Start a recorded patrol route of a channel.
Method	PUT
Input Data	None
Success Return Data	None
Note	The corresponding ID of RecordedPatrols is the recorded patrol route ID. Applicable to IPC, NVR, and VMS.
Status	

6. /LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/Stop

URL	/LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/Stop
Description	Stop a recorded patrol route of a channel.
Method	PUT
Input Data	None
Success Return Data	None
Note	The corresponding ID of RecordedPatrols is the recorded patrol route ID. Applicable to IPC, NVR, and VMS.
Status	

7. /LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/StartRecord

URL	/LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/StartRecord
Description	Start recording a patrol route for a channel.
Method	PUT
Input Data	None

Success Return Data	None
Note	The corresponding ID of RecordedPatrols is the recorded patrol route ID. Applicable to IPC, NVR, and VMS.
Status	

8. /LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/StopRecord

URL	/LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>/StopRecord
Description	Stop recording a patrol route for a channel.
Method	PUT
Input Data	None
Success Return Data	None
Note	The corresponding ID of RecordedPatrols is the recorded patrol route ID. Applicable to IPC, NVR, and VMS.
Status	

9. /LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>

URL	/LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols/<ID>
Description	Delete a recorded patrol route of a channel.
Method	DELETE
Input Data	None
Success Return Data	None
Note	The corresponding ID of RecordedPatrols is the recorded patrol route ID. Applicable to IPC and VMS.
Status	

10. /LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols

URL	/LAPI/V1.0/Channels/<ID>/PTZ/RecordedPatrols
Description	Get all the recorded patrol routes of a channel.
Method	GET
Input Data	None
Success Return Data	RecordedPatrolInfoList
Note	Applicable to IPC and NVR.
Status	

3.2.6 Json Block

1. PTZCommandInfo

PTZCommandInfo Json Block	{ "PTZCmd":, "Para1":, "Para2":, "Para3": }
----------------------------------	--

Param	Requirement	Type	Description	Example
PTZCmd	M	unsigned long	PTZ operation commands: 0x0101: Stop closing aperture 0x0102: Close aperture 0x0103: Stop opening aperture 0x0104: Open aperture 0x0201: Stop focusing near 0x0202: Focus near 0x0203: Stop focusing far 0x0204: Focus far 0x0301: Stop zooming in 0x0302: Zoom in 0x0303: Stop zooming out 0x0304: Zoom out 0x0402: Up 0x0404: Down 0x0502: Right 0x0504: Left 0x0702: Up left 0x0704: Down left 0x0802: Up right 0x0804: Down right 0x0901: Stop 0x0A01: Enable wiper 0x0A02: Disable wiper 0x0B01: Enable light 0x0B02: Disable light 0x0C01: Enable heater 0x0C02: Disable heater 0x0D01: Enable IR 0x0D02: Disable IR	1794 (0x0702)
Para1	C	unsigned long	PTZ command parameter 1, pan speed. Range: [1,9]. Only for use when PTZCmd is 0x402 – 0x804. When turning up (0x402) or	2

			turning down (0x404), the pan speed is 0. Optional when PTZCmd is not in the range 0x402–0x804.	
Para2	C	unsigned long	PTZ command parameter 2, tilt speed. Range: [1,9]. Only for use when PTZCmd is 0x402 – 0x804. When turning left (0x504) or turning right (0x502), the tilt speed is 0. Optional when PTZCmd is not in the range 0x402–0x804.	2
Para3	C	unsigned int	PTZ command parameter 3. Currently not in use. Enter 0.	0

2. PresetInfoList

PresetInfoList Json Block	{ "Nums": , "PresetInfos": [< PresetInfo >,< PresetInfo >,...] }
----------------------------------	---

Param	Requirement	Type	Description	Example
Nums	M	unsigned long	Number of presets. Max 255.	3
PresetInfos	C	array	Preset list. Optional when Nums is 0.	
PresetInfo	C	Json Block	Preset info. See PresetInfo Json Block. Optional when Nums is 0.	

3. PresetInfo

PresetInfo Json Block	{ "ID":, "Name": }
------------------------------	-----------------------------

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Preset ID, which is generated by the device when the preset is added. Starts from 1. Max 255.	1
Name	M	string	Preset name. Length: [1,64]. For NVR, the preset name is the default value and is not configurable.	"Preset1"

4. PTZCapabilityInfo

PTZCapInfo Json Block	{ "IsSupportPTZ": , "IsSupportRecordPatrol": }
----------------------------------	---

Param	Requirement	Type	Description	Example
IsSupportPTZ	M	boolean	Whether supports PTZ: 0: No 1: Yes	0
IsSupportRecordPatrol	M	boolean	Whether supports recorded patrol: 0: No 1: Yes	1
Remarks: Other parameters will not be released temporarily.				

5. PTZZoomAreaInfo

PTZZoomAreaInfoList Json Block	{ "CenterPointX":, "CenterPointY":, "LengthX":, "LengthY":, "Width": , "Height": }
---	---

Param	Requirement	Type	Description	Example
CenterPointX	M	unsigned long	x-coordinate of the center point of the zoomed area. Must not be greater than Width.	500
CenterPointY	M	unsigned long	y-coordinate of the center point of the zoomed area. Must not be greater than Height.	500
LengthX	M	unsigned long	Length of the zoomed area	0
LengthY	M	unsigned long	Width of the zoomed area	0
Width	M	unsigned long	Window length	1300
Height	M	unsigned long	Window height	1300

6. PatrolInfoList

PatrolInfoList Json Block	{ "Num": ,
--------------------------------------	---------------

	"PatrollInfos": [< PatrollInfo >, ...] }
--	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of existing patrol routes.	4
PatrollInfos	C	array	Patrol route info.	

7. PatrolInfo

PatrollInfo Json Block	{ "ID": , "Name": , "Num": , "Actions": [< Action >, ...] }
-------------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Patrol route ID (starts from 0). Set by using the retrieved ID.	0
Name	C	string	Patrol route name. Length range: [1,64]. Applicable to IPC and VMS.	
Num	M	unsigned long	Number of actions while operating PTZ	2
Actions	M	array	PTZ operation action info. See Action Json Block.	

8. Action

Action Json Block	{ "ID": , "Type": , "Para1": , "Para2": , "Para3": , "PresetID": , "Duration": , }
--------------------------	--

Param	Requirement	Type	Description	Example
ID	C	unsigned long	Starts from 0, up to (Num-1) in PatrollInfo . Applicable to NVR and VMS.	0
Type	M	unsigned long	Action type: 0x0302: Zoom to 0x0402: Turn up 0x0404: Turn down 0x0502: Turn right 0x0504: Turn left	0

			0x0602: Go to preset 0x0702: Turn up left 0x0704: Turn down left 0x0802: Turn up right 0x0804: Turn down right 0x0901: Stop NVR and VMS only support “Go to preset”.	
Para1	C	unsigned long	PTZ command parameter 1, pan speed. Range: [1,9]. Only for use when PTZCmd is 0x402 – 0x804. When turning up (0x402) or turning down (0x404), the pan speed is 0. Optional when PTZCmd is not in the range 0x402–0x804. Used by preset speed.	5
Para2	C	unsigned long	PTZ command parameter 2, tilt speed. Range: [1,9]. Only for use when PTZCmd is 0x402 – 0x804. When turning left (0x504) or turning right (0x502), the tilt speed is 0. Required when PTZCmd is 0x402 or 0x404. When PTZCmd is in the range 0x702-0x804, use Para1; Para2 data is invalid.	5
Para3	C	unsigned long	Lens’ zoom ratio*100 Required when Type is 0x302 or 0x304.	2
PresetID	C	unsigned long	Preset ID. Optional when Type is not 0x602.	1
Duration	C	unsigned long	Action duration (unit: millisecond) when the action type is not “Preset”. 0 means keeping the action. Range: [0,1800000]. NVR and VMS only supports prevision to seconds. For NVR and VMS, the value must be set to seconds in integer and then converted to milliseconds.	10

9. RecordedPatrolInfoList

RecordedPatrolInfoList Json Block	<pre>{ "Num": , "RecordedPatrolInfos": [<RecordedPatrolInfo>] }</pre>
--	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of recorded patrol routes.	1
RecordedPatrolInfos	C	array	Info about recorded patrol route.	

10. RecordedPatrolInfo

RecordedPatrolInfo Json Block	{ "ID": , "Name": }
--------------------------------------	------------------------------

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Recorded patrol route ID. Set by using the retrieved ID.	1
Name	C	string	Name of recorded patrol route. Applicable to VMS.	"PatrolInfo1"

3.3 Media

3.3.1 Capabilities

1. /LAPI/V1.0/Channels/<ID>/Media/Video/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Capabilities
Description	Get video capability of a video channel.
Method	GET
Input Data	None
Success Return Data	VideoCapabilityInfo
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Media/Audio/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Media/Audio/Capabilities
Description	Get audio capability of a video channel.
Method	GET
Input Data	None
Success Return Data	AudioCapabilityInfo
Note	Applicable to IPC.

3. /LAPI/V1.0/Channels/<ID>/Media/Snapshot/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Media/Snapshot/Capabilities
Description	Get snapshot capability of a video channel.
Method	GET
Input Data	None

Success Return Data	SnapshotCapabilities
Note	Applicable to IPC.
Status	

3.3.2 Video

1. /LAPI/V1.0/Channels/<ID>/Media/Video/Streams/DetailInfos

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/DetailInfos
Description	Get all video streams of a video channel.
Method	GET
Input Data	None
Success Return Data	VideoStreamInfoList
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/DetailInfos
Description	Set all video streams of a video channel.
Method	PUT
Input Data	VideoStreamInfoList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Media/Video/Mode

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Mode
Description	Get image collection mode.
Method	GET
Input Data	None
Success Return Data	VideoModeInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Mode
Description	Set image collection mode.
Method	PUT
Input Data	VideoModeInfo
Success Return Data	None
Note	Applicable to NVR and IPC.
Status	

3.3.3 Audio

1. /LAPI/V1.0/Media/Audio/Input

URL	/LAPI/V1.0/Media/Audio/Input
Description	Get audio input parameters.
Method	GET
Input Data	None
Success Return Data	AudioInputCfg
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Media/Audio/Input
Description	Set audio input parameters.
Method	PUT
Input Data	AudioInputCfg
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Media/Audio/Output

URL	/LAPI/V1.0/Media/Audio/Output
Description	Get audio output parameters.
Method	GET
Input Data	None
Success Return Data	AudioOutputCfg
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Media/Audio/Output
Description	Set audio output parameters
Method	PUT
Input Data	None
Success Return Data	AudioOutputCfg
Note	Applicable to IPC and NVR.

3.3.4 LiveStreamURL

1. /LAPI/V1.0/Channels/<ID>/Media/Video/Streams/<ID>/LiveStreamURL?TransType=<TransType>&TransProtocol=<TransProtocol>

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/<ID>/LiveStreamURL?TransType=<TransType>&TransProtocol=<TransProtocol>
Description	Get the live video URL of a video stream of a video channel.
Method	GET
Success Return Data	{ "LoginName": "PIN": "URL": }
Note	After getting the URL, use RTSP to get media streams. For IPC, the ID of the channels is always 0. Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
Streams/<ID>	M	unsigned long	Video stream index number, which starts from 0.	0
TransType	O	unsigned long	Stream transport type 0: Forward 1: Direct connection Applicable to VMS. Default: Forward	0
TransProtocol	C	unsigned long	Transport protocol type 0: TCP 1: UDP Applicable to VMS. Used with TransType.	1
LoginName	C	string	Device username encrypted using private algorithm. A decryption library is provided. Required when Transtype is 1.	sdfsdf
PIN	C	string	Device password encrypted using private algorithm. A decryption library is provided. Required when Transtype is 1.	sdfsdfjskdsdfsdfjskdsdfsdfjs
URL	M	string	Live video URL of the corresponding video stream.	rtsp://192.168.0.13:554/media/video1

3.3.5 Records

1. /LAPI/V1.0/Channels/<ID>/Media/Video/Streams/<ID>/Records?Begin=<Begin>&End=<End>&Limit=<Limit>&Offset=<Offset>&Types=<Types>&RelationOfTypes=<RelationOfTypes>&Position=<Position>

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/<ID>/Records?Begin=<Begin>&End=<End>&Limit=<Limit>&Offset=<Offset>&Types=<Types>&RelationOfTypes=<RelationOfTypes>&Position=<Position>
Description	Get recording files of a stream of a video channel during a time period.
Method	GET
Input Data	None
Success Return Data	RecordList
Note	Applicable to IPC, NVR, and VMS.
Status	

Param	Ranger	Type	Description	Example
Begin	M	unsigned long	Search start time (unit: second, UTC format)	1454428800
End	M	unsigned long	Search end time (unit: second, UTF format)	1454435960
Limit	C	unsigned long	Number of recordings to search. Range: [0, 100] Applicable to IPC only.	100
Offset	C	unsigned long	Specifies from which one to start searching. Applicable to IPC only.	100
Types	0	unsigned long	Type of recording to search. Described by bit order, from left to right are bit 0, bit 1, bit 2 ... bit 31, each representing a recording type. A recording type is valid when the corresponding bit is 1. 0: Manual recording (this recording type must not be mixed with other recording types and therefore are expressed using all 0s). BIT0: Common alarm (currently not applicable to NVR) BIT1: Motion detection BIT2: Input switch alarm BIT3: Video loss BIT4: Audio detection	Search motion detection recording: 00000000000000000000000000000000 00000000100 = 4, so fill this field with 4. Search motion detection recording and face detection recording: 00000000000000000000000000000000 00100000100 = 260, so fill this field with 260.

			BIT5: POS info (currently not applicable to NVR) BIT6: Scheduled recording BIT7: Face detection BIT8: Cross line detection BIT9: Intrusion detection All types of recordings will be searched when this field absent. Applicable to IPC and NVR.	
RelationOfTypes	C	unsigned long	Relationship between recording types: 0: AND 1: OR Optional when Types is absent or when involving one recording type only. Required when involving two or more recording types. Applicable to NVR only.	1
Position	0	unsigned long	Location of recording to search: 1: Local recording 2: Other device Applicable to VMS only.	1

2. /LAPI/V1.0/Channels/<ID>/Media/Video/Streams/RecordURL?

Begin=<Begin>&End=<End>&Types=<Types>&RelationOfTypes=<RelationOfTypes>&Position=<Position>&SessionID=<SessionID>&TransType=<TransType>

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/RecordURL?Begin=<Begin>&End=<End>&Types=<Types>&RelationOfTypes=<RelationOfTypes>&Position=<Position>&SessionID=<SessionID>&TransType=<TransType>
Description	Get the URL of certain types of recordings of a video channel during a time period.
Method	GET
Input Data	None
Success Return Data	<pre>{ "URL": , }</pre>
Note	The returned URL is only used for RTSP-based playback and download. Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
Begin	O	unsigned long	Recording start time (unit: second, UTC format) End time is required if there is a start time.	1454428800
End	O	unsigned long	Recording end time (unit: second, UTF)	1454435960

			format) Start time is required if there is an end time.	
Types	0	unsigned long	Type of recording to search. Described by bit order, from left to right are bit 0, bit 1, bit 2 ... bit 31, each representing a recording type. A recording type is valid when the corresponding bit is 1. 0: Manual recording (this recording type must not be mixed with other recording types and therefore are expressed using all 0s). BIT0: Common alarm (currently not applicable to NVR) BIT1: Motion detection BIT2: Input switch alarm BIT3: Video loss BIT4: Audio detection BIT5: POS info (currently not applicable to NVR) BIT6: Scheduled recording BIT7: Face detection BIT8: Cross line detection BIT9: Intrusion detection All types of recordings will be searched when this field absent.	Search motion detection recording: 00000000000000000000 0000000000100 = 4, so fill this field with 4. Search motion detection recording and face detection recording: 00000000000000000000 0000100000100 = 260, so fill this field with 260.
RelationOfTypes	0	unsigned long	Relationship between recording types: 0: AND 1: OR Optional if Types is absent or when involving one recording type only. Required when involving two or more recording types.	1
Position	0	unsigned long	Location of recording to play: 1: Local recording 2: Other device Currently applicable to VMS only.	1
SessionID	0	unsigned long	It is recommended that each playback stream has one corresponding session ID. Currently applicable to VMS only.	1
TransType	0	unsigned long	0: Forward 1: Direct connection	0
URL	M	string	Recording URL. Length range: [0,128].	rtsp://206.5.3.13:554/c2/replay/

3. /LAPI/V1.0/Channels/<ID>/Media/Video/Streams/Records/DailyDistribution?Year=<Year>&Month=<Month>&Position=<Position>

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/Records/DailyDistribution?Year=<Year>&Month=<Month>&Position=<Position>
Description	Get the distribution of a channel's recordings of a certain month (in unit of days).
Method	GET
Input Data	None
Success Return Data	{ "Nums": , "DailyStatuses": [<Status>, <Status>, <Status>, <Status>...] }
Note	Applicable to NVR.
Status	

Param	Requirement	Type	Description	Example
Year	M	unsigned long	Year.	2016
Month	M	unsigned long	Month, starts from 1.	9
Position	0	unsigned long	Location of recording to search: 1: Local recording 2: Other device Currently applicable to VMS only.	1
Nums	M	unsigned long	Actual number of days in the month.	30
DailyStatuses	C	Array	Daily recording status list. Optional when Nums is 0.	
Status	M	unsigned long	Recording status: 0: No recording 1: Event recording 2: Normal recording Event recording has higher priority level than normal recording.	1

4. /LAPI/V1.0/Channels/Media/Video/Streams/RecordFileSize

URL	/LAPI/V1.0/Channels/Media/Video/Streams/RecordFileSize
Description	Get the size of recording files of a certain period.
Method	POST
Input Data	RecordChannelInfoList
Success Return Data	RecordFileList
Note	Applicable to NVR.
Status	

3.3.6 Talk

1. /LAPI/V1.0/Channels/<ID>/Media/Talk

URL	/LAPI/V1.0/Channels/<ID>/Media/Talk
Description	Get audio URL for two-way audio with a video channel.
Method	GET
Success Return Data	{ "URL": }
Note	After getting the URL, use RTSP to get audio streams for two-way audio. For IPC, the ID of the channels is always 0. Applicable to IPC and VMS.
Status	

Param	Requirement	Type	Description	Example
URL	M	string	Corresponding live video stream URL of an audio stream.	rtsp://192.160.1.22:554/media/video1

3.3.7 OSD

1. /LAPI/V1.0/Channels/<ID>/Media/OSDs/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs/Capabilities
Description	Get OSD capability of a channel.
Method	GET
Input Data	None
Success Return Data	OSDCapabilities
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Media/OSDs/ContentStyle

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs/ContentStyle
Description	Get the OSD content style of a channel.
Method	GET
Input Data	None
Success Return Data	OSDContentStyle
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs/ContentStyle
Description	Set the OSD content style of a channel.
Method	PUT
Input Data	OSDContentStyle
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3. /LAPI/V1.0/Channels/<ID>/Media/OSDs/Contents

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs/Contents
Description	Get OSD contents.
Method	GET
Input Data	None
Success Return Data	OSDContent
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs/Contents
Description	Set OSD contents.
Method	PUT
Input Data	OSDContent
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

4. /LAPI/V1.0/Channels/<ID>/Media/OSDs

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs
Description	Get all OSD contents.
Method	GET
Input Data	None
Success Return Data	OSDList
Note	Applicable to NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Media/OSDs
Description	Set all OSD contents.
Method	PUT
Input Data	OSDList
Success Return Data	None
Note	Applicable to NVR.
Status	

3.3.8 Snapshot

1. /LAPI/V1.0/Channels/<ID>/Media/Video/Streams/<ID>/Snapshot

URL	/LAPI/V1.0/Channels/<ID>/Media/Video/Streams/<ID>/Snapshot
Description	Snapshot without preview.
Method	GET
Input Data	None
Success Return Data	Content-Length: 123131 Content-Type: image/jpeg Accept-Ranges: bytes Image data
Note	Applicable to IPC and NVR.
Status	

Param	Requirement	Type	Description	Example
Channels/<ID>	M	unsigned long	Video channel ID.	1
Streams/<ID>	M	unsigned long	The ID is the stream index (snapshot resolution of IPC is determined by snapshot configuration and so the index is invalid). 0: Main stream 1: Sub stream 2: Third stream	0

2. /LAPI/V1.0/Channels/<ID>/Alarm/SnapshotURL?Type=<Type>&Time=<Time>

URL	/LAPI/V1.0/Channels/<ID>/Alarm/SnapshotURL?Type=<Type>&Time=<Time>
Description	Get alarm snapshot URL.
Method	GET
Input Data	None
Success Return Data	SnapshotURLInfo
Note	Applicable to NVR and VMS.
Status	

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Video channel ID.	1
Type	M	string	Alarm type. Currently three types available: "MotionAlarmOn": Motion detection started "VideoLossAlarmOn": Video loss started "AbnormalAudioOn": Audio detection started	"MotionAlarmOn"
Time	M	unsigned long	Alarm occurrence time (UTC). Number of seconds since 0'o clock, Jan. 1, 1970	1476088399

3. /LAPI/V1.0/IO/InputSwitches/<ID>/Alarm/SnapshotURL?Time=<Time>

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/Alarm/SnapshotURL?Time=<Time>
Description	Get the URL of alarm snapshots triggered by an input switch.
Method	GET
Input Data	None
Success Return Data	SnapshotURLInfo
Note	Applicable to NVR and VMS.
Status	

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Alarm input ID. For NVR, the ID is channel ID*100+alarm input ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	1
Time	M	unsigned long	Alarm occurrence time (UTC). Number of seconds since 0'o clock, Jan. 1, 1970.	1476088399

4. /LAPI/V1.0/Channels/<ID>/Alarm/Snapshot?Name=<Name>&Size=<Size>

URL	/LAPI/V1.0/Channels/<ID>/Alarm/Snapshot?Name=<Name>&Size=<Size>
Description	Get alarm snapshot.
Method	GET
Input Data	None
Success Return Data	Content-Length: 44 Content-type: image/jpeg Accept-Ranges: bytes Image data
Note	Name and Size are returned by the interface used to get alarm image URLs. Applicable to NVR and VMS.
Status	

Param	Requirement	Type	Description	Example
ID	M	unsigned long	ID of the corresponding channel.	1
Name	M	string	Image name. Length: [1,64]	"aaa"
Size	M	unsigned long	Image size. Unit: Byte.	61440

5. /LAPI/V1.0/Channels/<ID>/Media/Snapshot/Info

URL	/LAPI/V1.0/Channels/<ID>/Media/Snapshot/Info
Description	Get snapshot parameters of a channel.
Method	GET
Input Data	None
Success Return Data	SnapshotInfo
Note	Applicable to IPC.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Media/Snapshot/Info
Description	Set snapshot parameters of a channel.
Method	PUT
Input Data	SnapshotInfo
Success Return Data	None
Note	Applicable to IPC.
Status	

3.3.9 Json Block

1. AudioCapabilityInfo

AudioCapabililInfo Json Block	<pre>{ "AudioInNum": , "AudioInModelInfoList": [<AudioModelInfo>, ...], "AudioInEncodeFormatNum":, "AudioInEncodeFormatInfoList": [<EncodeFormatInfo>,...] "SerialInNum": , "SerialInModelInfoList":[<AudioModelInfo>, ...] "SerialInEncodeFormatNum": , "SerialInEncodeFormatInfoList":[<EncodeFormatInfo>, ...] "AudioOutTypeNum": , "AudioOutTypeInfoList":[...], "SupportAudioOutGain", "SupportAlarmAudioOutGain": }</pre>
----------------------------------	---

Param	Requirement	Type	Description	Example
AudioInNum	M	unsigned long	Number of audio inputs supported.	2
AudioInModelInfoList	C	array	Audio input mode list	---
AudioModelInfo	M	Json Block	Audio input mode. See AudioModelInfo Json Block	---
AudioInEncodeFormat Num	M	unsigned long	Number of audio input encoding formats supported.	3

AudioInEncodeFormatInfoList	C	array	List of supported audio input encoding formats	---
EncodeFormatInfo	M	Json Block	Info about supported encoding formats. See EncodeFormatInfo Json Block	---
SerialInNum	M	unsigned long	Number of serial inputs supported. Used to connect external digital sound pick-up.	1
SerialInModeInfoList	C	array	Serial input mode list	---
SerialInEncodeFormatNum	M	unsigned long	Number of encoding formats supported by serial input	
SerialInEncodeFormatInfoList	M	array	List of encoding formats supported by serial input. See EncodeFormatInfo Json Block	
AudioOutTypeNum	M	unsigned long	Number of audio output device types supported: 0 means audio output parameters are not configurable.	2
AudioOutTypeInfoList	C	array	Audio output device type: 0: LINE 1: MIC 2: RS485 digital sound pick-up 3: Speaker Optional when AudioOutTypeNum is 0	[0, 1]
SupportAudioOutGain	C	boolean	Whether the device supports the configuration of audio gain 0: No 1: Yes When absent, it means the capability is not available.	1
SupportAlarmAudioOutGain	C	boolean	Whether the device supports the configuration of alarm output gain Present when supported by the device. (If this field is supported, it means audio output gain can be configured separately)	1

2. AudioModeInfo

AudioModeInfo Json Block	{ "Channel":, "ModeNum":,
---	---------------------------------

	"ModeList": }
--	------------------

Param	Requirement	Type	Description	Example
Channel	M	unsigned long	Audio channel ID	2
ModeNum	M	unsigned long	Number of audio modes	2
ModeList	C	array	Supported audio input modes. Optional when ModeNum is 0. 0: LINE 1: MIC 2: RS485 digital sound pick-up	[0, 1]

3. AudioInputCfg

AudioInputCfg Json Block	{ "IsMute";, "Type";, "EncodeFormat ";, "SampleRate";, "InputGain";, "NoiseReduction": { "Enabled";, "Mode";, "Strength": } "AudiolInputNum": "AudiolInputList": [< AudioInput >,< AudioInput >.....], "SerialInputNum": "SerialInputList": [< AudioInput >.....], }
-------------------------------------	---

Param	Requirement	Type	Description	Example
IsMute	M	boolean	Whether is muted. 0: No 1: Yes	0
Type	M	unsigned long	Audio collection port type. 0: Audio input 1: Serial port	
EncodeFormat	M	unsigned long	Audio encoding format. See EncodeFormatInfo .	2
SampleRate	M	unsigned long	Audio sampling rate. See EncodeFormatInfo .	0
InputGain	M	unsigned long	Audio gain. Range: [0,255].	128
NoiseReduction	M	Json Block	Noise reduction info.	1
Enabled	M	unsigned long	Enable or disable noise	0

			reduction. 0: Disable 1: Enable	
Mode	0	unsigned long	Reserved field.	
Strength	0	unsigned long	Reserved field.	1
AudioInputNum	M	unsigned long	Number of audio inputs.	
AudioInputList	C	array	Audio input list. Required when Type is 0.	
AudioInput	M	Json Block	Audio input channel info See AudioInput Json Block.	
SerialInputNum	M	unsigned long	Number of serial inputs.	1
SerialInputList	C	array	Serial audio input list. Required when Type is 1.	

4. AudioInput

AudioInput Json Block	{ "ID"; "Enabled"; "Mode": }
----------------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Audio channel ID, starts from 1.	1
Enabled	M	unsigned long	Enable or disable audio input channel.	1
Mode	M	unsigned long	Audio input mode. 0: LINE 1: MIC 2: RS485 digital sound pick-up	0

5. AudioOutputCfg

AudioOutputCfg Json Block	{ "Type"; "Gain"; "AlarmGain": }
--------------------------------------	--

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Audio output device type. See AudioCapabilityInfo .	
Gain	C	Unsigned long	Audio output gain Range: [0,255]. Required when	242

			SupportAudioOutGain is 1.	
AlarmGain	C	unsigned long	Alarm audio gain. Range: [0,255] Required when SupportAlarmAudioOutGain is 1.	255

6. ChannelRecordInfo

ChannelRecordInfo Json Block	{ "ChannelID": , "StreamID": , "Begin": , "End": }
---	---

Param	Requirement	Type	Description	Example
ChannelID	M	unsigned long	Video channel ID of recordings to search.	1
StreamID	M	unsigned long	Stream ID of requested recordings. 0: Main stream 1: Sub stream 2: Third stream	0
Begin	M	unsigned long	Start time of recording file (unit: second, UTC format)	1454428800
End	M	unsigned long	End time of recording file (unit: second, UTC format)	1454435960

7. EncodeFormatInfo

EncodeFormatInfo Json Block	{ "Type": , "Num": , "SampleList": }
--	--

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Audio encoding format 0 : MPEG1 1: G.711A 2: G.711U 3 : ADPCM 4: OGG 5: Auto 6: AAC-LC	2
Num	M	unsigned long	Number of supported sampling rates	3
SampleList	M	array	Sampling rate list 0: 8KHz	[1, 2, 6]

			1: 16KHz 2: 22.05KHz 3: 24KHz 4: 32KHz 5: 44.1KHz 6: 48KHz 7: 96KHz	
--	--	--	---	--

8. OSDCapabilities

OSDCapabilities Json Block	<pre> { "IsSupportCfg": , "SupportedOSDContentTypeNum": , "SupportedOSDContentTypeList": , "IsSupportFontSizeCfg": , "IsSupportFontColorCfg": , "MaxAreaNum": , "MaxOSDNum": , "MaxPerAreaOSDNum": , "SupportedTimeFormatNum": , "SupportedTimeFormatList": , "SupportedDateFormatNum": , "SupportedDateFormatList": } </pre>
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Param	Requirement	Type	Description	Example
IsSupportCfg	M	boolean	Whether is configurable: 0: No 1: Yes	0
SupportedOSDContentTypeNum	M	unsigned long	Number of OSD content types supported.	6
SupportedOSDContentTypeList	M	array	OSD that is configurable. OSD content type 0: Disable 1: Custom 2: Time and date 3: PTZ controller 4: PTZ coordinates 5: Patrol info 6: Zoom info 7: Preset info 8: Alarm info 9: Encoding info 10: Serial port OSD 11: PTZ direction info 12: Channel name 13: Test OSD	[1,2,3,10,16,17]

			14: People counting OSD 15: Network port OSD 16: Time 17: Date 18: Sensing OSD 19: Battery OSD 20: Scrolling characters 21: Logo OSD 22: Traffic monitoring (motor vehicle) 23: Traffic monitoring (non-motor vehicle) 24: Traffic monitoring (pedestrian) 25: Number of INFOSD types	
IsSupportFontSizeCfg	M	boolean	Whether OSD size is configurable.	0
IsSupportFontColorCfg	M	boolean	Whether OSD color is configurable.	0
MaxAreaNum	M	unsigned long	Maximum number of OSD areas.	8
MaxOSDNum	M	unsigned long	Maximum number of OSDs.	8
MaxPerAreaOSDNum	M	unsigned long	Maximum number of OSDs per area.	8
SupportedTimeFormatNum	M	unsigned long	Number of time OSD formats supported.	6
SupportedTimeFormatList	M	array	Time OSD formats. h/H=12/24 Hour; tt=A.M. or P.M.; mm=Minute; ss=Second; xxx=millisecond 0: HH:mm:ss (common in China, U.S., Great Britain) 1:hh:mm:ss t.t. (common in Spanish-speaking countries other than Spain) 2:hh:mm:ss tt (common in English-speaking countries other than the UK) 3:tt hh:mm:ss (common in Chinese (Singapore)) 4:Afternoon hh:mm:ss (common in Chinese (Taiwan)) 5:HH:mm:ss.xxx (millisecond) 6:hh:mm:ss.xxx tt (millisecond)	7
SupportedDateFormatNum	M	unsigned long	Number of date OSD formats supported.	6
SupportedDateFormatList	M	unsigned long	Date OSD format. Among which, MMMM means month, for example, June; dddd means day of week, for example, Mon Chinese 0-5 0: yyyy-MM-dd 1: MM-dd-yyyy 2: yyyyMMdd	[0,1,2,3,4,5]

			3: MMddyyyy 4: yyyyMMdd X (day of week) 5: MMddyyyy X (day of week) English 100-108 100: dd/MM/yyyy 101: MM/dd/yyyy 102: dd MMMM, yyyy 103: MMMM dd, yyyy 104: dddd, dd MMMM, yyyy 105: dddd, MMMM dd, yyyy 106: yyyy/MM/dd 107: yyyy, MMMM dd 108: dddd, yy, MM dd	
--	--	--	---	--

9. OSDContentStyle

OSDContentStyle Json Block	<pre>{ "FontStyle"; "FontSize"; "Color"; "DateFormat"; "TimeFormat"; "FontAlignList"; "Margin": }</pre>
---------------------------------------	---

Param	Requirement	Type	Description	Example
FontStyle	C	unsigned long	Font style. 0: Background 1: Stroke 2: Hollow 3: Normal Applicable to IPC.	3
FontSize	M	unsigned long	Font size. 0: Very large 1: Large 2: Medium 3: Small	1
Color	M	unsigned long	Color (decimal). An RGB color is calculated by combining the relative brightness of three primary colors: red, green, and blue, and the value range of each primary color is 0-255. Formula: (65536*Red + 256*Green + Blue). To set color #665432, this field should be 65536 * 102 (decimal number of 66) + 256 * 84	

			(decimal number of 54) + 50 (decimal number of 32) = 6706226 (actually it is decimal number of #665432) To set RGB(102, 84, 50), this field should be $65536 * 102 + 256 * 84 + 50 = 6706226$.	
DateFormat	M	unsigned long	Date format. See OSDCapabilities .	7
TimeFormat	M	unsigned long	Time format. See OSDCapabilities .	2
FontAlignList	C	array	Font alignment within the area. Fixed 8 areas. 0: Left alignment 1: Right alignment Applicable to IPC.	[0,1,1,0,0,0,0,0]
Margin	C	unsigned long	Number of characters with empty margins. 0: None 1: One-character width 2: Two-character width Applicable to IPC.	2

10. OSDContent

OSDContent Json Block	{ "Num": , "ContentList": [< OSDContentInfo > ,< OSDContentInfo > ,..... }
----------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of OSD contents.	8
ContentList	C	array	OSD content info list. Optional when Num is 0.	-
OSDContentInfo	M	Json Block	OSD content info. On NVR, channel name always uses the first OSD; date&time always uses the seconds OSD; people counting (if available, by querying capabilities) uses the third OSD, and other scenes names use the subsequent OSDs. See OSDContentInfo Json Block	-

11. OSDContentInfo

OSDContentInfo o Json Block	{ "ID": , "Enabled": , "Num": , "ContentInfo": [{
--	---

	<pre> "ContentType": , "Value": },.....] "Area": { "TopLeft": { "X": , "Y": } } } } </pre>
--	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	OSD ID. Range: [0,7].	0
Enabled	M	boolean	Enable or disable OSD. 0: Disable 1: Enable	1
Num	M	unsigned long	Number of OSDs in the area, limited by MaxPerAreaOSDNum in OSDCapabilities . For NVR, each area supports only one OSD.	2
ContentInfo	M	array	OSD content info.	-
ContentType	M	unsigned long	OSD content type. 0: Disable 1: Custom 2: Time and date 3: PTZ controller 4: PTZ coordinates 5: Patrol info 6: Zoom info 7: Preset info 8: Alarm info 9: Encoding info 10: Serial port OSD 11: PTZ direction info 12: Channel name 13: Test OSD 14: People counting OSD 15: Network port OSD 16: Time 17: Date 18: Sensing OSD 19: Battery OSD 20: Scrolling characters 21: Logo OSD 22: Traffic monitoring (motor vehicle) 23: Traffic monitoring (non-motor vehicle) 24: Traffic monitoring (pedestrian) 25: Number of INFOSD types	1
Value	M	string	OSD text info.	"OSD text"

Area	M	Json Block	OSD area info.	-
TopLeft	M	Json Block	Coordinates of area's top left corner.	
X	M	unsigned long	x-coordinate of OSD. Range: [0,9999]	200
Y	M	unsigned long	y-coordinate of OSD. Range: [0,9999]	200

12. OSDList

OSDList Json Block	{ "ContentStyle": < OSDContentStyle > , "Num": , "ContentList": [< OSDContentInfo > ,< OSDContentInfo > ,..... }
-------------------------------------	--

Param	Requirement	Type	Description	Example
ContentStyle	M	Json Block	OSD style. See OSDContentStyle Json Block.	-
Num	M	unsigned long	Number of OSD areas.	8
ContentList	C	array	Area and the corresponding OSD content list. See OSDContentInfo Json Block. Optional when Num is 0.	-

13. PicInfo

PicInfo Json Block	{ "URL": , "Name": , "Size": }
-------------------------------------	--

Param	Requirement	Type	Description	Example
URL	M	string	URL of alarm snapshots. Length range: [1,128].	http://206.5.3.32:80/LAPI/V1.0/Channels/1/Alarm/Snapshot
Name	M	string	Image name. Length: [1,64]	ubs_19_0_0_1_0_1_409_2795
Size	M	unsigned long	Image size. Unit: Byte.	64164

14. RecordList

RecordList Json Block	{ "Total": , "Offset": , "Nums": , }
--	--

	"RecordInfos": [< RecordInfo >, [< RecordInfo >, ...] }
--	--

Param	Requirement	Type	Description	Example
Total	C	unsigned long	Total number of recordings, returned for paginated search only. Applicable to IPC.	100
Offset	C	unsigned long	Specifies from where to start searching. Limit and Offset are used to limit the maximum number of records returned. These two parameters are either used simultaneously or be null simultaneously. Null: means up to 100 records will be returned. Applicable to IPC.	10
Nums	M	unsigned long	Number of recording files searched.	4
RecordInfos	C	Array	Recording info list. Optional when Nums is 0.	
RecordInfo	M	Json Block	See RecordInfo Json Block.	

15. RecordInfo

RecordInfo Json Block	{ "FileName": "Abc123", "RecordType": 4, "Begin": , "End": }
----------------------------------	---

Param	Requirement	Type	Description	Example
FileName	M	string	Recording file name. Range: [0, 64]. Not applicable to IPC.	Abc123
RecordType	M	unsigned long	Type of recording to search, handled by bit order, from left to right are bit 0, bit 1, bit 2 ... bit 31, each representing a recording type, and will be converted into the Ulong type after being assigned a value: 0: Manual recording. 1: Common alarm (currently not applicable to NVR) 2: Motion detection 3. Input switch (alarm input) 4. Video loss 5. Audio detection 6. POS info (currently not applicable to NVR) 7: Scheduled recording 8: Face detection 9: Cross line detection	Search motion detection recording: 00000000000000000000 0000000000100 = 4, so fill this field with 4. Search motion detection recording and face detection recording: 00000000000000000000 0000100000100 = 260, so fill this field with 260.

			10: Intrusion detection	
Begin	M	unsigned long	Start time of recording file (unit: second, UTC format)	1454428800
End	M	unsigned long	End time of recording file (unit: second, UTC format)	1454435960

16. RecordChannelInfoList

RecordChannelInfoList Json Block	{ "Num": , "ChannelInfoList": [< ChannelRecordInfo >, [< ChannelRecordInfo >, ...] }
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of video input channels.	3
ChannelInfoList	C	array	Video channel info list. Required when Num is greater than 0.	-
ChannelRecordInfo	C	Json Block	Video channel info. Required when Num is greater than 0.	See ChannelRecordInfo Json Block.

17. RecordFileList

RecordFileList Json Block	{ "Num": , "RecordFileInfoList": [< RecordFileInfo >, [< RecordFileInfo >, ...] }
--	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of recording files searched.	4
RecordFileInfos	C	array	Recording file info list. Required when Num is greater than 0.	-
RecordFileInfo	C	Json Block	Recording file info. Required when Num is greater than 0.	See RecordFileInfo Json Block.

18. RecordFileInfo

RecordFileInfo Json Block	{ "ChannelID": , "StreamID": , "Begin": , "End": , "FileSizeForRecord":
--	--

	}
--	---

Param	Requirement	Type	Description	Example
ChannelID	M	unsigned long	Video channel ID of recordings to search.	1
StreamID	M	unsigned long	Stream ID of requested recording: 0: Main stream 1: Sub stream 2: Third stream	0
Begin	M	unsigned long	Start time of recording file (unit: second, UTC format)	1454428800
End	M	unsigned long	End time of recording file (unit: second, UTC format)	1454435960
FileSizeForRecord	M	unsigned long	Recording file size (Unit: KB).	1024

19. ResolutionCapability

ResolutionCapability Json Block	{ "Width": , "Height": , "MinBitRate": , "MaxBitRate": , "DefaultBitRate": }
--	--

Param	Requirement	Type	Description	Example
Width	M	unsigned long	Image width.	0
Height	M	unsigned long	Image height.	1
MinBitRate	M	unsigned long	Minimum bit rate.	128
MaxBitRate	M	unsigned long	Maximum bit rate.	16384
DefaultBitRate	C	unsigned long	Default bit rate. Applicable to NVR.	2048

20. SnapshotCapabilities

SnapshotCapabilities Json Block	{ "SupportSnapshotCfg": , "PictureMaxSize": , "ResolutionNum": , "ResolutionCapabilityList": [< SnapshotResolutionCapability >, < SnapshotResolutionCapability >...] }
--	--

Param	Requirement	Type	Description	Example
SupportSnapshotCfg	M	boolean	Whether the device supports snapshot configuration 0: No	1

			1: Yes	
PictureMaxSize	M	unsigned long	Maximum image size of snapshot.	800
ResolutionNum	M	unsigned long	Number of resolutions supported.	4
ResolutionCapabilityList	C	array	Info list of supported snapshot resolutions. Optional when ResolutionNum is 0.	
SnapshotResolutionCapability	M	Json Block	Snapshot resolution. See SnapshotResolutionCapability Json Block.	

21. SnapshotResolutionCapability

SnapshotResolutionCapability Json Block	{ "Width": , "Height": }
---	-----------------------------------

Param	Requirement	Type	Description	Example
Width	M	unsigned long	Image width	1920
Height	M	unsigned long	Image height	1680

22. StreamCapability

StreamCapability Json Block	{ "ID": , "ResolutionNum": , "ResolutionCapabilityList": [< ResolutionCapability >, < ResolutionCapability >...] "MaxFrameRate": , "MaxMJPEGFramerate": , "SmartEncode": < SmartEncodeInfo > }
---------------------------------------	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Stream index 0: Main stream 1: Sub stream 2: Third stream	3
ResolutionNum	M	unsigned long	Number of resolutions supported.	4
ResolutionCapabilityList	M	array	Supported resolutions. Resolutions: Main stream >= Sub stream >= Third stream	
ResolutionCapability	M	Json Block	Resolutions	

bility			See ResolutionCapability Json Block.	
MaxFrameRate	C	unsigned long	Maximum frame rate at the current bit rate. Applicable to IPC.	25
MaxMJPEGFrameRate	C	unsigned long	Maximum MJPEG frame rate at the current bit rate. Applicable to IPC.	5
SmartEncode	M	Json Block	Extended image encoding modes. See SmartEncodeInfo Json Block.	

23. SmartEncodeInfo

SmartEncodeInfo Json Block	{ "H264SmartEncodeModeNum": , "H264SmartEncodeModeList": , "H265SmartEncodeModeNum": , "H265SmartEncodeModeList": , }
---------------------------------------	--

Param	Requirement	Type	Description	Example
H264SmartEncodeModeNum	M	unsigned long	Number of H.264 smart encoding modes supported.	2
H264SmartEncodeModeList	M	array	H.264 extended image encoding modes. 0: off (disabled) 1: Basic (basic mode) 2: Advanced (advanced mode)	[1,2]
H265SmartEncodeModeNum	M	unsigned long	Number of H.265 smart encoding modes supported.	2
H265SmartEncodeModeList	M	array	H.265 extended image encoding modes. 0: off (disabled) 1: Basic (basic mode) 2: Advanced (advanced mode)	[1,2]

24. SnapshotURLInfo

SnapshotURLInfo Json Block	{ "Num": , "PicInfos": [< PicInfo >,.....] }
---------------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of URLs.	2
PicInfos	C	array	Alarm snapshot info. Optional when Num is	-

			0.	
PicInfo	M	Json block	Alarm snapshot info.	See PicInfo Json Block

25. SnapshotResolutionCapability

SnapshotResolutionCapability Json Block	{ "Width": , "Height": }
---	-----------------------------------

Param	Requirement	Type	Description	Example
Width	M	unsigned long	Image width	1920
Height	M	unsigned long	Image height	1680

26. SnapshotCapabilities

SnapshotCapabilities Json Block	{ "SupportSnapshotCfg": , "PictureMaxSize": , "ResolutionNum": , "ResolutionCapabilityList": [< SnapshotResolutionCapability > , < SnapshotResolutionCapability > ...] }
---------------------------------	--

Param	Requirement	Type	Description	Example
SupportSnapshotCfg	M	boolean	Whether the device supports snapshot configuration 0: No 1: Yes	1
PictureMaxSize	M	unsigned long	Maximum image size of snapshot.	800
ResolutionNum	M	unsigned long	Number of resolutions supported.	4
ResolutionCapabilityList	C	array	Info list of supported snapshot resolutions. Optional when ResolutionNum is 0.	
SnapshotResolutionCapability	M	Json Block	Snapshot resolution. See ResolutionCapability Json Block.	

27. SnapshotInfo

SnapshotInfo Json Block	{ "Enabled": ,
-------------------------	-------------------

	<pre> “Resolution”:{ “Width”:, “Height”:, }, “PictureMaxSize”: } </pre>
--	---

Param	Requirement	Type	Description	Example
Enabled	M	Boolean	Enable or disable snapshot: 0: Disable 1: Enable	0
Resolution	M	Json block	Snapshot resolution	
Width	M	unsigned long	Image width of snapshot. Unit: Pixel. For snapshot resolutions available, see snapshot capability.	1920
Height	M	unsigned long	Image height of snapshot. Unit: Pixel. For snapshot resolutions available, see snapshot capability.	1080
PictureMaxSize	M	unsigned long	Maximum image size. Unit: KB The value of this parameter cannot be greater than the maximum value of image size in snapshot capability.	10

28. VideoCapabilityInfo

VideoCapability Info Json Block	<pre> { “IsSupportCfg”: , “IsSupportSmoothLevel”: , “IsSupportImageFormat”: , “EncodeFormatNum”: , “EncodeFormatList”: , “MinIFrameInterval”: , “MaxIFrameInterval”: , “StreamCapabilityNum”: , </pre>
--	--

	<pre> "StreamCapabilityList": [<StreamCapability>, <StreamCapability> ...] "VideoModeNum": , "VideoModeInfoList": [<VideoModeInfo>, <VideoModeInfo> ...] "GOPTypeNum":, "GOPTypeList ":[] } </pre>
--	---

Param	Requirement	Type	Description	Example
IsSupportCfg	M	boolean	Whether is configurable: 0: No 1: Yes	1
IsSupportSmoothLevel	M	boolean	Whether supports stream smoothness settings. 0: No 1: Yes	0
IsSupportImageFormat	M	boolean	Whether image format is configurable: 0: No 1: Yes	0
EncodeFormatNum	M	unsigned long	Number of supported video encoding formats.	
EncodeFormatList	C	array	Supported video encoding formats. 0: MJPEG 1: H.264 2: H.265 3: JPEG	[1,2]
MinIFrameInterval	M	unsigned long	Minimum I-frame interval.	5
MaxIFrameInterval	M	unsigned long	Maximum I-frame interval.	250
StreamCapabilityNum	M	unsigned long	Number of video streams supported.	
StreamCapabilityList	C	array	List of supported stream capabilities.	
StreamCapability	M	Json Block	Stream capability info. See StreamCapability Json Block.	
VideoModeNum	M	unsigned long	Number of supported video modes.	3
VideoModeInfoList	C	Json Block	Video mode list.	
VideoModeInfo	M	Json Block	Video mode info. See VideoModeInfo Json Block.	
GOPTypeNum	M	unsigned long	Number of supported GOP types.	
GOPTypeList	C	array	List of GOP types supported. 0: IP 1: IBP 2: IBBP 3: I Optional when GOPTypeNum is 0.	[0, 3]

29. VideoModeInfo

VideoModeInfo	{
----------------------	---

Json Block	<pre> "Resolution": { "Width": , "Height": }, "FrameRate": </pre>
-------------------	---

Param	Requirement	Type	Description	Example
Resolution	M	Json Block	Resolution.	
Width	M	unsigned long	Image width.	1920
Height	M	unsigned long	Image height.	1080
FrameRate	M	unsigned long	Maximum frame rate in the current mode.	25

30. VideoStreamInfoList

VideoStreamInfoList Json Block	<pre> { "Num":, "VideoStreamInfos":[<VideoStreamInfo>,<VideoStreamInfo>,...] } </pre>
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of video streams	1
VideoStreamInfos	C	array	Video stream info list. Optional when Num is 0.	-
VideoStreamInfo	M	Json Block	Info about a video stream. See VideoStreamInfo Json Block.	-

31. VideoStreamInfo

VideoStreamInfo Json Block	<pre> { "ID":, "MainStreamType":, "Enabled":, "VideoEncodeInfo":<VideoEncodeInfo> } </pre>
---------------------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Video stream index number, starts from 0. 0: Main stream 1: Sub stream 2: Third stream	0
MainStreamType	C	unsigned long	0: Main stream of scheduled recording 1: Main stream of event recording Valid when ID is 0.	0

			Applicable to NVR only.	
Enabled	M	boolean	Enable or disable encoding for the video stream. 0: Disable 1: Enable	1
VideoEncodeInfo	M	Json Block	Info about video encoding. See VideoEncodeInfo Json Block	-

32. VideoEncodeInfo

VideoEncodeInfo Json Block	{ "EncodeFormat";, "Resolution": { "Width";, "Height"; }, "BitRate";, "BitRateType";, "FrameRate";, "GOPType";, "IFrameInterval";, "ImageQuality";, "SmoothLevel";, "SVCMODE";, "SmartEncodeMode"; }
---------------------------------------	---

Param	Requirement	Type	Description	Example
EncodeFormat	M	unsigned long	Video encoding format. 0: MJPEG 1: H.264 2: H.265	2
Resolution	M	Json Block	Resolution.	
Width	M	unsigned long	Image width.	1920
Height	M	unsigned long	Image height.	1080
BitRate	M	unsigned long	Bit rate.	4096
BitRateType	M	unsigned long	Bit rate type. 0: CBR 1: VBR	0
FrameRate	M	unsigned long	Frame rate	25
GOPType	C	unsigned long	GOP mode. 0: IP 1: IBP 2: IBBP 3: I Applicable to IPC.	0
IFrameInterval	M	unsigned long	I frame interval. The range depends on	50

			capability.	
ImageQuality	C	unsigned long	Image quality. Range: [0 9]. 9 is the highest. Valid only when BitrateType is VBR. Optional when invalid.	9
SmoothLevel	M	unsigned long	Stream smoothness level. Range: [1,9]. 1 is the lowest.	5
SVCMode	C	boolean	Enable or disable SVC: 0: Disable 1: Enable Applicable to IPC.	0
SmartEncodeMode	M	unsigned long	Smart encoding format. 0: off (disabled) 1: Basic (basic mode) 2: Advanced (advanced mode)	2

3.4 Image

3.4.1 Capabilities

1. /LAPI/V1.0/Channels/<ID>/Image/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Image/Capabilities
Description	Get image parameters of a channel.
Method	GET
Input Data	None
Success Return Data	ImageCapabilities
Note	Applicable to IPC and NVR.
Status	

3.4.2 Enhance

1. /LAPI/V1.0/Channels/<ID>/Image/Enhance

URL	/LAPI/V1.0/Channels/<ID>/Image/Enhance
Description	Get image enhancement parameters of a channel.
Method	GET
Input Data	None
Success Return Data	ImageEnhanceInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Image/Enhance
------------	--

Description	Set image enhancement parameters of a channel.
Method	PUT
Input Data	ImageEnhanceInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3.4.3 Focus

1. /LAPI/V1.0/Channels/<ID>/Image/FocusInfo

URL	/LAPI/V1.0/Channels/<ID>/Image/FocusInfo
Description	Get focus parameters of a video channel.
Method	GET
Input Data	None
Success Return Data	FocusInfo
Note	Applicable to IPC.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Image/FocusInfo
Description	Set focus parameters of a video channel.
Method	PUT
Input Data	FocusInfo
Success Return Data	None
Note	Applicable to IPC.
Status	

3.4.4 LampCtrl

1. /LAPI/V1.1/Channels/<ID>/Image/LampCtrl

URL	/LAPI/V1.1/Channels/<ID>/Image/LampCtrl
Description	Get illuminator parameters of a video channel.
Method	GET
Input Data	None
Success Return Data	LampCtrlInfo
Note	Applicable to IPC.

URL	/LAPI/V1.1/Channels/<ID>/Image/LampCtrl
Description	Set illuminator parameters of a video channel.
Method	PUT
Input Data	LampCtrlInfo
Success Return Data	None
Note	Applicable to IPC.

3.4.5 Exposure

1. /LAPI/V1.0/Channels/<ID>/Image/Advanced/Exposure

URL	/LAPI/V1.0/Channels/<ID>/Image/Advanced/Exposure
Description	Get image exposure parameters.
Method	GET
Input Data	None
Success Return Data	ExposureInfo
Note	Applicable to IPC.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Image/Advanced/Exposure
Description	Set image exposure parameters.
Method	PUT
Input Data	ExposureInfo
Success Return Data	None
Note	Applicable to IPC.
Status	

3.4.6 Illumination

1. /LAPI/V1.0/Channels/<ID>/Image/Illumination

URL	/LAPI/V1.0/Channels/<ID>/Image/Illumination
Description	Get image illumination level.
Method	GET
Input Data	None
Success Return Data	IlluminationInfo
Note	Applicable to IPC.
Status	

3.4.7 Reset

1. /LAPI/V1.0/Channels/<ID>/Image/ImageParamReset

URL	/LAPI/V1.0/Channels/<ID>/Image/ImageParamReset
Description	Set default image parameter values.
Method	PUT
Input Data	ImageParamResetInfo
Success Return Data	None
Note	Applicable to IPC.

3.4.8 Json Block

1. DayNightCapabilitiesInfo

DayNightCapabilitiesInfo Json Block	{ "SupportDayNightCfg";, "DayNightModeNum";, "DayNightModeList";, "SupportSensitivityCfg";, "SupportSwitchingCfg"; }
-------------------------------------	--

Param	Requirement	Type	Description	Example
SupportDayNightCfg	M	boolean	Whether day/night mode is configurable: 0: No 1: Yes	1
DayNightModeNum	M	unsigned long	Number of day/night mode types available.	
DayNightModeList	C	array	Day/night mode types: 0: Auto 1: Day--color 2: Night--black & white 3: Trigger day to night Optional when DayNightModeNum is 0.	[0,1,2]
SupportSensitivityCfg	M	boolean	Whether day/night mode sensitivity is configurable: 0: No 1: Yes	1
SupportSwitchingCfg	M	boolean	Whether day/night mode switching time is configurable: 0: No 1: Yes	1

2. DayNightInfo

DayNightInfo Json Block	{ "Mode";, "Sensitivity";, "Time"; }
-------------------------	--

Param	Requirement	Type	Description	Example
Mode	C	unsigned long	Day/night mode type.	2

			See DayNightCapabilitiesInfo . Required when supported by image capability.	
Sensitivity	C	unsigned long	Day/night mode sensitivity. Valid when day/night mode is “auto”. Range: [0, 9]. Required when supported by image capability.	4
Time	C	unsigned long	Day/night mode switching time. Valid when day/night mode is “auto”. Range: [3, 120]. Unit: Second. Required when supported by image capability.	3

3. EnhanceCapabilitiesInfo

EnhanceCapabilitiesInfo Json Block	{ “ImageRotationModeNum”: , “ImageRotationModeList”: , “IsSupportSharpness”:, “IsSupport2DNoiseReduce”:, “IsSupport3DNoiseReduce”: }
---------------------------------------	--

Param	Requirement	Type	Description	Example
ImageRotationModeNum	M	unsigned long	Number of image rotation modes supported.	3
ImageRotationModeList	C	array	Info list of supported image rotation modes: 0: Normal 1: Flip vertically 2: Flip horizontally 3: Rotate 180° clockwise 4: Rotate 90° clockwise 5: Rotate 90° counterclockwise Optional when ImageRotationModeNum is 0.	[1,3,5]
IsSupportSharpness	M	boolean	Whether sharpness is supported 0: No 1: Yes	1
IsSupport2DNoiseReduce	M	boolean	Whether 2D noise reduction is supported. 0: No 1: Yes	0
IsSupport3DNoiseReduce	M	boolean	Whether 3D noise reduction is supported. 0: No 1: Yes	0

			Applicable to IPC and VMS.	
--	--	--	----------------------------	--

4. ExposureCapabilitiesInfo

ExposureCapabilitiesInfo	{
o Json Block	"SupportCompensation";, "ExposureModeNum";, "ExposureModeList";, "SupportIrisCfg";, "IrisRangeNum";, "IrisRangeList";, "MinGain";, "MaxGain";, "SupportHLCCfg";, "SupportMeteringCfg";, "MeteringModeNum";, "MeteringModeList";, "Shutter": < ShutterCapabilitiesInfo >, "WideDynamic"< WideDynamicCapabilitiesInfo >, "DayNight": < DayNightCapabilitiesInfo >
	}

Param	Requirement	Type	Description	Example
SupportCompensationCfg	M	boolean	Whether exposure compensation is configurable. 0: No 1: Yes	1
ExposureModeNum	M	unsigned long	Number of exposure modes available.	
ExposureModeList	C	array	Exposure modes: 0: Automatic exposure 1: Custom exposure 2: Shutter priority 3: Iris priority 4: Gain priority 5: Indoor 50Hz 6: Indoor 60Hz 7: Manual exposure 8: Low motion blur Optional when ExposureModeNum is 0.	[0,1,3,4,6,7]
SupportIrisCfg	M	boolean	Whether iris is configurable. 0: No 1: Yes	1
IrisRangeNum	M	unsigned long	Number of iris values available.	
IrisRangeList	C	array	Iris values:	[160, 200, 240]

			160, 200, 240, 280, 340, 400, 480, 560, 680, 800, 960, 1100, 1400, 1600, 2200 Optional when IrisRangeNum is 0.	
MinGain	M	unsigned long	Minimum gain.	1
MaxGain	M	unsigned long	Maximum gain.	100
SupportHLCCfg	M	boolean	Whether HLC sensitivity is configurable. 0: No 1: Yes	1
SupportMeteringCfg	M	boolean	Whether metering control is configurable. 0: No 1: Yes	1
MeteringModeNum	M	unsigned long	Number of metering control modes available.	
MeteringModeList	C	array	Metering control modes: 0: Center-weighted average metering 1: Evaluative metering 2: HLC 3: Vehicle metering 4: Face metering 5: Spot metering Optional when MeteringModeNum is 0.	[0,1,2,3,4,5]
Shutter	M	Json block	Shutter See ShutterCapabilitiesInfo Json Block.	
WideDynamic	M	Json block	WDR See WideDynamicCapabilitiesInfo Json Block.	
DayNight	M	Json block	Day/night mode See DayNightCapabilitiesInfo Json Block.	

5. ExposureInfo

ExposureInfo Block	Json	{ "Mode": , "CompensationLevel":, "HLCsensitivity":, "IrisInfo": < IrisInfo >, "ShutterInfo": < ShutterInfo >, "GainInfo": < GainInfo >, "WideDynamic": < WideDynamicInfo >, "Metering": < MeteringInfo >, "DayNight": < DayNightInfo > }
-----------------------	------	---

Param	Requirement	Type	Description	Example
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Mode	M	unsigned long	Exposure mode. See ExposureCapabilitiesInfo	0
Compensation Level	C	unsigned long	Exposure compensation level. Valid when exposure mode is not "Manual". Range: [-100,100]. Required when supported by image capability.	0
HLCsensitivity	C	unsigned long	HLC sensitivity Valid when the scene is Road HLC or Park HLC. Range: [1, 9]. Required when supported by image capability.	
IrisInfo	C	Json Block	Iris info. See IrisInfo . Required when supported by image capability.	
ShutterInfo	C	Json Block	Shutter info. See ShutterInfo . Required when supported by image capability.	
GainInfo	M	Json Block	Gain info. See GainInfo .	
WideDynamic	C	Json Block	WDR info See WideDynamicInfo Required when supported by image capability.	
Metering	C	Json Block	Metering info See MeteringInfo . Valid when the scene is not Road HLC and Park HLC. Required when supported by image capability.	
DayNight	C	Json Block	Day/night mode info. See DayNightInfo Required when supported by image capability.	

6. FocusCapabilitiesInfo

FocusCapabilitiesInfo Json Block	{ "SupportFocusCfg":, "FocusModeNum":, "FocusModeList":, "FocusSceneNum":,
--	--

	"FocusSceneList": }
--	------------------------

Param	Requirement	Type	Description	Example
SupportFocusCfg	M	boolean	Whether focus parameters are configurable: 0: No 1: Yes	1
FocusModeNum	M	unsigned long	Number of focus modes available.	
FocusModeList	C	array	List of focus modes available. 0: Auto focus 1: Manual focus 2: One-key focus 3: One-key focus (IR) 4: One-key focus (locked) One-click focus: The device triggers automatic focus once when rotating, zooming or calling a preset. One-click focus (IR): This mode achieves optimal effects when IR is turned on in a low-light environment (e.g., during night-time or in a dark room). One-click focus (lock): Automatic focus is performed once after the camera finishes zooming. Other actions cannot trigger focus. Optional when FocusModeNum is 0.	[1,2,3,4]
FocusSceneNum	M	unsigned long	Number of focus scenes available	
FocusSceneList	C	array	Focus scenes available: 0: Normal 1: Ultra-long distance 2: Indoor The Normal mode is applicable to general scenes such as road, parks, etc. The Ultra-long distance mode is applicable to long distant road scenes, for example, when the camera mounting height is greater than 30m. The Indoor mode is applicable to indoor scenes. Optional when FocusSceneNum is 0.	[1,2]

7. FocusInfo

FocusInfo Json Block	{ "Mode": , "Scene":
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	}
--	---

Param	Requirement	Type	Description	Example
Mode	M	unsigned long	Focus mode. See FocusInfoCapabilitiesInfo .	1
Scene	C	unsigned long	Focus scene. See FocusInfoCapabilitiesInfo . Required when supported by image capability.	5

8. GainInfo

GainInfo Json Block	{ "Gain":, "MinGain":, "MaxGain": }
---------------------	---

Param	Requirement	Type	Description	Example
Gain	C	unsigned long	Gain (unit: db). Valid in manual exposure mode. Required when supported by image capability. See image capability for the enumerations and range.	0
MinGain	C	unsigned long	Minimum gain. Valid in custom exposure mode. Must not be greater than the maximum gain. Required when supported by image capability. See image capability for the range.	0
MaxGain	C	unsigned long	Maximum gain. Valid in custom exposure mode. Must not be less than the minimum gain. Required when supported by image capability. See image capability for the range.	30

9. ImageCapabilities

ImageCapabilities Json Block	{ "IsSupportCfg":, "SupportCorrectCfg":, "SupportIlluminationCfg", "SupportEPTZ":, "Enhance": < EnhanceCapabilitiesInfo >, "Focus":< FocusCapabilitiesInfo >, "LampCtrl":< LampCtrlCapabilitiesInfo >, "Exposure": < ExposureCapabilitiesInfo >, }
---------------------------------	---

	<pre> "WhiteBalance":<WhiteBalanceCapabilitiesInfo>, "Scene":<SceneCapabilitiesInfo> } </pre>
--	---

Param	Requirement	Type	Description	Example
IsSupportCfg	M	boolean	Whether the device supports image parameter configuration: 0: No 1: Yes	1
SupportCorrectCfg	M	boolean	Whether the device supports image correction. 0: No 1: Yes Applicable to IPC only.	1
SupportIlluminationCfg	M	boolean	Whether the device supports obtaining illumination level. 0: No 1: Yes Applicable to IPC only.	1
SupportEPTZ	C	boolean	Whether the device supports ePTZ auto tracking. 0: No 1: Yes	1
Enhance	M	Json block	Image enhancement capability parameter	See EnhanceCapabilitiesInfo Json Block.
Focus	M	Json block	Focus capability parameter Applicable to IPC only.	See FocusCapabilitiesInfo Json Block.
LampCtrl	M	Json block	Illuminator capability parameter Applicable to IPC only.	See LampCtrlCapabilitiesInfo Json Block.
Exposure	M	Json block	Exposure capability parameter Applicable to IPC only.	See ExposureCapabilitiesInfo Json Block.
WhiteBalance	M	Json block	White balance capability parameter Applicable to IPC only.	See WhiteBalanceCapabilitiesInfo Json Block.
Scene	M	Json block	Scene capability parameter Applicable to IPC only.	See SceneCapabilitiesInfo Json Block.

10. IrisInfo

IrisInfo Json Block	<pre> { "Iris":, "MinIris":, </pre>
---------------------	---

	<pre> "MaxIris": } </pre>
--	---------------------------

Param	Requirement	Type	Description	Example
Iris	C	unsigned long	Iris Optional in iris priority or manual exposure mode. Required when supported by image capability. See image capability for the enumerations.	160
MinIris	C	unsigned long	Minimum iris. Valid in custom exposure mode. Must not be greater than the maximum iris. See iris capability for the enumerations. Required when supported by image capability.	160
MaxIris	C	unsigned long	Maximum iris. Valid in custom exposure mode. Must not be less than the minimum iris. See iris capability for the enumerations. Required when supported by image capability.	200

11. IlluminationInfo

IlluminationInfo Json Block	<pre> { "Value": } </pre>
--------------------------------	-------------------------------

Param	Requirement	Type	Description	Example
Value	C	unsigned long	Illumination level value, normalized to the range [0 - 255]. The higher the level, the higher the overall brightness of the scene. Required when supported by image capability.	1

12. ImageParamResetInfo

ImageParamResetInfo Json Block	<pre> { "Type": } </pre>
-----------------------------------	------------------------------

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Restore default parameter values 0: Video parameters 1: Image parameters	1

13. ImageEnhanceInfo

ImageEnhanceInfo Json Block	{ "Brightness": , "Contrast": , "Saturation": , "Sharpness": , "ImageRotation": , "2DNoiseReduce": , "3DNoiseReduce": }
--------------------------------	---

Param	Requirement	Type	Description	Example
Brightness	M	unsigned long	Brightness. Range: [0,255]	128
Contrast	M	unsigned long	Contrast. Range: [0,255].	128
Saturation	M	unsigned long	Saturation. Range: [0,255].	128
Sharpness	C	unsigned long	Sharpness. Range: [0,255]. Required when supported by image capability.	128
ImageRotation	C	unsigned long	Image rotation modes. Required when supported by image capability.	1
2DNoiseReduce	C	unsigned long	2D NR. Range: [0,255]. Required when supported by image capability.	128
3DNoiseReduce	C	unsigned long	3D noise reduction. Required when supported by image capability.	128

14. LampInfo

LampInfo Json Block	{ "LampType": , "LampCtrlModelListNum": , "LampCtrlModelList": , "SupportNearLampCfg": , "SupportMiddleLampCfg": , "SupportFarLampCfg": , "SupportSuperFarLampCfg": , "SupportLaserAngleCfg": , "LaserAngleMin": , "LaserAngleMax": }
---------------------	--

Param	Requirement	Type	Description	Example
LampType	M	unsigned long	Illuminator types:	1

			0: No illuminator 1: White light illuminator 2: IR illuminator 3: Laser 4: Housing IR illuminator	
LampCtrlMode ListNum	M	unsigned long	Number of control modes available.	
LampCtrlMode List	C	array	Control modes available: 0: Global mode 1: Overexposure restrain mode 2: Preset-road mode 3: Manual mode 4: Preset-park mode 5: Auto sensitivity control mode 6: Indoor mode 7: Manual-always on mode 8: Nighttime video detection mode Global mode and Overexposure restrain mode: Illuminator on/off is linked to day/night mode, and illumination intensity is calculated by AE algorithm. Preset-road mode or Preset-park mode: Turning illuminator on or off is linked to day/night mode, and the intensity of illumination is calculated based on the current zoom ratio and PT angle in combination with preset table file. Manual mode: Linked to day/night mode, and illumination intensity is determined by values configured in the user interface. Manual-always on mode: Not linked with day/night mode, and illuminator is always on. Nighttime video detection mode: Linked with day/night mode and smart virtual iris status. When day/night mode is “nigh” and smart virtual iris is “on”, the illuminator is on; otherwise, the illuminator is off. Illumination intensity level is set on the GUI. Optional when LampCtrlModeListNum is 0.	0
SupportNearLampCfg	M	boolean	Whether near-range illuminator is configurable: 0: No 1: Yes	0

SupportMiddleLampCfg	M	boolean	Whether medium-range illuminator is configurable: 0: No 1: Yes	0
SupportFarLampCfg	M	boolean	Whether long-range illuminator is configurable: 0: No 1: Yes	0
SupportSuperFarLampCfg	M	boolean	Whether ultra-long-range illuminator is configurable: 0: No 1: Yes	1
SupportLaserAngleCfg	M	boolean	Whether laser angle is configurable: 0: No 1: Yes	0
LaserAngleMin	C	unsigned long	Minimum laser angle. Optional when SupportLaserAngleCfg is 0.	
LaserAngleMax	C	unsigned long	Maximum laser angle. Optional when SupportLaserAngleCfg is 0.	

15. LampCtrlCapabilitiesInfo

LampCtrlCapabilitiesInfo Json Block	{ "LampNum": "LampInfos": [< LampInfo >,< LampInfo >,...] }
-------------------------------------	--

Param	Requirement	Type	Description	Example
LampNum	M	unsigned long	Number of illuminator types available. 0 means illuminator is not supported.	
LampInfos	C	array	Illuminator info list. Optional when LampNum is 0.	
LampInfo	M	Json Block	Illuminator info.	See LampInfo Json Block.

16. LampCtrlInfo

LampCtrlInfo Json Block	{ "Enable":, "Type": , "Mode": , "Num": , "LampList": [< LampCtrlInfoList >,< LampCtrlInfoList >,...] }
-------------------------	---

	}
--	---

Param	Requirement	Type	Description	Example
Enable	M	boolean	Enable or disable illuminator: 0: Disable 1: Enable	1
Type	M	unsigned long	Illuminator type. See LampCtrlCapabilitiesInfo).	1
Mode	M	unsigned long	Illuminator control mode. See LampCtrlCapabilitiesInfo).	1
Num	M	unsigned long	Number of illuminator groups.	1
LampList	C	array	Illuminator parameter list. Optional when Num is 0.	
LampCtrlInfoList	M	Json Block	Illuminator parameter. See LampCtrlInfoList Json.	

17. LampCtrlInfoList

LampCtrlInfoList Json Block	{ "Type": , "Num": , "PropertyList": [< PropertyCtrlInfoList >, < PropertyCtrlInfoList >,]" }
---------------------------------------	---

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Illuminator type. See LampCtrlCapabilitiesInfo	0
Num	M	unsigned long	Number of illuminator groups available.	1
PropertyList	C	array	Illuminator attribute list. Optional when ListNum is 0.	
PropertyCtrlInfo	M	Json Block	Illuminator attribute parameter. See PropertyCtrlInfoList Json.	

18. MeteringInfo

MeteringInfo Json Block	{ "Mode": , "RefBrightness": "HoldTime": "Area": { "TopLeft": { "X": , "Y": }, " BottomRight": { "X": , "Y":
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	<pre> } } }</pre>
--	-------------------------------

Param	Requirement	Type	Description	Example
Mode	C	unsigned long	Metering control mode. Valid when not in manual exposure mode. See ExposureCapabilitiesInfo for the enumerations Required when supported by image capability.	1
RefBrightness	C	unsigned long	Brightness of face. Valid in face metering mode. Range: [0, 100].	1
HoldTime	C	unsigned long	Minimum duration. Valid in face metering mode. Unit: minute. Range: [0, 60].	1
MeteringArea	C	Json block	Metering area. Valid in evaluative metering mode or spot metering mode.	
TopLeft	M	Json block	Coordinates of top left corner.	(25, 25)
X	M	unsigned long	x-coordinate of top left corner(proportion). Evaluative metering range: [0100].	25
Y	M	unsigned long	y-coordinate of top left corner(proportion): Evaluative metering range: [0, 100]	25
BottomRight	M	Json block	Coordinates of bottom right corner.	(75, 75)
X	M	unsigned long	x-coordinate of bottom right corner(proportion): Evaluative metering range: [0, 100] Must not be less than the x-coordinate of top left corner.	75
Y	M	unsigned long	y-coordinate of bottom right corner(proportion): Evaluative metering range: [0, 100] Must not be less than the y-coordinate of top left corner.	75

19. PropertyCtrlInfoList

PropertyCtrlInfoList Json Block	<pre> { "Index": , "Property": , "Level" }</pre>
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Param	Requirement	Type	Description	Example
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Index	M	unsigned long	Index of the illuminator group. Range: [0-3].	0
Property	M	unsigned long	Illumination distance: 0: Near-range illuminator 1: Medium-range illuminator 2: Long-range illuminator 3: Ultra-long-range illuminator 4: Laser	0
Level	M	unsigned long	Range for near-range, medium-range, long-range, and ultra-long-range illuminator: [0-1000]. A higher level means higher illumination intensity. For laser range see illuminator capability. A smaller laser angle means higher illumination intensity.	5

20. SceneCapabilitiesInfo

SceneCapabilitiesInfo Json Block	{ "SupportSceneCfg"; "SceneTemplateNum"; "SceneTypeNum"; "SceneTypeList"; "TriggerNum"; "EnvironmentTypeNum"; "EnvironmentTypeList"; }
-------------------------------------	--

Param	Requirement	Type	Description	Example
SupportSceneCfg	M	boolean	Whether scene is configurable: 0: No 1: Yes	1
SceneTemplateNum	C	unsigned long	Number of scene templates available. Optional when scene is not configurable.	1
SceneTypeNum	C	unsigned long	Number of scene types available. Optional when scene is not configurable.	1
SceneTypeList	C	array	Scene types: 0: Custom 1: Indoor 2: Common 3: Vehicle license plate 4: Starlight 5: Test 6: Photographic 7: HLC 8: WDR	[2, 1, 5, 7, 12, 4, 13, 0]

			9: Standard 10: Bright 11: Vivid 12: Park HLC 13: Face 14: Clarity 15: Mixed traffic Optional when SceneTemplateNum is 0.	
TriggerNum	C	unsigned long	Number of triggers available. Optional when scene is not configurable.	4
EnvironmentTypeNum	C	unsigned long	Number of environment types. Optional when scene is not configurable.	1
EnvironmentTypeList	C	array	Environment types: 0: Ambient illumination 1: PTZ tilt angle Optional when EnvironmentTypeNum is 0.	[0]

21. ShutterCapabilitiesInfo

ShutterCapabilities Json Block	<pre> { "SupportShutterTimeCfg": , "ShtterTimeUnit": , "ShutterTimeRangeNum": , "ShutterTimeRangeList": , "SupportSlowShutterCfg":, "SlowShutterTimeRangeNum":, "SlowShutterTimeRangeList": } </pre>
-----------------------------------	--

Param	Requirement	Type	Description	Example
SupportShutterTimeCfg	M	boolean	Whether shutter time is configurable: 0: No 1: Yes	1
ShtterTimeUnit	M	unsigned long	Shuttle time unit: 0: Microsecond 1: Second	1
ShutterTimeRangeNum	M	unsigned long	Number of shutter time values available.	
ShutterTimeRangeList	C	array	Shutter time values: 0: Automatic 1: 1/1 2: 1/2	[0,4,8,10]

			3: 1/3 4: 1/4 6: 1/6 8: 1/8 10: 1/10 12: 1/12 15: 1/15 20: 1/20 25: 1/25 30: 1/30 50: 1/50 60: 1/60 100: 1/100 120: 1/120 150: 1/150 180: 1/180 200: 1/200 250: 1/250 500: 1/500 1000: 1/1000 2000: 1/2000 4000: 1/4000 8000: 1/8000 50000: 1/50000 100000: 1/100000 Optional when ShutterTimeRangeNum is 0.	
SupportSlowShutterCfg	M	boolean	Whether slow shutter is configurable: 0: No 1: Yes	1
SlowShutterTimeRangeNum	M	unsigned long	Number of slow shutter time values available.	
SlowShutterTimeRangeList	C	array	Slow shutter time values: 0: Automatic 1: 1/1 2: 1/2 3: 1/3 4: 1/4 6: 1/6 8: 1/8 10: 1/10 12: 1/12 15: 1/15 20: 1/20 25: 1/25 Optional when SlowShutterTimeRangeNum is 0.	[0,4,8,10]

22. ShutterInfo

ShutterInfo Block	Json	{ "Shutter":, "MinShutter":, "MaxShutter":, "IsEnableSlowShutter":, "SlowestShutter": }
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Param	Requirement	Type	Description	Example
Shutter	C	unsigned long	Shutter time. Valid in iris priority mode, manual exposure mode, or low motion blur mode. Required when supported by image capability. See image capability for the enumerations and units.	0
MinShutter	C	unsigned long	Minimum shutter time. Valid in custom exposure mode. Must not be greater than the maximum shutter time. See shutter capability for the enumerations. Required when supported by image capability.	0
MaxShutter	C	unsigned long	Maximum shutter time. Valid in custom exposure mode. Must not be less than the minimum shutter time. See shutter capability for the enumerations. Required when supported by image capability.	30
IsEnableSlowShutter	C	boolean	Enable or disable slow shutter. 0: Disable 1: Enable Valid when not in iris priority mode. Required when supported by image capability.	1
SlowestShutter	C	unsigned long	Slowest shutter. Valid when slow shutter is enabled. Required when supported by image capability. See image capability for the enumerations.	20

23. WhiteBalanceCapabilitiesInfo

WhiteBalanceCapabilitiesInfo	{
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Json Block	<pre> “WhiteBalanceModeNum”:, “WhiteBalanceModeList” } </pre>
------------	---

Param	Requirement	Type	Description	Example
WhiteBalanceModeNum	M	unsigned long	Number of white balance modes available.	4
WhiteBalanceModeList	M	array	Info list of supported white balance modes: 0: Auto 1: Fine-tune 2: Indoor 3: Outdoor 4: Auto mercury lamp 5: Night fine-tune 6: Auto sodium lamp 7: Locked 8: Adaptive night fine-tune 9: Auto 2	[0,1,2,3]

24. WideDynamicCapabilitiesInfo

WideDynamicCapabilitiesInfo Json Block	<pre> { “SupportWideDynamicCfg”:, “WideDynamicModeNum”:, “WideDynamicModeList”: } </pre>
--	--

Param	Requirement	Type	Description	Example
SupportWideDynamicCfg	M	boolean	Whether WDR is configurable: 0: No 1: Yes	1
WideDynamicModeNum	M	unsigned long	Number of WDR modes available.	
WideDynamicModeList	C	array	WDR modes: 0: Disable 1: Enable 2: Auto Optional when WideDynamicModeNum is 0.	[0,1]

25. WideDynamicInfo

WideDynamicInfo Json Block	<pre> { “Mode”:, “Level”:, </pre>
----------------------------	---

	"OpenSensitivity":, "CloseSensitivity":, "AntiFlicker": }
--	--

Param	Requirement	Type	Description	Example
Mode	M	unsigned long	WDR mode. See WideDynamicCapabilitiesInfo	1
Level	M	unsigned long	WDR level. Valid when WDR is enabled and the exposure mode is one of the following: Automatic, Custom, Shutter priority, Indoor 50Hz, Indoor 60Hz, Low motion blur. Range: [1, 9].	1
OpenSensitivity	M	unsigned long	Sensitivity for enabling WDR. Valid when WDR mode is Auto. Range: [1, 9].	1
CloseSensitivity	M	unsigned long	Sensitivity for disabling WDR. Valid when WDR mode is Auto. Range: [1, 9].	1
AntiFlicker	M	Boolean	Anti-flicker (suppress WDR stripes) 0: Disable 1: Enable When enabled, this feature can eliminate stripes in images.	1

3.5 Storage

3.5.1 Capabilities

1. /LAPI/V1.0/Storage/Capabilities

URL	/LAPI/V1.0/Storage/Capabilities
Description	Get device storage capability.
Method	GET
Input Data	None
Success Return Data	StorageCapabilities
Notes	Applicable to NVR.
Status	

3.5.2 Containers

1. /LAPI/V1.0/Storage/Containers/DetailInfos

URL	/LAPI/V1.0/Storage/Containers/DetailInfos
Description	Get storage container info list.
Method	GET
Input Data	None
Success Return Data	ContainerInfoList
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/Storage/Containers/DetailInfos
Description	Set all storage container info.
Method	PUT
Input Data	<pre>{ "Num":, "ContainerInfoList": [{ "ID": , "Type": , "Property": , "UsageType": , "GroupID": }, { "ID": , "Type": , "Property": , "UsageType": , "GroupID": },] }</pre>
Success Return Data	None
Note	Applicable to NVR and VMS.
Status	

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of storage containers to modify.	3
ContainerInfoList	M	array	Storage container info list.	-
ID	M	unsigned long	Storage container ID. Read-only.	1
Type	M	unsigned long	Storage container type. Read-only. 0: Local HDD 1: Disk enclosure-1 HDD 2: Disk enclosure-2 HDD 3: RAID	0

			4: NAS 5: SAN 6: eSATA 7: SD	
Property	M	unsigned long	Storage container property. 0: Read/write 1: Read-only 2: Redundant. Valid only when the device supports redundancy. Applicable to NVR.	0
GroupID	0	unsigned long	Disk group ID. Applicable to NVR.	1
UsageType	0	unsigned long	Usage. 0: Recording/snapshot 1: Backup (valid only for NAS and eSATA) Absent when the storage container is SD card. Applicable to NVR.	0

3.5.3 Schedule

1. /LAPI/V1.0/Channels/<ID>/Storage/Schedule/Record

URL	/LAPI/V1.0/Channels/<ID>/Storage/Schedule/Record
Description	Get recording schedule of a video channel.
Method	GET
Input Data	None
Success Return Data	RecordScheduleInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Storage/Schedule/Record
Description	Set recording schedule of a video channel.
Method	PUT
Input Data	RecordScheduleInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Storage/Schedule/Picture

URL	/LAPI/V1.0/Channels/<ID>/Storage/Schedule/Picture
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Description	Get snapshot schedule of a video channel.
Method	GET
Input Data	None
Success Return Data	PictureScheduleInfo
Note	Applicable to NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Storage/Schedule/Picture
Description	Set snapshot schedule of a video channel.
Method	PUT
Input Data	PictureScheduleInfo
Success Return Data	None
Note	Applicable to NVR.
Status	

3.5.4 Json Block

1. StorageCapabilities

StorageCapabilities Json Block	{ "IsSupportRedundancy": , "IsSupporteSATA": }
---------------------------------------	---

Param	Requirement	Type	Description	Example
IsSupportRedundancy	M	boolean	Whether redundancy is supported. 0: No 1: Yes	1
IsSupporteSATA	M	boolean	Whether eSATA is supported. 0: No 1: Yes	0

2. ContainerInfoList

ContainerInfoList Json Block	{ "LocalHDDNum" : , "LocalHDDLlist": [< ContainerInfo >, < ContainerInfo >,.....], "SDNum" : , "SDLlist": [< ContainerInfo >, < ContainerInfo >,.....], "ArrayNum" : , "ArrayList": [< ContainerInfo >, < ContainerInfo >,.....], "ExtendCabinet1HDDNum" : , }
-------------------------------------	--

	<pre> "ExtendCabinet1HDDList": [<ContainerInfo>, <ContainerInfo>,.....], "ExtendCabinet2HDDNum" : , "ExtendCabinetHDDList": [<ContainerInfo>, <ContainerInfo>,.....], "NASNum" : , "NASList": [<ExContainerInfo>, <ExContainerInfo>,.....], "eSATANum" : , "eSATAList": [<ExContainerInfo>, <ExContainerInfo>,.....] } </pre>
--	---

Param	Requirement	Type	Description	Example
LocalHDDNum	M	unsigned long	Number of local HDDs.	4
LocalHDDList	C	array	Local HDD info list. Optional when LocalHDDNum is 0.	-
SDNum	M	unsigned long	Number of SD cards. Reserved. Not applicable to NVR and VMS.	0
SDList	C	array	SD card info list. Optional when SDNum is 0.	-
ArrayNum	M	unsigned long	Number of RAIDs.	1
ArrayList	C	array	Array info list. Optional when ArrayNum is 0.	-
ExtendCabinet1HDDNum	M	unsigned long	Number of HDDs in storage expansion enclosure -1.	2
ExtendCabinet1HDDList	C	array	Expansion disk enclosure- 1 info list. Optional when ExtendCabinetHDD1Num is 0.	-
ExtendCabinet2HDDNum	M	unsigned long	Number of HDDs in storage expansion enclosure -2.	2
ExtendCabinet2HDDList	C	array	Expansion disk enclosure- 2 info list. Optional when ExtendCabinet2HDDNum is 0.	-
ContainerInfo	C	Json Block	Container info. See ContainerInfo Json Block.	-
NASNum	M	unsigned long	Number of NAS. Applicable to NVR.	2
NASList	C	array	NAS info list. Optional when NASNum is 0.	-
eSATANum	M	unsigned long	Number of eSATA disks Applicable to NVR.	2
eSATAList	C	array	eSATA info list. Optional when eSATANum is 0.	-

ExContainerInfo	M	Json Block	Extended info about storage container. See ExContainerInfo Json Block.	-
-----------------	---	------------	--	---

3. ContainerInfo

ContainerInfo Json Block	{ "ID": , "RemainCapacity": , "TotalCapacity": , "Manufacturer": , "Status": , "Property": , "FormatProgress": , "GroupID": }
---	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Storage container ID.	1
RemainCapacity	M	unsigned long	Remaining space of storage container (MB).	1907730
TotalCapacity	M	unsigned long	Total capacity of storage container (MB).	1907730
Manufacturer	O	string	Manufacturer name. Length: [1,64].	er000VX000-9YW1CV12
Status	M	unsigned long	Storage container status. 0: No HDD/idle 1: Not formatted 2: Formatting 3: Healthy 4: Sleep 5: Abnormal 6: Switching 7: Unmounted (for NAS HDDs, applicable to NVR)	3
Property	C	unsigned long	Disk property. 0: Read/write 1: Read-only 2: Redundant Optional when Status is 0. Applicable to NVR.	0
FormatProgress	O	unsigned long	Formatting progress. Percentage. Required when Status is 2. Applicable to NVR.	0
GroupID	O	unsigned long	Disk group ID.	1

4. ExContainerInfo

ExContainerInfo Json Block	<pre>{ "ID": , "RemainCapacity": , "TotalCapacity": , "AddressType": , "IPAddress": , "Path": , "UsageType": , "Status": , "Property": , "FormatProgress": , "GroupID": }</pre>
-----------------------------------	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Storage container ID.	1
RemainCapacity	M	unsigned long	Remaining space of storage disk (MB).	1907730
TotalCapacity	M	unsigned long	Total capacity of storage disk (MB).	1907730
Status	M	unsigned long	Storage disk status. 0: No HDD/idle 1: Not formatted 2: Formatting 3: Healthy 4: Sleep 5: Abnormal 6: Switching 7: Unmounted (for NAS HDDs, not applicable to VMS)	3
Property	M	unsigned long	Disk property: 0: Read/write 1: Read-only 2: Redundant	0
FormatProgress	C	unsigned long	Progress. Percentage. Required when Status is 2.	0
AddressType	C	unsigned long	IP address type. 0: IPv4 (default) 1: IPv6 2: Domain name 3: IPv4 and IPv6 Currently only IPv4 is supported. Required when the storage type is NAS.	0
IPAddress	C	string	Server IP address. Length: [0,64]. Required when the storage type is	"192.168.100.30"

			NAS.	
Path	C	string	Storage path on NAS server. Length: [1,256].	"/volume2/123456"
UsageType	M	unsigned long	Usage. 0: Recording/snapshot 1: Backup	0
GroupID	C	unsigned long	Disk group ID. The default is 1.	1

5. RecordScheduleInfo

RecordScheduleInfo Json Block	<pre>{ "Enabled": , "IsRedundantStorage": , "RecordRule": { "PreRecordTime": , "PostRecordTime": , }, "WeekPlan":<WeekPlanInfo> }</pre>
--------------------------------------	---

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable schedule: 0: Disable 1: Enable	1
IsRedundantStorage	C	unsigned long	Enable or disable redundant recording. 0: Disable 1: Enable. When enabled, an extra copy of recordings will be stored on the redundant HDD. Absent when the device does not support redundant recording storage. Not applicable to IPC.	1
RecordRule	M	Json Block	Video recording rules.	-
PreRecordTime	M	unsigned long	Length of pre-alarm recording. Unit: Second. Range: 0, 5, 10, 20, 30, 60	5
PostRecordTime	M	unsigned long	Length of post-alarm recording. Unit: Second. Range: 5, 10, 30, 60, 120, 300, 600	10

WeekPlan	M	Json Block	Scheduled time list.	-
WeekPlanInfo	M	Json Block	Detailed schedule info. See Json Block.	-

6. PictureScheduleInfo

PictureScheduleInfo Json Block	<pre>{ "Enabled": , "IsRedundantStorage": , "WeekPlan":<WeekPlanInfo> }</pre>
---------------------------------------	---

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable schedule: 0: Disable 1: Enable	1
IsRedundantStorage	C	unsigned long	Enable or disable redundant snapshot 0: Disable 1: Enable. When enabled, an extra copy of snapshots will be stored on the redundant HDD. Absent when the device does not support redundant recording storage. Applicable to NVR.	1
WeekPlan	M	Json Block	Scheduled time list	-
WeekPlanInfo	M	Json Block	Detailed arming schedule info. See WeekPlanInfo Json Block.	-

3.6 Alarm

3.6.1 Capabilities

1. /LAPI/V1.0/Channels/<ID>/Alarm/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Alarm/Capabilities
Description	Get alarm capability of a channel.
Method	GET
Input Data	None

Success Return Data	AlarmCapabilities
Note	Applicable to IPC and NVR.
Status	

3.6.2 MotionDetection

1. /LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/AreaType

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/AreaType
Description	Get the motion detection area type of a video channel.
Method	GET
Input Data	None
Success Return Data	MotionDetectionAreaType
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/AreaType
Description	Set the motion detection area type of a video channel.
Method	PUT
Input Data	MotionDetectionAreaType
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Rectangle

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Rectangle
Description	Get all motion detection rectangular areas of a video channel.
Method	GET
Input Data	None
Success Return Data	MotionDetectionRectangleAreaInfoList
Note	Applicable to IPC.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Rectangle
Description	Set all motion detection rectangular areas of a video channel.
Method	PUT
Input Data	MotionDetectionRectangleAreaInfoList
Success Return Data	None
Note	Applicable to IPC.
Status	

3. /LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Rectangle/<ID>

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Rectangle/<ID>
Description	Get a motion detection rectangular area of a video channel.
Method	GET
Input Data	None
Success Return Data	MotionDetectionRectangleAreaInfo
Note	Applicable to IPC.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Rectangle/<ID>
Description	Set a motion detection rectangular area of a video channel.
Method	PUT
Input Data	MotionDetectionRectangleAreaInfo
Success Return Data	None
Note	Applicable to IPC.
Status	

4. /LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Grid

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Grid
Description	Get all motion detection grid areas of a video channel.
Method	GET
Input Data	None
Success Return Data	MotionDetectionGridAreaInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/Areas/Grid
Description	Set all motion detection grid areas of a video channel.
Method	PUT
Input Data	MotionDetectionGridAreaInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

5. /LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/LinkageActions
Description	Get linkage actions of motion detection.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC and NVR.

Status	
---------------	--

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/LinkageActions
Description	Set linkage actions of motion detection.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

6. /LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/WeekPlan
Description	Get arming schedule of motion detection.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/MotionDetection/WeekPlan
Description	Set arming schedule of motion detection.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3.6.3 VideoLoss

1. /LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/Rule

URL	/LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/Rule
Description	Get video loss alarm rules of a video channel.
Method	GET
Input Data	None
Success Return Data	VideoLossRuleInfo
Note	Applicable to NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/Rule
Description	Set video loss alarm rules of a video channel.
Method	PUT

Input Data	VideoLossRuleInfo
Success Return Data	None
Note	Applicable to NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/LinkageActions
Description	Get linkage actions of video loss alarm.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/LinkageActions
Description	Set linkage actions of video loss alarm.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to NVR.
Status	

3. /LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/WeekPlan
Description	Get arming scheduling of video loss alarm.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/VideoLoss/WeekPlan
Description	Set arming schedule of video loss alarm.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to NVR.
Status	

3.6.4 TamperDetection

1. /LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/Rule
Description	Get tampering detection rules of a video channel.
Method	GET
Input Data	None
Success Return Data	TamperDetectionRuleInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/Rule
Description	Set tampering detection rules of a video channel.
Method	PUT
Input Data	TamperDetectionRuleInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/LinkageActions
Description	Get linkage actions of tampering detection alarm.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/LinkageActions
Description	Set linkage actions of tampering detection alarm.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Linkage actions of tampering detection alarm do not support snapshot and pop-up alarm window. Applicable to IPC and NVR.
Status	

3. /LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/WeekPlan
Description	Get arming schedule of tampering detection alarm.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/TamperDetection/WeekPlan
Description	Set arming schedule of tampering detection alarm.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3.6.5 AudioDetect

1. /LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/Rule
Description	Get audio detection rules.
Method	GET
Input Data	None
Success Return Data	AudioDetectInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/Rule
Description	Set audio detection rules.
Method	PUT
Input Data	AudioDetectInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/LinkageActions
------------	--

Description	Get linkage actions of audio detection.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/LinkageActions
Description	Set linkage actions of audio detection.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3. /LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/WeekPlan
Description	Get arming schedule of audio detection.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Alarm/AudioDetection/WeekPlan
Description	Set arming schedule of audio detection.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3.6.6 HumanShapeDetection

1. /LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/Rule
Description	Get human body detection rules.
Method	GET
Input Data	None
Success Return Data	HumanShapeDetectionRuleInfo

Note	Applicable to IPC and NVR.
-------------	----------------------------

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/Rule
Description	Set human body detection rules.
Method	PUT
Input Data	HumanShapeDetectionRuleInfo
Success Return Data	None
Note	Applicable to IPC and NVR.

2. /LAPI/V1.1/Channels/<ID>/Alarm/HumanShapeDetection/Areas

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/Areas
Description	Get all rectangular detection areas of human body detection.
Method	GET
Input Data	None
Success Return Data	HumanShapeDetectAreaInfoList
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/Areas
Description	Set all rectangular detection areas of human body detection.
Method	PUT
Input Data	HumanShapeDetectAreaInfoList
Success Return Data	None
Note	Applicable to IPC and NVR.

3. /LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/LinkageActions
Description	Get linkage actions of human body detection.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/LinkageActions
Description	Set linkage actions of human body detection.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC.

4. /LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/WeekPlan
Description	Get arming schedule of human body detection.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Alarm/HumanShapeDetection/WeekPlan
Description	Set arming schedule of human body detection.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.

3.6.7 Json Block

1. AlarmCapabilities

AlarmCapabilities Json Block	<pre>{ "MotionDetection": { "IsSupportCfg": , "SupportedTypeNum":, "SupportedTypeList":[...] , "RectangleAreaNum ": , "BlockWidth": , "BlockHeight": }, "TamperDetection": { "IsSupportCfg": , "TamperRECTAreaNum": }, "AudioDetection": { "SupportCfg": } "HumanShapeDetection": { "SupportCfg": , "RectangleAreaNum": , "LinkagePlanCfg": { "SupportAudioLinkagePlan":, "SupportLightLinkagePlan": } } }</pre>
---------------------------------	--

	} }
--	--------

Param	Requirement	Type	Description	Example
MotionDetection	M	Json Block	Motion detection capability info.	-
IsSupportCfg	M	boolean	Whether is configurable: 0: No 1: Yes	1
SupportedTypeNum	C	unsigned long	Number of motion detection types supported. Applicable to IPC.	2
SupportedTypeList	C	array	Supported motion detection types 0: Rectangular area 1: Grid area Applicable to IPC.	[0, 1]
RectangleAreaNum	C	unsigned long	Number of rectangular detection areas supported. Optional when rectangular area is not supported. Applicable to IPC.	
BlockWidth	M	unsigned long	Maximum value of grid width.	22
BlockHeight	M	unsigned long	Maximum value of grid height.	18
TamperDetection	M	Json Block	Tampering detection capability info.	-
IsSupportCfg	M	boolean	Whether the connected device supports configuration: 0: No 1: Yes	1
TamperRECTAreaNum	M	unsigned long	Number of tampering detection areas that can be drawn. Currently fixed to 1.	1
AudioDetection	M	Json Block	Audio detection capability info.	
SupportCfg	M	boolean	Whether is configurable: 0: No 1: Yes	1
HumanShapeDetection	M	Json Block	Human body detection capability info.	-
SupportCfg	M	boolean	Whether is configurable: 0: No 1: Yes	1
RectangleAreaNum	M	unsigned long	Number of human body detection areas that can be drawn. Currently fixed to 1.	1
LinkagePlanCfg	C	Json Block	Arming schedule of linkage actions. Required when supported.	
SupportAudioLinkagePlan	M	boolean	Whether audio linkage schedule is supported. 0: No	1

			1: Yes	
SupportLightLinkage Plan	M	boolean	Whether light linkage schedule is supported. 0: No 1: Yes	1

2. HumanShapeDetectionRuleInfo

HumanShapDetectionRuleInfo Json Blocok	{ "Enabled":, "DetectDisplayEnabled": }
---	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable human body detection 0: Disable 1: Enable	1
DetectDisplayEnabled	C	boolean	Enable or disable bounding box of human body detection 0: Disable 1: Enable	1

3. HumanShapeDetectAreaInfoList

HumanShapeDetectAreaInfoList Json Block	{ "Num": , "PolygonInfoList":[< HumanShapeDetectPolygonInfo >,] }
--	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of detection areas.	1
PolygonInfoList	C	array	Detection area info list. Optional when Num is 0.	-
HumanShapeDetecPolygonInfo	M	Json Block	Detection area info. See HumanShapeDetectPolygonInfo Json Block	-

4. HumanShapeDetectPolygonInfo

HumanShapeDetectPolygonInfo Json Block	{ "ID": , "Enabled": , "DisplayEnabled":,
---	--

	<pre> "Sensitivity": , "TargetSize": , "PointNum": , "PointList": [{ "X": , "Y": }, ...] </pre>
--	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Detection area ID, starts from 0. Use the /LAPI/V1.1/Channels/<ID>/Alarm/Capabilities interface to obtain the maximum number allowed.	1
Enabled	M	boolean	Enable or disable detection area: 0: Disable 1: Enable	1
DisplayEnabled	M	boolean	Enable or disable detection area display 0: Disable 1: Enable Indicates whether to show the detection box.	1
Sensitivity	M	unsigned long	Motion detection sensitivity. The greater the value, the higher the sensitivity. Range: [1,100]	50
TargetSize	M	unsigned long	Object size. Range: [1,100]	1
PointNum	M	unsigned long	Number of vertices of the polygon detection area. Range: [0,6]	4
PointList	C	array	Coordinates of vertices of the detection area. Optional when PointNum is 0.	
X	M	unsigned long	x-coordinate. Unit: ‰. Range: [0,9999]	0
Y	M	unsigned long	y-coordinate. Unit: ‰. Range: [0,9999]	0

5. LinkageActionList

LinkageActionList Json Block	<pre> { "Num": , "Actions": [<LinkageActionInfo>, ...] } </pre>
---------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of linkage actions.	8
Actions	C	array	Linkage action info list. Optional when Num is	-

			0.	
LinkageActionInfo	M	Json Block	Linkage action. See LinkageActionInfo Json Block.	-

6. LinkageActionInfo

LinkageActionInfo Json Block	{ "Enabled": , "ActID": , "ActParam": }
---------------------------------	---

Param	Requirement	Type	Description	Example
Enabled	C	boolean	Enable or disable linkage action. Required when ActID is greater than or equal to 28. 0: Disable 1: Enable (For compatibility, any value that is less than 28, even when present, is invalid)	1
ActID	M	unsigned long	See the table below for ActIDs.	5
ActParam	C	Json Block	See the table below for the correspondence relationship between ActID and ActParam.	

ActID	Description	IPC ActParam	NVR ActParam
0	NVR preview (applicable to NVR)	Not Support	ChannelActParamInfo
1	Buzzer (not available to IPC)	Not Support	EnabledActParamInfo
2	E-mail (applicable to NVR)	Not Support	EnabledActParamInfo
3	Recording	Not Support	ChannelActParamInfo
4	Go to PTZ preset	PresetActParamInfo	PresetActParamInfo
5	Output switch	OutputSwitchActParamInfo	OutputSwitchActParamInfo
6	Snapshot	Null	ChannelActParamInfo
7	Pop-up alarm window (applicable to NVR)	Not Support	EnabledActParamInfo
8	Central recording	Null	Not Support

9	Start local recording	Null	Not Support
10	Stop local recording	Null	Not Support
11	Snapshot and then upload via FTP	Null	Not Support
12	Snapshot and then send via e-mail	Null	Not Support
13	Snapshot and then upload via email and FTP	Null	Not Support
14	Smart snapshot and upload	Null	Not Support
15	Upload close-up image of face	Null	Not Support
16	Alarm notification	Null	Not Support
28	Sound alarm	AudioLinkagePlanInfo	AudioLinkagePlanInfo
29	Light alarm	LightLinkagePlanInfo	LightLinkagePlanInfo

7. AudioLinkagePlanInfo

AudioLinkagePlanInfo Json Block	<pre>{ "TimeMode": , "AudioCustomInfo": <AudioCustomInfo> }</pre>
--	---

Param	Requirement	Type	Description	Example
TimeMode	M	unsigned long	Type of linkage time period. 0: Custom time period	0
AudioCustomInfo	C	Json Block	Detailed arming schedule info when the linkage action is sound alarm. Required when in PUT messages and TimeMode is 0, or when in GET messages. See AudioCustomInfo Json Block.	

8. AudioCustomInfo

AudioCustomInfo Json Block	<pre>{ "Num": , "Days": [<AudioCustomDayPlanInfo>, ...] }</pre>
-----------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of scheduled days IPC: Fixed to 7 (seven days a week).	7
Days	M	array	Arming schedule of every day in the	

			week.	
AudioCustomDayPlanInfo	M	Json Block	Detailed info about arming schedule of every day. See AudioCustomDayPlanInfo Json Block.	

9. AudioCustomDayPlanInfo

AudioCustomDayPlanInfo Json Block	{ "ID": , "Num": , "TimeSectionInfos": [< AudioTimeSectionInfo >, ...] }
-----------------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of days 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday 7: Sunday	1
Num	M	unsigned long	Number of time periods per day. IPC: Fixed to 4.	4
TimeSectionInfos	M	array	Time periods of arming schedule. Time periods on the same day must not overlap.	
AudioTimeSectionInfo	M	Json Block	Detailed audio arming schedule. See AudioTimeSectionInfo .	

10. AudioTimeSectionInfo

AudioTimeSectionInfo Json Block	{ "Enabled": , "Begin": , "End": , "AudioActParamInfo":< AudioActParamInfo > }
------------------------------------	---

Param	Requirement	Type	Description	Example
Enabled	M	unsigned long	Enable or disable arming time periods. 0: Disable	0

			1: Enable	
Begin	M	string	Start time. Format: hh:mm:ss Length range: [0,31]	00:00:00
End	M	string	End time. Format: hh:mm:ss Length range: [0,31] The end time must be later than the start time.	23:59:59
AudioActParamInfo	M	Json Block	Sound alarm configuration. See AudioActParamInfo Json Block.	

11. LightLinkagePlanInfo

LightLinkagePlanInfo Json Block	<pre>{ "TimeMode": , "LightCustomInfo": <LightCustomInfo> }</pre>
---------------------------------	---

Param	Requirement	Type	Description	Example
TimeMode	M	unsigned long	Type of linkage time period. 0: Custom time period	0
LightCustomInfo	C	Json Block	Detailed arming schedule info when the linkage action is light alarm. Required when in PUT messages and TimeMode is 0, or when in GET messages. See LightCustomInfo Json Block.	

12. LightCustomInfo

LightCustomInfo Json Block	<pre>{ "LightActParamInfo":<LightActParamInfo>, "LightWeekPlanInfo":<LightWeekPlanInfo> }</pre>
----------------------------	---

Param	Requirement	Type	Description	Example
LightActParamInfo	M	Json Block	Light alarm configuration. See LightActParamInfo Json Block.	
LightWeekPlanInfo	M	Json Block	Detailed info about light arming schedule. See LightWeekPlanInfo Json Block	

13. LightWeekPlanInfo

LightWeekPlanInfo Json Block	{ "Num": , "Days": [< LightDayPlanInfo >, ...] }
------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of scheduled days IPC: Fixed to 7 (seven days a week).	7
Days	M	array	Arming schedule of every day in the week.	
LightDayPlanInfo	M	Json Block	Detailed info about daily arming schedule. See LightDayPlanInfo Json Block	

14. LightDayPlanInfo

LightDayPlanInfo Json Block	{ "ID": , "Num": , "TimeSectionInfos": [< LightTimeSectionInfo >, ...] }
-----------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of days 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday 7: Sunday	1
Num	M	unsigned long	Number of time periods per day IPC: Fixed to 4.	4
TimeSectionInfos	M	array	Time periods of arming schedule. Time periods on the same day must not overlap.	
LightTimeSectionInfo	M	Json Block	Detailed light arming schedule. See LightTimeSectionInfo .	

15. LightTimeSectionInfo

LightTimeSectionInfo Json Block	{ "Enabled": , "Begin": , "End": }
------------------------------------	--

Param	Requirement	Type	Description	Example
Enabled	M	unsigned long	Enable or disable arming time periods. 0: Disable 1: Enable	0
Begin	M	string	Start time. Format: hh:mm:ss Length range: [0,31]	00:00:00
End	M	string	End time. Format: hh:mm:ss Length range: [0,31] The end time must be later than the start time.	23:59:59

16. EnabledActParamInfo

EnabledActParamInfo Json Block	{ "Enabled": }
-------------------------------------	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	1: Linkage action enabled 0: Linkage action disabled	1

17. ChannelActParamInfo

ChannelActParamInfo Json Block	{ "Num": , "IDs": [<ID>,<ID>...] }
-------------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of video channels.	
IDs	C	array	Video channel IDs.	
ID	C	unsigned long	Video channel ID	1

18. PresetActParamInfo

PresetActParamInfo Json Block	{ "Num":, "ChannelPresetList": [< ChannelPreset >, < ChannelPreset >...] }
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of linkage actions.	2
ChannelPresetList	C	array	Info list of linked presets. Optional when Num is 0.	
ChannelPreset	C	Json block	Channel and preset ID.	See ChannelPreset Json block

19. ChannelPreset

ChannelPreset Json Block	{ "ChannelID":, "PresetID": }
--	--

Param	Requirement	Type	Description	Example
ChannelID	C	unsigned long	Video channel ID. Optional for IPC.	1
PresetID	M	unsigned long	Preset ID.	1

20. OutputSwitchActParamInfo

OutputSwitchActParamInfo Json Block	{ "Num":, "IDs":[<ID>,<ID>...] }
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of linked output switches.	M
IDs	C	array	List of linked output switches.	C
ID	C	unsigned long	Output switch channel ID. For NVR, the ID is video channel ID*100 + alarm output ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	1

21. DayPlanInfo

DayPlanInfo Json Block	{ "ID": , "Num": , "TimeSectionInfos": [< TimeSectionInfo >, ...] }
---	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of days 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday 7: Sunday 8: Holiday Remarks: 8 is applicable to NVR and VMS.	1
Num	M	unsigned long	Number of time periods per day. NVR: Max 8. IPC: Max 4.	8
TimeSectionInfos	C	array	Time periods of arming schedule. Time periods on the same day must not overlap. Optional when Num is 0.	
TimeSectionInfo	C	Json Block	Detailed arming schedule. See TimeSectionInfo Json Block. Optional when Num is 0.	

22. WeekPlanInfo

WeekPlanInfo Json Block	{ "Enabled": , "Num": , "Days": [< DayPlanInfo >, ...] }
--	--

Param	Requirement	Type	Description	Example
Enabled	C	boolean	Enable or disable arming schedule. 0: Disable 1: Enable Applicable to IPC only. Default: Enable.	1
Num	M	unsigned long	Number of scheduled days. NVR: Max 8 (7 days of a week plus holiday).	8

			IPC: Max 7 (7 days of a week).	
Days	M	array	Arming schedule of every day in the week.	
DayPlanInfo	M	Json Block	Detailed info about daily arming schedule. See DayPlanInfo Json Block	

23. TimeSectionInfo

TimeSectionInfo Json Block	{ "Begin": , "End": , "ArmingType": }
-----------------------------------	---

Param	Requirement	Type	Description	Example
Begin	M	string	Start time. Format: hh:mm:ss Length range: [0,31]. NVR does not support ss (seconds).	00:00:00
End	M	string	End time. Format: hh:mm:ss Length range: [0,31]. NVR does not support ss (seconds).	24:00:00
ArmingType	C	unsigned long	Arming type. 0: Normal 1: Motion detection 2: Alarm 3: Motion detection and alarm 4: Motion detection or alarm 5: No schedule 10: Event Recording and snapshot support all; other linkage actions support Normal only. For IPC this field is optional because IPC only supports Normal (default setting). Not applicable to IPC.	

24. VideoLossRuleInfo

VideoLossRuleInfo Json Block	{ "Enabled": }
-------------------------------------	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable 0: Disable	1

			1: Enable	
--	--	--	-----------	--

25. TamperDetectionRuleInfo

TamperDetectionRuleInfo Block	Json	{ "Enabled": , "Sensitivity": , "Duration": }
--	-------------	---

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	50
Duration	C	unsigned long	Duration (unit: second). Range: [0, 10]. Applicable to IPC.	1

26. MotionDetectionAreaType

MotionDetectionAreaType Block	Json	{ "Type": }
--	-------------	-------------------

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Area type. Currently two types: 0: Rectangular area 1: Grid area	1

27. MotionDetectionRectangleAreaInfo

MotionDetectionRectangleAreaInfo Block	Json	{ "ID": , "Enabled": , "Sensitivity": , "Duration": , "TargetSize": , "Area": { "TopLeft": { "X": 0, "Y": 0 }, }, "BottomRight": { "X": 10,
---	-------------	--

	<pre> "Y": 10 } } } </pre>
--	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Rectangle area ID, starts from 0. Use the /LAPI/V1.0/Channels/<ID>/Alarm/Capabilities interface to get the maximum number of areas allowed.	1
Enabled	M	boolean	Enable or disable area. 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Motion detection sensitivity. The greater the value, the higher the sensitivity. Range: [1,100].	50
Duration	C	unsigned long	Duration. Range: [1,100]. Applicable to IPC.	5
TargetSize	C	unsigned long	Object size. Range: [1,100]. Applicable to IPC.	1
RectangleArea	M	Json Block	Rectangular area info.	-
TopLeft	M	Json Block	Coordinates of the rectangular area's top left corner.	-
X	M	unsigned long	x-coordinate. Range: [0,9999].	0
Y	M	unsigned long	y-coordinate. Range: [0,9999].	0
BottomRight	M	Json Block	Coordinates of the rectangular area's bottom right corner.	-
X	M	unsigned long	x-coordinate. Range: [0,9999].	10
Y	M	unsigned long	y-coordinate. Range: [0,9999].	10

28. MotionDetectionRectangleAreaInfoList

MotionDetectionRectangleAreaInfoList Block	Json	<pre> { "Num": , "RectangleAreas": [<MotionDetectionRectangleAreaInfo>,] } </pre>
---	-------------	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of rectangular areas.	1
RectangleAreas	C	array	Rectangular area info list. Optional when Num is 0.	-
MotionDetectionRectangleAreaInfo	M	Json Block	Rectangular area info. See MotionDetectionRectangleAreaInfo Json Block.	-

29. MotionDetectionGridAreaInfo

MotionDetectionGridAreaInfo Json Block	<pre>{ "Enabled": , "Sensitivity": , "Grid": }</pre>
--	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable area. 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Motion detection sensitivity. The greater the value, the higher the sensitivity. Range: [1,100].	50
Grid	M	string	Detection area. Length range: [0, 256] 1. The entire screen is divided into 22x18 grids, starting from the upper left corner to the lower right corner (from left to right, top to bottom), a total of 396 grids, numbered sequentially. Then the 8-bit number (maximum 255) represented by binary bits indicates the selection status of the motion detection area of 8 grids: 1 means “selected”, 0 means “not selected” (from low order to high order). This converts all grids into 50 arrays of 8-bit numbers. 2. The array is first compressed by the PackBits algorithm, and then encoded with Base64 to obtain the compressed string.	"DWAAAYAA Bj/5mP/mA AAY/gAgP/4 A//77//vv/+ +//74ABvgB G+AAb4ABv h/2+H/YAf8 A"

30. AudioDetectInfo

AudioDetectInfo Json Block	<pre>{ "Enabled": , "DetectType": , "DiffValue": , "Threshold": }</pre>
--	---

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable audio detection: 0: Disable 1: Enable	1

DetectType	M	unsigned long	Audio detection type: 0: Sudden rise 1: Sudden fall 2: Sudden change 3: Threshold	0
DiffValue	C	unsigned long	Difference value, which is used when DetectType is “Sudden Rise”, “Sudden Fall”, “Sudden Change”. Range: [0,400]. An alarm is triggered when the value of sudden rise, sudden fall, or sudden change exceeds the threshold.	100
Threshold	C	unsigned long	Threshold. Valid when audio detection type is “Threshold”. Range: [0,400]. An alarm is triggered when the sound volume exceeds the threshold.	100

3.7 IO

3.7.1 InputSwitch

1. /LAPI/V1.0/IO/InputSwitches/BasicInfos?[DevID=<DevID>]&[OrgID=<OrgID>]

URL	/LAPI/V1.0/IO/InputSwitches/BasicInfos?[DevID=<DevID>]&[OrgID=<OrgID>]
Description	Get detailed info list of multiple alarm input channels.
Method	GET
Input Data	None
Success Return Data	InputSwitchBasicInfoList
Note	Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
DevID	0	unsigned long	Device ID. Null means searching all devices. Applicable to VMS.	21
OrgID	0	unsigned long	Organization ID. Null means searching all organizations. Applicable to VMS.	55

2. /LAPI/V1.0/IO/InputSwitches/<ID>/BasicInfos

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/BasicInfos
------------	---

Description	Get an alarm input channel.
Method	GET
Input Data	None
Success Return Data	InputSwitchBasicInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/BasicInfos
Description	Set an alarm input channel.
Method	PUT
Input Data	InputSwitchBasicInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Alarm input channel ID. For NVR, the ID is channel ID*100+alarm input ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	108

3. /LAPI/V1.0/IO/InputSwitches/<ID>/LinkageActions

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/LinkageActions
Description	Get linkage actions of an input switch.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/LinkageActions
Description	Set linkage actions of an input switch.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

4. /LAPI/V1.0/IO/InputSwitches/<ID>/WeekPlan

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/WeekPlan
Description	Get arming schedule of input switch.

Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/IO/InputSwitches/<ID>/WeekPlan
Description	Set arming schedule of input switch.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3.7.2 OutputSwitch

1. /LAPI/V1.0/IO/OutputSwitches/BasicInfos?[DevID=<DevID>]&[OrgID=<OrgID>]

URL	/LAPI/V1.0/IO/OutputSwitches/BasicInfos?[DevID=<DevID>]&[OrgID=<OrgID>]
Description	Get detailed info list of multiple alarm output channels.
Method	GET
Input Data	None
Success Return Data	OutputSwitchBasicInfoList
Note	Applicable to IPC, NVR, and VMS.
Status	

Param	Requirement	Type	Description	Example
DevID	0	unsigned long	Device ID. Null means searching all devices. Applicable to VMS.	21
OrgID	0	unsigned long	Organization ID. Null means searching all organizations. Applicable to VMS.	55

2. /LAPI/V1.0/IO/OutputSwitches/<ID>/BasicInfos

URL	/LAPI/V1.0/IO/OutputSwitches/<ID>/BasicInfos
Description	Get info about alarm output channel.
Method	GET
Input Data	None
Success Return Data	OutputSwitchBasicInfo
Note	Applicable to IPC, NVR, and VMS.
Status	

URL	/LAPI/V1.0/IO/OutputSwitches/<ID>/BasicInfos
Description	Set info about alarm output channel.
Method	PUT
Input Data	OutputSwitchBasicInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Alarm output channel ID.	99

3. /LAPI/V1.0/IO/OutputSwitches/<ID>/WeekPlan

URL	/LAPI/V1.0/IO/OutputSwitches/<ID>/WeekPlan
Description	Get arming schedule of a device's output switch.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/IO/OutputSwitches/<ID>/WeekPlan
Description	Set arming schedule of a device's output switch.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

4. /LAPI/V1.0/IO/OutputSwitches/ManualAlarm

URL	/LAPI/V1.0/IO/OutputSwitches/ManualAlarm
Description	Manually control output switch alarm.
Method	PUT
Input Data	SetOutputSwitchManualAlarmList
Success Return Data	None
Note	Applicable to NVR and VMS.
Status	

5. /LAPI/V1.0/IO/OutputSwitches/AlarmStatus

URL	/LAPI/V1.0/IO/OutputSwitches/AlarmStatus
Description	Get alarm status of output switch.
Method	GET
Input Data	None
Success Return Data	OutputSwitchAlarmStatusList
Note	Applicable to IPC, NVR, and VMS.
Status	

3.7.3 Json Block

1. InputSwitchBasciInfo

InputSwitchDetailInfo Json Block	{ "ID": , "Name": , "Status": , "GBID": , "VMSCCommon": , "RunMode": , "Enabled": }
-------------------------------------	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Alarm input channel ID. For NVR, the ID is the video channel ID*100+alarm output ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	1
Name	C	string	Channel name. Length: [1, 64]. Currently not applicable to NVR.	"IP Camera 01"
Status	C	unsigned long	Channel status 0: Offline 1: Online Applicable to VMS	0
GBID	C	string	GB/T28181 resource ID Length range: [0, 20]. Applicable to VMS and IPC.	"34000000002 000000010"
VMSCCommon	C	Json Block	Common VMS information	See VMSCCommon Json Block.
RunMode	M	unsigned long	Alarm type 1: Normally open 2: Normally closed	1

Enabled	M	boolean	Enable or disable alarm 0: Disable 1: Enable	1
---------	---	---------	--	---

2. InputSwitchBasicInfoList

InputSwitchBasicInfoList Json Block	{ "Num": , "BasicInfos": [< InputSwitchBasicInfo >,.....] }
--	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of alarm input channels.	4
BasicInfos	C	array	Detailed info list of alarm input channels. Optional when Num is 0.	-
InputSwitchBasicInfo	M	Json Block	Detailed info about alarm input channel. See InputSwitchBasicInfo Json Block.	-

3. OutputSwitchBasicInfo

OutputSwitchBasicInfo Json Block	{ "ID": , "Name": , "Status": , "VMSCCommon": , "RunMode": , "Duration": }
---	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Output switch channel ID. For NVR, the ID is video channel ID*100+alarm output ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	1
Name	C	string	Channel name. Length: [1, 64]. Currently not applicable to NVR.	"O_AlarmOutput_2"
Status	C	unsigned long	Channel status. 0: Offline 1: Online Applicable to VMS.	0

VMSCommon	C	Json Block	Common VMS information	See VMSCommon Json Block
RunMode	M	unsigned long	Alarm type. 1: Normally open 2: Normally closed	1
Duration	M	unsigned long	Duration. Range: NVR, VMS and ordinary IPC: [5, 3600]. Unit: second. LPR camera: [500, 3600000]. Unit: millisecond. Smart camera: [1, 3600]. Unit: second. All the ranges need to be converted to time values in unit of millisecond.	6

4. OutputSwitchBasicInfoList

OutputSwitchBasicInfoList Json Block	{ "Num": , "BasicInfos": [< OutputSwitchBasicInfo >,.....] }
--	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of alarm output channels.	4
BasicInfos	C	array	Detailed info list of alarm output channels. Optional when Num is 0.	-
OutputSwitchBasicInfo	M	Json Block	Detailed info about alarm output channel. See OutputSwitchBasicInfo Json Block .	-

5. SetOutputSwitchManualAlarmList

SetOutputSwitchManualAlarmList Json Block	{ "Num": , "IDs": [<ID>, ...], "AlarmAction": }
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of output switch channels whose alarm status has changed.	4

IDs	C	array	Output switch ID list. Optional when Num is 0.	-
ID	M	unsigned long	Output switch ID. For NVR, the value is video channel ID*100 + alarm output ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	1
AlarmAction	M	unsigned long	Alarm action of output switch. 0: Close alarm 1: Trigger alarm	1

6. OutputSwitchStatus

OutputSwitchStatus Json Block	{ "ID": , "Status": }
--------------------------------------	--------------------------------

Param	Requirement	Type	Description	Example
ID	M	unsigned long	ID is channel ID*100+alarm output ID. The channel ID of the device itself is 0. For example, the first output switch ID on IPC is 0*100+1=1.	1
Status	M	unsigned long	Output switch status. 0: Close, closed circuit 1: Open, open circuit	0

7. OutputSwitchStatusList

OutputSwitchStatusList Json Block	{ "Nums":, "Statuses":[< OutputSwitchStatus >, < OutputSwitchStatus >...] }
--	--

Param	Requirement	Type	Description	Example
Nums	M	unsigned long	Number of output switches.	1
Statuses	C	array	Output switch list. Optional when Nums is 0.	-
OutputSwitchStatus	M	Json Block	Output switch info. See OutputSwitchStatus Json Block.	-

8. OutputSwitchAlarmStatusList

OutputSwitchAlarmStatusList Json Block	{ "Num": , "AlarmStatusList": [< AlarmStatusInfo >, ...] }
---	---

	}
--	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of alarm output switch channels	1
AlarmStatusList	C	array	List of alarm status of output switch. Optional when Num is 0.	-
AlarmStatusInfo	C	Json Block	Status of output switch alarm. See AlarmStatusInfo Json Block	-

9. AlarmStatusInfo

AlarmStatusInfo Json Block	{ "ID": , "AlarmStatus": }
-----------------------------------	-------------------------------------

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Output switch ID. For NVR, the ID is video channel ID*100 + alarm output ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	1
AlarmStatus	M	unsigned long	Alarm status of output switch. 0: No alarm 1: Alarm	1

3.8 Smart

3.8.1 Capabilities

1. /LAPI/V1.0/Channels/<ID>/Smart/Capabilities

URL	/LAPI/V1.0/Channels/<ID>/Smart/Capabilities
Description	Get channel's smart capability.
Method	GET
Input Data	None
Success Return Data	SmartCapInfo
Note	Applicable to IPC.

3.8.2 CrossLineDetection

1. /LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Rule
Description	Get cross line detection rules of a video channel.
Method	GET
Input Data	None
Success Return Data	CrossLineDetectionRuleInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Rule
Description	Set cross line detection rules of a video channel.
Method	PUT
Input Data	CrossLineDetectionRuleInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Areas

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Areas
Description	Get all cross line detection areas of a video channel.
Method	GET
Input Data	None
Success Return Data	CrossLineDetectionAreaInfoList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Areas
Description	Set all cross line detection areas of a video channel.
Method	PUT
Input Data	CrossLineDetectionAreaInfoList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3. /LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Areas/<ID>

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Areas/<ID>
Description	Get a cross line detection area of a video channel.
Method	GET

Input Data	None
Success Return Data	CrossLineDetectionLineInfo
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/Areas/<ID>
Description	Set a cross line detection area of a video channel.
Method	PUT
Input Data	CrossLineDetectionLineInfo
Success Return Data	None
Note	Applicable to IPC and NVR.

4. /LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/LinkageActions
Description	Get linkage actions of cross line detection.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/LinkageActions
Description	Set linkage actions of cross line detection.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

5. /LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/WeekPlan
Description	Get arming schedule of cross line detection.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/CrossLineDetection/WeekPlan
Description	Set arming schedule of cross line detection.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None

Note	Applicable to IPC and NVR.
Status	

3.8.3 IntrusionDetection

1. /LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Rule
Description	Get intrusion detection rules of a video channel.
Method	GET
Input Data	None
Success Return Data	IntrusionDetectionRuleInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Rule
Description	Set intrusion detection rules of a video channel.
Method	PUT
Input Data	IntrusionDetectionRuleInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

2. /LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Areas

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Areas
Description	Get all intrusion detection areas of a video channel.
Method	GET
Input Data	None
Success Return Data	IntrusionDetectionAreaInfoList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Areas
Description	Set all intrusion detection areas of a video channel.
Method	PUT
Input Data	IntrusionDetectionAreaInfoList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3. /LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Areas/<ID>

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Areas/<ID>
------------	--

Description	Get area info of intrusion detection.
Method	GET
Input Data	None
Success Return Data	IntrusionDetectionPolygonInfo
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/Areas/<ID>
Description	Set area info of intrusion detection.
Method	PUT
Input Data	IntrusionDetectionPolygonInfo
Success Return Data	None
Note	Applicable to IPC and NVR.

4. LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/LinkageActions
Description	Get linkage actions of intrusion detection.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/LinkageActions
Description	Set linkage actions of intrusion detection.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

5. /LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/WeekPlan
Description	Get arming schedule of intrusion detection.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC and NVR.
Status	

URL	/LAPI/V1.0/Channels/<ID>/Smart/IntrusionDetection/WeekPlan
Description	Set arming scheduling of intrusion detection.

Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC and NVR.
Status	

3.8.4 LeaveZone

1. /LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Rule

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Rule
Description	Get leave area detection rules of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>LeaveZoneRuleInfo</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Rule
Description	Set leave area detection rules of a video channel.
Method	PUT
Input Data	<u>LeaveZoneRuleInfo</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

2. /LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Areas

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Areas
Description	Get all leave area detection areas of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>LeaveZoneAreaInfoList</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Areas
Description	Set all leave area detection areas of a video channel.
Method	PUT
Input Data	<u>LeaveZoneAreaInfoList</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

3. /LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Areas/<ID>

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Areas/<ID>
Description	Get area info of leave area detection.
Method	GET
Input Data	None
Success Return Data	<u>ZonePolygonInfo</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/Areas/<ID>
Description	Set area info of leave area detection.
Method	PUT
Input Data	<u>ZonePolygonInfo</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

4. /LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/LinkageActions
Description	Get linkage actions of leave area detection.
Method	GET
Input Data	None
Success Return Data	<u>LinkageActionList</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/LinkageActions
Description	Set linkage actions of leave area detection.
Method	PUT
Input Data	<u>LinkageActionList</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

5. /LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/WeekPlan
Description	Get arming schedule of leave area detection.
Method	GET
Input Data	None
Success Return Data	<u>WeekPlanInfo</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/LeaveZone/WeekPlan
------------	---

Description	Set arming schedule of leave area detection.
Method	PUT
Input Data	<u>WeekPlanInfo</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

3.8.5 AccessZone

1. /LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Rule

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Rule
Description	Get enter area detection rules of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>AccessZoneRuleInfo</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Rule
Description	Set enter area detection rules of a video channel.
Method	PUT
Input Data	<u>AccessZoneRuleInfo</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

2. /LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Areas

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Areas
Description	Get all enter area detection areas of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>AccessZoneAreaInfoList</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Areas
Description	Set all enter area detection areas of a video channel.
Method	PUT
Input Data	<u>AccessZoneAreaInfoList</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

3. /LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Areas/<ID>

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Areas/<ID>
Description	Get area info of enter area detection.
Method	GET
Input Data	None
Success Return Data	<u>ZonePolygonInfo</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/Areas/<ID>
Description	Set area info of enter area detection.
Method	PUT
Input Data	<u>ZonePolygonInfo</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

4. /LAPI/V1.0/Channels/<ID>/Smart/AccessZone/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/LinkageActions
Description	Get linkage actions of enter area detection.
Method	GET
Input Data	None
Success Return Data	<u>LinkageActionList</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/LinkageActions
Description	Set linkage actions of enter area detection.
Method	PUT
Input Data	<u>LinkageActionList</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

5. /LAPI/V1.0/Channels/<ID>/Smart/AccessZone/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/WeekPlan
Description	Get arming schedule of enter area detection.
Method	GET
Input Data	None
Success Return Data	<u>WeekPlanInfo</u>
Note	Applicable to IPC and NVR.

URL	/LAPI/V1.0/Channels/<ID>/Smart/AccessZone/WeekPlan
-----	--

Description	Set arming schedule of enter area detection.
Method	PUT
Input Data	<u>WeekPlanInfo</u>
Success Return Data	None
Note	Applicable to IPC and NVR.

3.8.6 FaceDetection

1. /LAPI/V1.0/Channels/<ID>/Smart/FaceEnable

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceEnable
Description	Get enabled/disabled status of face related smart functions (face and body) of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>FaceEnableInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceEnable
Description	Enable/disable face related smart functions (face and body) of a video channel.
Method	PUT
Input Data	<u>FaceEnableInfo</u>
Success Return Data	None
Note	Applicable to IPC.

2. /LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Rule
Description	Get face detection rules of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>FaceDetectionRuleInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Rule
Description	Set face detection rules of a video channel.
Method	PUT
Input Data	<u>FaceDetectionRuleInfo</u>
Success Return Data	None
Note	Applicable to IPC.

3. /LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Areas/Detections

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Areas/Detections
Description	Get all face detection areas of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>FaceDetectionAreaList</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Areas/Detections
Description	Set all face detection areas of a video channel.
Method	PUT
Input Data	<u>FaceDetectionAreaList</u>
Success Return Data	None
Note	Applicable to IPC.

4. /LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Areas/Detections/<ID>

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Areas/Detections/<ID>
Description	Get face detection areas of a video channel.
Method	GET
Input Data	None
Success Return Data	<u>FaceDetectionPolygonInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/Areas/Detections/<ID>
Description	Set face detection areas of a video channel.
Method	PUT
Input Data	<u>FaceDetectionPolygonInfo</u>
Success Return Data	None
Note	Applicable to IPC.

5. /LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/LinkageActions
Description	Get linkage actions of face detection.
Method	GET
Input Data	None
Success Return Data	<u>LinkageActionList</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/LinkageActions
Description	Set linkage actions of face detection.

Method	PUT
Input Data	<u>LinkageActionList</u>
Success Return Data	None
Note	Applicable to IPC.

6. /LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/WeekPlans

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/WeekPlans
Description	Get arming schedule of face detection.
Method	GET
Input Data	None
Success Return Data	<u>WeekPlanInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/WeekPlans
Description	Set arming schedule of face detection.
Method	PUT
Input Data	<u>WeekPlanInfo</u>
Success Return Data	None
Note	Applicable to IPC.

7. /LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/FaceCount/Reset

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/FaceCount/Reset
Description	Get configuration of face counting reset.
Method	GET
Input Data	None
Success Return Data	<u>CountResetInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/FaceDetection/FaceCount/Reset
Description	Set configuration of face counting reset
Method	PUT
Input Data	<u>CountResetInfo</u>
Success Return Data	None
Note	Applicable to IPC.

3.8.7 HybridDetection

1. /LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/Rule

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/Rule
Description	Get mixed traffic detection rules.

Method	GET
Input Data	None
Success Return Data	<u>HybridDetectionRuleInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/Rule
Description	Set mixed traffic detection rules.
Method	PUT
Input Data	<u>HybridDetectionRuleInfo</u>
Success Return Data	None
Note	Applicable to IPC.

2. /LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/Areas

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/Areas
Description	Get areas of mixed traffic detection.
Method	GET
Input Data	None
Success Return Data	<u>HybridDetectionAreaCfg</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/Areas
Description	Set areas of mixed traffic detection.
Method	PUT
Input Data	<u>HybridDetectionAreaCfg</u>
Success Return Data	None
Note	Applicable to IPC.

3. /LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/LinkageActions

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/LinkageActions
Description	Get linkage actions of mixed traffic detection.
Method	GET
Input Data	None
Success Return Data	LinkageActionList
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/LinkageActions
Description	Set linkage actions of mixed traffic detection.
Method	PUT
Input Data	LinkageActionList
Success Return Data	None
Note	Applicable to IPC.

4. /LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/WeekPlan

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/WeekPlan
Description	Get arming schedule of mixed traffic detection.
Method	GET
Input Data	None
Success Return Data	WeekPlanInfo
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/HybridDetection/WeekPlan
Description	Set arming schedule of mixed traffic detection.
Method	PUT
Input Data	WeekPlanInfo
Success Return Data	None
Note	Applicable to IPC.

3.8.8 AttributeCollect

1. /LAPI/V1.0/Channels/<ID>/Smart/AttributeCollect

URL	/LAPI/V1.0/Channels/<ID>/Smart/AttributeCollect
Description	Get attribute collection info.
Method	GET
Input Data	None
Success Return Data	<u>AttributeCollectInfo</u>
Note	Applicable to IPC.

URL	/LAPI/V1.0/Channels/<ID>/Smart/AttributeCollect
Description	Set attribute collection info.
Method	PUT
Input Data	<u>AttributeCollectInfo</u>
Success Return Data	None
Note	Applicable to IPC.

2. /LAPI/V1.0/Channels/<ID>/Smart/AttributeParam

URL	/LAPI/V1.0/Channels/<ID>/Smart/AttributeParam
Description	Get attribute parameters (province and first letter of license plate, and pupillary distance).
Method	GET
Input Data	None
Success Return Data	<u>AttributeParamInfo</u>

Note	Applicable to IPC.
URL	/LAPI/V1.0/Channels/<ID>/Smart/AttributeParam
Description	Set attribute parameters (province and first letter of license plate, and pupillary distance).
Method	PUT
Input Data	<u>AttributeParamInfo</u>
Success Return Data	None
Note	Applicable to IPC.

3.8.9 Json Block

1. SmartCapInfo

SmartCapInfo Json Block	{ "IntrusionDetection": <IntrusionDetectionCapInfo>, "CrossLineDetection": <CrossLineDetectionCapInfo>, "AccessZone": <AccessZoneCapInfo>, "LeaveZone": <LeaveZoneCapInfo> }
----------------------------	---

Param	Requirement	Type	Description	Example
IntrusionDetection	M	Json Block	Intrusion detection capability. See IntrusionDetectionCapInfo Json Block.	--
CrossLineDetection	M	Json Block	Cross line detection capability. See CrossLineDetectionCapInfo Json Block.	--
AccessZone	M	Json Block	Enter area detection capability. See AccessZoneCapInfo Block.	-
LeaveZone	M	Json Block	Leave area detection capability. See LeaveZoneCapInfo Block.	

2. IntrusionDetectionCapInfo

IntrusionDetectionCapInfo Json Block	{ "SupportCfg":, "Mode":, "LinkagePlanCfg": { "SupportAudioLinkagePlan":, "SupportLightLinkagePlan": } }
---	--

Param	Requirement	Type	Description	Example
SupportCfg	M	boolean	Whether intrusion detection is supported. 0: No 1: Yes	0
Mode	0	unsigned long	Smart mode: 0: Ordinary 1: Deep Learning	0
LinkagePlanCfg	C	Json Block	Arming schedule of linkage actions. Required when supported.	
SupportAudioLinkagePlan	M	boolean	Whether audio linkage schedule is supported. 0: No 1: Yes	1
SupportLightLinkagePlan	M	boolean	Whether light linkage schedule is supported. 0: No 1: Yes	1

3. CrossLineDetectionCapInfo

CrossLineDetectionCapInfo Json Block	<pre>{ "SupportCfg": , "Mode":, "LinkagePlanCfg": { "SupportAudioLinkagePlan":, "SupportLightLinkagePlan": } }</pre>			
---	--	--	--	--

Param	Requirement	Type	Description	Example
SupportCfg	M	boolean	Whether cross line detection is supported: 0: No 1: Yes	0
Mode	0	unsigned long	Smart mode: 0: Ordinary 1: Deep Learning	0
LinkagePlanCfg	C	Json Block	Arming schedule of linkage actions.	

			Required when supported.	
SupportAudioLinkagePlan	M	boolean	Whether audio linkage schedule is supported. 0: No 1: Yes	1
SupportLightLinkagePlan	M	boolean	Whether light linkage schedule is supported. 0: No 1: Yes	1

4. AccessZoneCapInfo

AccessZoneCapInfo Json Block	{ "SupportCfg"; "Mode"; "LinkagePlanCfg": { "SupportAudioLinkagePlan"; "SupportLightLinkagePlan"; } }
---------------------------------	---

Param	Requirement	Type	Description	Example
SupportCfg	M	boolean	Whether enter area detection is supported. 0: No 1: Yes	0
Mode	0	unsigned long	Smart mode: 0: Ordinary 1: Deep Learning	0
LinkagePlanCfg	C	Json Block	Arming schedule of linkage actions. Required when supported.	
SupportAudioLinkagePlan	M	boolean	Whether audio linkage schedule is supported. 0: No 1: Yes	1
SupportLightLinkagePlan	M	boolean	Whether light linkage schedule is supported. 0: No 1: Yes	1

5. LeaveZoneCapInfo

LeaveZoneCapInfo	{
------------------	---

Json Block	<pre> "SupportCfg"; "Mode": , "LinkagePlanCfg": { "SupportAudioLinkagePlan"; "SupportLightLinkagePlan": } </pre>
------------	--

Param	Requirement	Type	Description	Example
SupportCfg	M	boolean	Whether leave area detection is supported. 0: No 1: Yes	0
Mode	0	unsigned long	Smart mode: 0: Ordinary 1: Deep Learning	0
LinkagePlanCfg	C	Json Block	Arming schedule of linkage actions. Required when supported.	
SupportAudioLinkagePlan	M	boolean	Whether audio linkage schedule is supported. 0: No 1: Yes	1
SupportLightLinkagePlan	M	boolean	Whether light linkage schedule is supported. 0: No 1: Yes	1

6. CrossLineDetectionAreaInfoList

CrossLineDetectionAreaInfoList Json Block	<pre> { "Num": , "LineInfoList": [<CrossLineDetectionLineInfo>,....] } </pre>
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Maximum number of areas that can be drawn. The default is 4 for cross line detection.	4
LineInfoList	C	array	Info list of detection area parameters. Optional when Num is 0.	--
CrossLineDetectionLineInfo	M	Json Block	Detection area parameter info. See CrossLineDetectionLineInfo	--

			Json Block.	
--	--	--	-------------	--

7. CrossLineDetectionLineInfo

CrossLineDetectionLineInfo Json Block	<pre>{ "ID": , "Enabled": , "Sensitivity": , "Direction": , "StartPoint": { "X": , "Y": }, "EndPoint": { "X": , "Y": } "Priority":, "Num":, "DetectTargetList": [<DetectTargetInfo>,<DetectTargetInfo>,...] }</pre>
--	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of detection area, starts from 0. Up to 4 areas are allowed.	1
Enabled	M	boolean	Enable or disable detection area. 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
Direction	M	unsigned long	Direction 0: A < -> B, bidirectional between A and B 1: B -> A, from B to A 2: A -> B, from A to B	1
StartPoint	M	Json Block	Coordinates of start point of detection line.	
EndPoint	M	Json Block	Coordinates of end point of detection line.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000
Priority	0	unsigned long	Priority level High: 0	0

			Medium: 1 Low: 2	
Num	C	unsigned long	Number of detection objects. Required when object parameters are present.	3
DetectTargetList	C	array	Detection object list. Required when object parameters are present.	
DetectTargetInfo	M	Json Block	Detection object info. See DetectTargetInfo Json Block.	

8. DetectTargetInfo

DetectTargetInfo Json Block	{ "Enabled":, "Type":, "ObjectMaxSize":< ObjectSize >, "ObjectMinSize":< ObjectSize > }
-----------------------------	--

Param	Requirement	Type	Description	Example
Enabled	M	unsigned long	Enable or disable detection 0: Disable 1: Enable	0
Type	M	unsigned long	Detection type Motor vehicle: 0 Non-motor vehicle: 1 Pedestrian: 2	0
ObjectMaxSize	M	Json Block	Maximum object size. See ObjectSize Json Block.	--
ObjectMinSize	M	Json Block	Minimum object size. See ObjectSize Json Block.	--

9. ObjectSize

ObjectSize Json Block	{ "Width":, "Height": }
-----------------------	----------------------------------

Param	Requirement	Type	Description	Example
Width	M	unsigned long	Object width. Range: [0,9999].	5000
Height	M	unsigned long	Object height. Range: [0,9999].	5000

10. IntrusionDetectionAreaInfoList

IntrusionDetectionAreaInfoList Json Block	{ "Num": , "PolygonInfoList": [< IntrusionDetectionPolygonInfo >,...] }
---	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Maximum number of areas that can be drawn. The default is 4 for intrusion detection.	4
PolygonInfoList	C	array	Info list of detection area parameters. Optional when Num is 0.	--
IntrusionDetectionPolygonInfo	M	Json Block	Detection area parameter info. See IntrusionDetectionPolygonInfo Json Block.	--

11. IntrusionDetectionPolygonInfo

IntrusionDetectionPolygonInfoJson Block	{ "ID": , "Enabled": , "Sensitivity": , "Percentage": , "TimeThreshold": , "PointNum": , "PointList": [{ "X": , "Y": }], "Priority": , "Num": , "DetectTargetList": [< DetectTargetInfo >,...] }
---	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of detection area, starts from 0. Up to 4 areas are allowed.	1
Enabled	M	boolean	Enable or disable detection area. 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
Percentage	M	unsigned long	Percentage. Range:	1

			[1,100]	
TimeThreshold	M	unsigned long	Time threshold. Range: [1, 10]	1
PointNum	M	unsigned long	Number of vertices of the polygon detection area. Range: [0,6]	4
PointList	C	array	Coordinates of vertices of the detection area. Optional when PointNum is 0.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000
Priority	C	unsigned long	Priority level High: 0 Medium: 1 Low: 2	0
Num	C	unsigned long	Number of detection object types.	3
DetectTargetList	C	array	Detection object list. Required when object parameters are present.	
DetectTargetInfo	M	Json Block	Detection object info. See DetectTargetInfo Json Block.	

12. LeaveZoneRuleInfo

LeaveZoneRuleInfoJson Block	{ "Enabled": }
-----------------------------	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable leave area detection. 0: Disable 1: Enable	1

13. LeaveZoneAreaInfoList

LeaveZoneAreaInfoList Json Block	{ "Num": , "PolygonInfoList": [< ZonePolygonInfo >,...] }
-------------------------------------	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Maximum number of areas that can be drawn. The default is 4 for leave area detection.	4
PolygonInfoList	C	array	Detection area parameter list. Optional when Num is 0.	
ZonePolygonInfo	M	Json Block	Detection area parameter info. See ZonePolygonInfo Json Block.	

14. IntrusionDetectionRuleInfo

IntrusionDetectionRuleInfo JsonBlock	{ "Enabled": }
---	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable intrusion detection. 0: Disable 1: Enable	1

15. ZonePolygonInfo

ZonePolygonInfo Block	Json Block	{ "ID": , "Enabled": , "Sensitivity": , "PointNum": , "PointList": [{ "X": , "Y": }, ...], "Priority": , "Num": , "DetectTargetList": [< DetectTargetInfo >,< DetectTargetInfo >,...] }
--------------------------	---------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of detection area, starts from 1, and the	1

			maximum is the value of Nums in Areas.	
Enabled	M	boolean	Enable or disable detection area. 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
PointNum	M	unsigned long	Number of vertices of the polygon detection area.	4
PointList	C	array	Coordinates of vertices of the detection area. Optional when PointNum is 0.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000
Priority	0	unsigned long	Priority level High: 0 Medium: 1 Low: 2	0
Num	M	unsigned long	Number of detection objects.	3
DetectTargetList	C	array	Detection object list. Required when object parameters are present.	
DetectTargetInfo	M	Json Block	Detection object info. See DetectTargetInfo Json Block.	

16. AccessZoneRuleInfo

AccessZoneRuleInfo Json Block	{ "Enabled": }
----------------------------------	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable enter area detection. 0: Disable 1: Enable	1

17. AccessZoneAreaInfoList

AccessZoneAreaInfoList Json Block	{ "Num": , "PolygonInfoList": [ZonePolygonInfo ,...] }
--------------------------------------	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Maximum number of areas that can be drawn. The default is 4 for enter area detection.	4
PolygonInfoList	C	array	Detection area parameter list. Optional when Num is 0.	
ZonePolygonInfo	M	Json Block	Detection area parameter info. See ZonePolygonInfo Json Block.	

18. HybridDetectionRuleInfo

HybridDetectionRuleInfo Json Block	<pre>{ "Enabled":, "ObjectCount":<ObjectCountInfo>, "Num":, "DetectTargetList":[{" "Enabled":, "Type":, "ObjectMaxSize": { "Width":, "Height": }, "ObjectMinSize":{ "Width":, "Height": } },...] }</pre>			
--	--	--	--	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable mixed traffic detection. 0: Disable 1: Enable	1
Num	M	unsigned long	Number of detection object types.	1
ObjectCount	M	Json Block	Object counting information.	See ObjectCountInfo Json Block
DetectTargetList	C	array	Detection object info. Optional when the Num is 0.	--
Enabled	M	boolean	Enable or disable	0

			detection. 0: Disable 1: Enable	
Type	M	unsigned long	Detection type. 0: Motor vehicle 1: Non-motor vehicle 2: Pedestrian	
ObjectMaxSize	M	Json Block	Max object size.	--
Width	M	unsigned long	Object width. Range: [0,9999].	5000
Height	M	unsigned long	Object height. Range: [0,9999].	5000
ObjectMinSize	M	Json Block	Minimum object size.	--
Width	M	unsigned long	Object width. Range: [0,9999].	5000
Height	M	unsigned long	Object height. Range: [0,9999].	5000

19. HybridDetectionAreaCfg

HybridDetectionAreaCfg Json Block	<pre>{ "IsEnableEntireImage";, "Sensitivity";, "PointNum";, "PointList": [{ "X": , "Y": },...] }</pre>
--	--

Param	Requirement	Type	Description	Example
IsEnableEntireImage	M	boolean	Enable or disable full screen detection (detect entire image). 0: Full screen detection is disabled (detection of specified areas) 1: Full screen detection is enabled	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
PointNum	M	unsigned long	Number of vertices of the detection area.	4
PointList	C	array	Coordinates of vertices of the detection area. Optional when the number of vertices is 0.	

X	M	unsigned long	x-coordinate. Range: [0,9999]	5000
Y	M	unsigned long	y-coordinate. Range: [0,9999]	5000

20. AttributeCollectInfo

AttributeCollectInfo Json Block	<pre>{ "AttributeAll":, "FaceAttribute": { "IsEnableGender": , "IsEnableAge": , "IsEnableGlasses": , "IsEnableMaskFlag": , "IsEnableTemp": }, "PedestrianAttribute": { "IsEnableGender": , "IsEnableAge": , "IsEnableMaskFlag": , "IsEnableUpperColor": , "IsEnableLowerColor": , "IsEnableUpClothesTexture": , "IsEnableUpperClothingStyle": , "IsEnableLowerClothingStyle": , "IsEnableShoes": , "IsEnableBodyAngle": , "IsEnableBagInfo": , "IsEnableHairStyle": , "IsEnableDirection": }, "NonMotorVehicleAttribute": { "IsEnableGender": , "IsEnableAge": , "IsEnableUpperColor": , "IsEnableUpperClothingStyle": , "IsEnableNonVehicleType": , "IsEnableDirection": , "IsEnableSpeed": , "IsEnablePlateNumber": , "IsEnablePlateType": , "IsEnablePlateColor": }, "VehicleAttribute": { "IsEnableLogo": , "IsEnableVehicleType": , "IsEnableVehicleColor": ,</pre>
--	--

	<pre>"IsEnablePlateNumber": , "IsEnablePlateType": , "IsEnablePlateColor": , "IsEnableDirection": , "IsEnableSpeed": , "IsEnableSunVisor": , "IsEnableSeatBelt": , "IsEnableDriverMobile": , "IsEnableAim": , "IsEnablePendant": } }</pre>
--	---

Param	Requirement	Type	Description	Example
AttributeAll	0	boolean	Whether collect all attributes. 0: Not all attributes 1: All attributes	0
FaceAttribute	M	Json Block	Face attribute enablement/disablement info	--
IsEnableGender	M	boolean	Enable or disable gender detection. 0: Disable 1: Enable	0
IsEnableAge	M	boolean	Enable or disable age detection. 0: Disable 1: Enable	0
IsEnableGlasses	M	boolean	Enable or disable glasses detection. 0: Disable 1: Enable	0
IsEnableMarkFlag	M	Boolean	Enable or disable mask detection. 0: Disable 1: Enable	0
IsEnableTemp	M	Boolean	Enable or disable temperature detection. 0: Disable 1: Enable	0
PedestrianAttribute	M	Json Block	Pedestrian attribute enablement/disablement info	--
IsEnableGender	M	boolean	Enable or disable gender detection. 0: Disable 1: Enable	0
IsEnableAge	M	boolean	Enable or disable age detection. 0: Disable 1: Enable	0
IsEnableMarkFlag	M	boolean	Enable or disable mask detection. 0: Disable 1: Enable	0
IsEnableUpperColor	M	boolean	Enable or disable coat color detection. 0: Disable 1: Enable	0

IsEnableLowerColor	M	boolean	Enable or disable trousers color detection. 0: Disable 1: Enable	0
IsEnableUpClothesTexture	M	boolean	Enable or disable coat texture detection. 0: Disable 1: Enable	0
IsEnableUpperClothingStyle	M	boolean	Enable or disable coat style detection. 0: Disable 1: Enable	0
IsEnableLowerClothingStyle	M	boolean	Enable or disable trousers style detection. 0: Disable 1: Enable	0
IsEnableShoes	M	boolean	Enable or disable shoes detection. 0: Disable 1: Enable	0
IsEnableBodyAngle	M	boolean	Enable or disable body angle detection. 0: Disable 1: Enable	0
IsEnableBagInfo	M	boolean	Enable or disable bag detection. 0: Disable 1: Enable	0
IsEnableHairStyle	M	boolean	Enable or disable hairstyle detection. 0: Disable 1: Enable	0
IsEnableDirection	M	boolean	Enable or disable movement direction detection. 0: Disable 1: Enable	0
NonMotorVehicleAttributeInfo	M	Json Block	Non-motor vehicle attribute enablement/disablement info	--
IsEnableGender	M	boolean	Enable or disable gender detection. 0: Disable 1: Enable	0
IsEnableAge	M	boolean	Enable or disable age detection. 0: Disable 1: Enable	0

IsEnableUpperColor	M	boolean	Enable or disable coat color detection. 0: Disable 1: Enable	0
IsEnableUpperClothingStyle	M	boolean	Enable or disable coat style detection. 0: Disable 1: Enable	0
IsEnableNonVehicleType	M	boolean	Enable or disable non-motor vehicle type detection. 0: Disable 1: Enable	0
IsEnableDirection	M	boolean	Enable or disable movement direction detection. 0: Disable 1: Enable	0
IsEnableSpeed	M	boolean	Enable or disable movement speed detection. 0: Disable 1: Enable	0
VehicleAttribute	M	Json Block	Motor vehicle attribute enablement/disablement info	--
IsEnableVehicleType	M	boolean	Enable or disable vehicle type detection. 0: Disable 1: Enable	0
IsEnableVehicleColor	M	boolean	Enable or disable vehicle color detection. 0: Disable 1: Enable	0
IsEnableLogo	M	boolean	Enable or disable vehicle brand detection. 0: Disable 1: Enable	0
IsEnableDirection	M	boolean	Enable or disable driving direction detection. 0: Disable 1: Enable	0
IsEnableSpeed	M	boolean	Enable or disable driving speed detection. 0: Disable 1: Enable	0
IsEnablePlateNumber	M	boolean	Enable or disable plate number detection. 0: Disable	0

			1: Enable	
IsEnablePlateType	M	boolean	Enable or disable plate type detection. 0: Disable 1: Enable	0
IsEnablePlateColor	M	boolean	Enable or disable plate color detection. 0: Disable 1: Enable	0
IsEnablePlateNumber	M	boolean	Enable or disable plate number detection. 0: Disable 1: Enable	0
IsEnablePlateType	M	boolean	Enable or disable plate type detection. 0: Disable 1: Enable	0
IsEnablePlateColor	M	boolean	Enable or disable plate color detection. 0: Disable 1: Enable	0
IsEnableSunVisor	M	boolean	Enable or disable sun visor detection. 0: Disable 1: Enable	0
IsEnablePendant	M	boolean	Enable or disable pendant detection. 0: Disable 1: Enable	0
IsEnableAim	M	boolean	Enable or disable inspection sticker detection. 0: Disable 1: Enable	0
IsEnableSeatBelt	M	boolean	Enable or disable seat belt detection. 0: Disable 1: Enable	0
IsEnableDriverMobile	M	boolean	Enable or disable driver's phone usage detection. 0: Disable 1: Enable	0

21. AttributeParamInfo

AttributeParamInfo Json Block	{ "PupilDistance":,
--	------------------------

	<pre> "PlateParam":{ "PlateProvince":, "PlateLetter": } } </pre>
--	--

Param	Requirement	Type	Description	Example
PupilDistance	M	unsigned long	Pupillary distance Range: 40-250. Default: 60. Unit: px	60
PlateParam	M	Json Block	Plate parameter	--
PlateProvince	M	unsigned long	Default province of plate 0: Undefined 1~43	1
PlateLetter	M	unsigned long	Default letter of plate 'A'-'H' > 0 - 7 'J'-'N' > 8 - 12 'P'-'Z' > 13 - 23 'O' > 24	1

22. AudioActParamInfo

AudioActParamInfo Json Block	<pre> { "AudioFileID": , "WarnCount": } </pre>
---------------------------------	--

Param	Requirement	Type	Description	Example
AudioFileID	M	unsigned long	Audio file ID. Used to identify different audio files.	1
WarnCount	M	unsigned long	How many times to play audio files repeatedly. Range: [1,9]	2

23. AudioFileInfo

AudioFileInfo Json Block	<pre> { "ID": , "Status": , "InstallType":, "FileName": } </pre>
-----------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Audio file ID. Used to identify different audio files.	1
Status	M	unsigned long	Audio file status.	0

			0: Not in use 1: In use	
InstallType	M	unsigned long	Installation type: 0: Built-in 1: External Built-in: Internal alarm audio file of the device. External: Audio file imported by user.	1
FileName	M	string	Audio file name Max 256 bytes. Required when getting info about alarm audio files	"AudioFile.pcm"

24. CalibrationLineInfo

CalibrationLineInfo Json Block	<pre>{ "Left": { "X": , "Y": }, "Right": { "X": , "Y": }, }</pre>
---	---

Param	Requirement	Type	Description	Example
Left	M	Json Block	Coordinates of the left point of the calibration line	--
Right	M	Json Block	Coordinates of the right point of the calibration line	--
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000

25. CalibrationPointsInfo

CalibrationPoints Json Block	<pre>{ "DomeID";, "SceneNum";, "SceneCalibrationPointList": [<SceneCalibrationPointInfo>, ...] }</pre>
---	--

Param	Requirement	Type	Description	Example
DomeID	M	unsigned long	Dome camera ID.	1
SceneNum	M	unsigned long	Number of scenes.	1
SceneCalibrationPointInfo	M	array	Info about all scene calibration points.	-

26. CalibrationPointInfo

CalibrationPointInfo Json Block	{ "BoxPoint":<BoxPointInfo>, "DomePoint":<DomePointInfo> }
---	---

Param	Requirement	Type	Description	Example
BoxPoint	M	Json Block	Calibration point of box camera.	-
DomePoint	M	Json Block	Calibration point of dome camera.	-

27. CaptureInfo

CaptureInfo Json Block	{ "CaptureNums"; "CaptureType"; "Interval"; }
----------------------------------	---

Param	Requirement	Type	Description	Example
CaptureNums	M	unsigned long	Number of snapshots. Range: [3,6]	4
CaptureType	M	array	Type of captured image: 1: Panoramic 2: Close-up 4: Features	[1, 2, 1, 2]
Interval	M	array	Time interval of image capture (unit: second)	[120, 120, 120]

28. ChannelPreset

ChannelPreset Json Block	{ "ChannelID";
------------------------------------	-------------------

	"PresetID"
	}

Param	Requirement	Type	Description	Example
ChannelID	C	unsigned long	Video channel ID Optional for IPC.	1
PresetID	M	unsigned long	Preset ID	1

29. CompareInfo

CompareInfo Json Block	{ "Similarity":, "MemberInfo": <MemberInfo>, "FaceAttr":<FaceAttr>, "PersonAttr":<PersonAttr>, "SnapshotImage": <SnapshotImage > }
-----------------------------------	--

Param	Requirement	Type	Description	Example
Similarity	C	unsigned long	Matching degree. Required for match alarms.	80
MemberInfo	C	JsonBlock	Person info. See MemberInfo JsonBlock. Required for match alarms.	--
FaceAttr	C	Json Block	Face attributes. See <u>FaceAttr</u> Json Block. Applicable to VMS when querying structured data.	--
PersonAttr	C	Json Block	Person attributes. See <u>PersonAttr</u> Json Block. Applicable to VMS when querying structured data.	--
SnapshotImage	M	JsonBlock	Snapshot image. See SnapshotImage JsonBlock.	--

30. CompassInfo

CompassInfo Json Block	{ "Angle": }
-----------------------------------	--------------------

Param	Requirement	Type	Description	Example
Angle	M	float	Angle between the electronic compass and true north (unit: degree). Two decimal places.	50.12

31. CrossLineDetectionCapInfo

CrossLineDetectionCapInfo Json Block	{ "SupportCfg": , "Mode": }
---	--------------------------------------

Param	Requirement	Type	Description	Example
SupportCfg	M	boolean	Whether cross line detection is supported: 0: No 1: Yes	0
Mode	C	unsigned long	Smart mode: 0: Ordinary 1: Deep Learning	0

32. CrossLineDetectionInfo

CrossLineDetectionInfo Json Block	{ "Rule":<RuleInfo>, "Areas": { "Nums": , "LineInfos": [<LineInfo>,...] }, "LinkageActions":<LinkageActionInfoList>, "WeekPlan":<WeekPlanInfo> }
--	--

Param	Requirement	Type	Description	Example
Rule	M	Json Block	Cross line detection rules. See RuleInfoJsonBlcok.	--
RuleInfo	M	Json Block	Detailed cross line detection rules.	--
Areas	M	Json Block	Info about cross line detection areas. See AreaInfo Json Block.	--
LinkageActions	M	Json Block	Linkage actions of cross line detection. See LinkageActionInfoList Json Block.	--
WeekPlan	M	Json Block	Arming scheduling of cross line detection. See WeekPlanInfo Json Block.	--

33. CrossLineDetectionRuleInfo

CrossLineDetectionRuleInfo Json Block	{ "Enabled": }
--	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable cross line detection. 0: Disable 1: Enable	1

34. CrossLineDetectionLineInfo

CrossLineDetectionLineInfo Json Block	{ "ID": , "Enabled": , "Sensitivity": , "Direction": , "StartPoint": { "X": , "Y": }, "EndPoint": { "X": , "Y": } "Priority":, "Num":, "DetectTargetList": [<DetectTargetInfo>,<DetectTargetInfo>,...] }
--	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of detection area, starts from 0. Up to 4 areas are allowed.	1
Enabled	M	boolean	Enable or disable detection area. 0: Disable 1: Enable	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
Direction	M	unsigned long	Direction 0: A < - > B, bidirectional between A and B 1: B -> A, from B to A	1

			2: A -> B, from A to B	
StartPoint	M	Json Block	Coordinates of start point of detection line.	
EndPoint	M	Json Block	Coordinates of end point of detection line.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000
Priority	C	unsigned long	Priority level High: 0 Medium: 1 Low: 2	0
Num	C	unsigned long	Number of detection objects. Required when object parameters are present.	3
DetectTargetList	C	array	Detection object list. Required when object parameters are present.	
DetectTargetInfo	M	Json Block	Detection object info. See <u>DetectTargetInfo</u> Json Block.	

35. CrossLineDetectionAreaInfoList

CrossLineDetectionAreaInfoList Json Block	{ "Num": , "LineInfoList": [<CrossLineDetectionLineInfo>,...] }
--	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Maximum number of areas that can be drawn. The default is 4 for cross line detection.	4
LineInfoList	C	array	Info list of detection area parameters. Optional when Num is 0.	--
CrossLineDetectionLineInfo	M	Json Block	Detection area parameter info. See <u>CrossLineDetectionLineInfo</u> Json Block.	--

36. CrossLineDetectionInfoList

CrossLineDetectionInfoList Json Block	{ "Total": , }
--	----------------------

	<pre> "Num": , "CrossLineDetectionInfoList":[<PicDataInfo>, ...] } </pre>
--	---

Param	Ranger	Type	Description	Example
Total	M	unsigned long	Total number of cross line detections.	1000
Num	M	unsigned long	Number of cross line detections currently returned.	2
CrossLineDetectionInfoList	C	array	Image results of cross line detection. Optional when Num is 0. See PicDataInfoJson Block.	-

37. CtrlFaceInfo

CtrlFaceInfo Json Block	<pre> { "ID";, "Timestamp";, "CapSrc";, "FeatureNum": , "FeatureList":[{ "FeatureVersion";, "Feature": },...], "PanolImage":<FileInfo>, "FaceImage":<FileInfo>, "FaceArea": { "LeftTopX";, "LeftTopY";, "RightBottomX";, "RightBottomY"; } } </pre>
------------------------------------	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Record ID	0
Timestamp	O	unsigned long	Collection time (unit: second, UTC format) Optional when absent.	
CapSrc	M	unsigned long	Source of collection 1. Face info collected by face recognition terminal. 2. Access card info collected by card reader. 3. ID card info collected by card reader. 4. Pass-through info collected by gate For FaceInfo, choose 1.	
FeatureNum	0	unsigned long	Number of semi-structured features.	

			Range: [0, 2]. Optional when there is no semi-structured feature.	
FeatureList	0	array	List of semi-structured features. Optional when there is no semi-structured feature.	
FeatureVersion	0	string	Version number of the algorithm that extracts semi-structured face features, for example, "ISFRFR259.2.0". Range: [0, 20].	
Feature	0	string	Base64 encoding. Extracted face features. Used to assist the backend server to perform face comparison. Currently 512Bytes before encryption.	
PanorImage	0	Json Block	Panoramic image (original image). Optional. Note: For PTS, the file size range is [0,1M].	
FacelImage	0	Json Block	Close-up image of face. Optional. Note: For PTS, the file size range is [0,256K].	
FaceArea	M	Json Block	Coordinates of face area in the panoramic image Position of the face in the panoramic image. Coordinates: (0-10000). The top left and bottom right points of the rectangle: "138,315,282,684"	
LeftTopX	M	unsigned long	x-coordinate of upper-left corner	
LeftTopY	M	unsigned long	y-coordinate of upper-left corner	
RightBottomX	M	unsigned long	x-coordinate of lower-right corner	
RightBottomY	M	unsigned long	y-coordinate of lower-right corner	

38. CtrlCardInfo

CtrlCardInfo Json Block	<pre>{ "ID";, "Timestamp";, "CapSrc";, "CardType";, "CardID": , "CardStatus": , "Name";, "Gender";, "Ethnicity";, "Birthday";, "ResidentialAddress";, "IdentityNo";, "IssuingAuthority";, "IssuingDate";,</pre>
--	---

	"ValidDateStart";, "ValidDateEnd";, "IDImage":<FileInfo> }
--	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Record ID	
Timestamp	O	unsigned long	Collection time (unit: second, UTF format) Optional when absent.	
CapSrc	M	unsigned long	Source of collection 1. Face info collected by face recognition terminal. 2. Access card info collected by card reader. 3. ID card info collected by card reader. 4. Pass-through info collected by gate For CardInfo choose 2 or 3.	
CardType	M	unsigned long	0: ID card 1: Access card	
CardID	C	string	Field of access card: Physical card number. Max. 18 digits.	
CardStatus	C	unsigned long	Field of access card. Card status: 1: Valid 0: Invalid	
Name	C	string	ID card field: Name. Range: [1,63].	
Gender	C	unsigned long	Field of ID card. 0: Unknown gender 1: Male 2: Female 9: Unspecified gender	
Ethnicity	C	unsigned long	Field of ID card. Refer to GB/T 3304 for the names of nationalities of China in romanization with codes.	
Birthday	C	string	Field of ID card: Date of birth. Format: YYYYMMDD.	
ResidentialAddress	C	string	Field of ID card: Address.	
IdentityNo	C	string	Field of ID card: ID card number. Max. 18 bits.	
IssuingAuthority	C	string	Field of ID card: Issuing authority, generally in format of XX Province XX City XX District (County) Public Security Bureau.	
IssuingDate	C	string	Field of ID card. Date of issuance. Format: YYYYMMDD.	
ValidDateStart	C	string	Field of ID card. Start time of validity period of ID card. Format: YYYYMMDD.	
ValidDateEnd	C	string	Field of ID card. End time of validity period of ID card. Format: YYYYMMDD.	
IDImage	C	Json Block	Field of ID card: ID card photo. Note: For PTS, the file size range is [0,32K].	

39. CtrlGateInfo

CtrlGateInfo Json Block	{ "ID";, "Timestamp";, "CapSrc";, "InPersonCnt";, "OutPersonCnt": }
--	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Record ID	
Timestamp	M	unsigned long	Collection time (unit: second, UTC format) Optional when absent.	
CapSrc	M	unsigned long	Source of collection 1. Face info collected by face recognition terminal. 2. Access card info collected by card reader. 3. ID card info collected by card reader. 4. Pass-through info collected by gate For GateInfo, choose 4.	
InPersonCnt	M	unsigned long	Number of people that have entered	
OutPersonCnt	M	unsigned long	Number of people that have left	

40. CtrlLibMatchInfo

CtrlLibMatchInfo Json Block	{ "ID";, "LibID";, "LibType";, "MatchStatus";, "MatchPersonID";, "MatchFaceID";, "MatchPersonInfo":{ "PersonCode";, "PersonName";, "Gender";, "CardID";, "IdentityNo": } }
--	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Record ID	
LibID	M	unsigned long	Library ID	
LibType	O	unsigned long	Library type. Optional.	
MatchStatus	M	unsigned long	Matching status. 1: Authentication succeeded. 2: Authentication failed (comparison failed). 3: Authentication failed (comparison succeeded, not within monitoring period). 41: Library image collected successfully 42: Failed to collect library image	
MatchPersonID	M	unsigned long	Matching person ID	
MatchFaceID	M	unsigned long	Matching face ID	
MatchPersonInfo	O	Json Block	Matching person info. Optional.	
PersonCode	O	string	Person ID. Can be student ID or employee ID. Range: [1, 15]	"3060911012"
PersonName	O	string	Member name. Range: [1,63].	
Gender	O	unsigned long	Gender. 0: Unknown 1: Male 2: Female 9: Unspecified	
CardID	O	string	Access card number. Empty if not available.	
IdentityNo	O	string	ID card number. Empty if not available.	

41. CountResetInfo

CountResetInfo Json Block	{ "ResetEnabled"; "TimingResetEnabled"; "ResetTime": }
--	--

Param	Requirement	Type	Description	Example
ResetEnabled	M	boolean	Whether to reset counter immediately. 0: No.	0

			1: Yes	
TimingResetEnabled	M	boolean	Whether timed reset is enabled. 0: Disable 1: Enable	0
ResetTime	C	string	Scheduled reset time. Required when timed reset is enabled. Reset counter at a set time of the day (24-hour format). Format: xx:xx:xx.	"23:30:59"

42. CustomInfo

CustomInfo Json Block	{ "ID"; "Value": }
----------------------------------	-----------------------------

Param	Requirement	Type	Description	Example
ID	M	unsigned long	ID of customer attribute name. Starts from 0.	1
Value	M	string	Custom attribute value. Range: [1,63].	"Custom attribute value"

43. CustomAttrInfo

CustomAttrInfo Json Block	{ "ID"; "Enabled"; "Name": }
--------------------------------------	--

Param	Requirement	Type	Description	Example
ID	M	unsigned long	ID of customer attribute name. Starts from 1.	1
Enabled	M	boolean	Enable or disable custom attribute 0: Disable 1: Enable	1
Name	M	string	Custom attribute name. Range: [1,63].	"Custom attribute title"

44. CustomAttrList

CustomAttrList	{
-----------------------	---

Json Block	<pre> "Num"; "CustomAttrList":[<CustomAttrInfo>,...] } </pre>
-------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of custom attributes.	5
CustomAttrList	C	Array	Custom attribute list. This field will not be returned when Num is 0.	-
CustomAttrInfo	C	Json Block	Custom attribute info. See CustomAttrInfo JsonBlock.	-

45. DatabaseInfo

DatabaseInfo Json Block	<pre> { "ID"; "Name"; "IsMonitored"; "MemberNum"; } </pre>
------------------------------------	--

Param	Requirement	Type	Description	Example
ID	C	unsigned long	Library ID. Required when using the GET method.	3
Name	M	string	Library name. Length range: [1,63]	"List"
IsMonitored	C	boolean	Whether monitoring is enabled. Required when using the GET method. 0: No 1: Yes	0
MemberNum	C	unsigned long	Total number of members in the face library. Required when using the GET method.	100

46. DatabaseList

DatabaseList Json Block	<pre> { "DatabaseNum"; "DatabaseList":[<DatabaseInfo>,...] } </pre>
------------------------------------	--

Param	Requirement	Type	Description	Example
DatabaseNum	M	unsigned long	Number of libraries.	3
DatabaseList	C	array	Library list. Optional when DatabaseNum is 0.	-
DatabaseInfo	M	Json Block	Library info. See DatabaseInfo Json Block.	-

47. DatabaseImportInfo

DatabaseImportInfo Json Block	{ "ID"; "ImportedNum": }
--	-----------------------------------

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Face library ID.	1
ImportedNum	M	unsigned long	Number of imported members	1000

48. DayPlanInfo

DayPlanInfo Json Block	{ "ID"; "Num"; "TimeSectionInfos": [<TimeSectionInfo>, ...] }
---	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of days 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday 7: Sunday; 8: Holiday;	1
Num	M	unsigned long	Number of time periods per day NVR: Max 8 IPC: Max 4	8
TimeSectionInfos	M	array	Arming schedule info. Time periods on the same day must not overlap.	
TimeSectionInfo	M	Json Block	Detailed arming schedule. See TimeSectionInfo.	

49. DetectionSceneInfo

DetectionSceneInfo Json Block	{ "SceneType": }
--	------------------------

Param	Requirement	Type	Description	Example
SceneType	M	unsigned long	Scene of smart detection. 0: Indoor scene 1: Outdoor scene	0

50. DetectTargetInfo

DetectTargetInfo Json Block	{ "Enabled";, "Type";, "ObjectMaxSize":<ObjectSize>, "ObjectMinSize":<ObjectSize> }
--	--

Param	Requirement	Type	Description	Example
Enabled	M	unsigned long	Enable or disable detection 0: Disable 1: Enable	0
Type	M	unsigned long	Detection type Motor vehicle: 0 Non-motor vehicle: 1 Pedestrian: 2	0
ObjectMaxSize	M	Json Block	Maximum object size. See ObjectSize Json Block.	--
ObjectMinSize	M	Json Block	Minimum object size. See ObjectSize Json Block.	--

51. FaceAngleFilterInfo

FaceAngleFilterInfo Json Block	{ "Enabled";, "RollAngleLeft";, "RollAngleRight";, "YawAngleLeft";, "YawAngleRight";, "PitchAngleUp";, "PitchAngleDown"; }
---	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Whether face angle filtering is enabled. 0: Disabled (default) 1: Enabled	0
RollAngleLeft	M	unsigned short	Left roll angle, rotated left around Z-axis of the right-handed coordinate system. Unit: Degree Range: [0,90] 0°: The face is horizontal and front-facing 90°: The maximum angle a left-facing face can be detected.	40
RollAngleRight	M	unsigned short	Right roll angle, rotated right around Z-axis of the right-handed coordinate system. Unit: Degree Range: [0,90] 0°: The face is horizontal and front-facing 90°: The maximum angle a right-facing face can be detected.	40
YawAngleLeft	M	unsigned short	Left yaw angle, rotated left around Y-axis of the right-handed coordinate system. Unit: Degree Range: [0,90] 0°: The face is horizontal and front-facing 90°: The maximum angle a left-facing face can be detected.	40
YawAngleRight	M	unsigned short	Right yaw angle, rotated right around Y-axis of the right-handed coordinate system. Unit: Degree Range: [0,90] 0°: The face is horizontal and front-facing 90°: The maximum angle a right-facing face can be detected.	40
PitchAngleUp	M	unsigned short	Positive pitch angle, rotated upward around X-axis of the right-handed coordinate system. Unit: Degree	40

			Range: [0,90] 0°: The face is horizontal and front-facing 90°: The maximum angle an upward-facing face can be detected.	
PitchAngleDown	M	unsigned short	Negative pitch angle, rotated downward around X-axis of the right-handed coordinate system. Unit: Degree Range: [0,90] 0°: The face is horizontal and front-facing 90°: The maximum angle an downward-facing face can be detected.	40

52. FaceAttr

Faceattr Json Block	<pre> { "Gender"; "AgeRange"; "EthicCode"; "GlassFlag"; "GlassesStyle"; "GlassesColor"; "HairStyle"; "HairColor"; "FaceStyle"; "SkinColor"; "EyebrowStyle"; "WrinklePouch"; "NoseStyle"; "Beard "; "LipStyle"; "MustacheStyle"; "MaskFlag"; "MaskColor"; "HatFlag"; "HatStyle"; "HatColor"; "Scarf"; "ScarfColor"; "CoatColor"; "AcneStain"; "FreckleBirthmark"; </pre>
--------------------------------------	---

	"ScarDimple"; "FacialFeature"; "Temperature": }
--	--

Param	Requirement	Type	Description
Gender	C	unsigned long	Gender. Optional when detection is disabled. 0: Unknown 1: Male 2: Female
AgeRange	C	unsigned long	Age range. Optional when detection is disabled. 0: Unknown 1: Child 2: Juvenile 3: Youth 4: Middle-aged 5: Senior
GlassFlag	C	unsigned long	Whether glasses are detected. Optional when detection is disabled. 0: Unknown 1: No 2: Yes
GlassesStyle	C	unsigned long	Glasses style. Optional when GlassFlag is 0. 0: Unknown 1: Common glasses 2: Sunglasses 99: Other
HairStyle	C	unsigned long	Hairstyle. Optional when detection is disabled. See definitions of person info elements.
HairColor	C	unsigned long	Hair color. Optional when detection is disabled. See definitions of person info elements.
FaceStyle	C	unsigned long	Face style. Optional when detection is disabled. See definitions of person info elements.
SkinColor	C	unsigned long	Skin color. Optional when detection is disabled. See definitions of person info elements.
EyebrowStyle	C	unsigned long	Eyebrow style. Optional when detection is disabled. See definitions of person info elements.
WrinklePouch	C	unsigned long	Wrinkle and pouch. Optional when detection is disabled. See definitions of person info elements.
NoseStyle	C	unsigned long	Nose style. Optional when detection is disabled. See definitions of person info elements.
Beard	C	unsigned long	Beard. Optional when detection is disabled. See definitions of person info elements.
LipStyle	C	unsigned long	Lip style. Optional when detection is disabled. See definitions of person info elements.

MustacheStyle	C	unsigned long	Mustache style. Optional when detection is disabled. See definitions of person info elements.
MaskFlag	C	unsigned long	Whether mask is detected. Optional when detection is disabled. 0: Unknown 1: No 2: Yes
MaskColor	C	unsigned long	Mask color. Optional when detection is disabled. See definitions of person info elements.
HatFlag	C	unsigned long	Whether hat is detected. Optional when detection is disabled. 0: Unknown 1: No 2: Yes
HatStyle	C	unsigned long	Hat style. Optional when detection is disabled. See definitions of person info elements.
HatColor	C	unsigned long	Hat color. Optional when detection is disabled. See definitions of person info elements.
Scarf	C	unsigned long	Whether scarf is detected. Optional when detection is disabled. 0: Unknown 1: No 2: Yes
ScarfColor	C	unsigned long	Scarf color. Optional when detection is disabled. See definitions of person info elements.
CoatColor	C	unsigned long	Coat color. Optional when detection is disabled. See definitions of person info elements.
AcneStain	C	unsigned long	Acne and stain. Optional when detection is disabled. See definitions of person info elements.
FreckleBirthmark	C	unsigned long	Freckle and birthmark. Optional when detection is disabled. See definitions of person info elements.
ScarDimple	C	unsigned long	Scar and dimple. Optional when detection is disabled. See definitions of person info elements.
FacialFeature	C	unsigned long	Facial features. Optional when detection is disabled. See definitions of person info elements.
Temperature	C	float	Body temperature. Optional when detection is disabled. Unit: Celsius degree Precision: 2 decimal places

53. FaceAttributeAnalysisInfo

FaceAttributeAnalysisInfo Json Block	{ "Enabled";, "SexEnabled";, "AgeEnabled";, "GlassEnabled": }
---	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable face attribute analysis. 0: Disable 1: Enable When enabled, face attributes can be pushed.	0
SexEnabled	C	boolean	Enable or disable gender detection. 0: Disable 1: Enable Applicable to IPC.	1
AgeEnabled	C	boolean	Enable or disable age detection. 0: Disable 1: Enable Applicable to IPC.	1
GlassEnabled	C	boolean	Enable or disable glasses detection. 0: Disable 1: Enable Applicable to IPC.	1

54. FaceCalibrationSizeInfo

FaceCalibrationSizeInfo Json Block	{ "WidthMin";, "HeightMin";, "WidthMid";, "HeightMid";, "WidthMax";, "HeightMax": }
---	--

Param	Requirement	Type	Description	Example
WidthMin	M	unsigned long	Minimum face width that can be	625

			detected. Range: [104,2604]. Unit: ‰. The displayed image is converted to the unit of pixels based on the actual image size.	
HeightMin	C	unsigned long	Minimum face height that can be detected. Range: [104,2604]. Unit: ‰. The displayed image is converted to the unit of pixels based on the actual image size.	1111
WidthMid	C	unsigned long	Moderate face width that can be detected. Range: [104,2604]. Unit: ‰. The displayed image is converted to the unit of pixels based on the actual image size.	1093
HeightMid	C	unsigned long	Moderate face height that can be detected. Range: [104,2604]. Unit: ‰. The displayed image is converted to the unit of pixels based on the actual image size.	1944
WidthMax	M	unsigned long	Maximum face width that can be detected. Range: [104,2604]. Unit: ‰. The displayed image is converted to the unit of pixels based on the actual image size.	1562
HeightMax	C	unsigned long	Maximum face height that can be detected. Range: [104,2604]. Unit: ‰. The displayed image is converted to the unit of pixels based on the actual image size.	2777

55. FaceCountInfo

FaceCountInfo Json Block	{ "Enabled"; "Direction": }
-------------------------------------	--------------------------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable people/face counting 0: Disable 1: Enable	0
Direction	C	unsigned long	People/face counting direction: 0: In 1: Out Applicable to IPC.	1

56. FaceDetectionCalPolygonInfo

FaceDetectionCalPolygonInfo Json Block	{ "ID";, "PointNum";, "Points": [{ "X";, "Y"; },...] }
---	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of calibrated area, starts from 0, and the maximum is the value of Num in Calibrations.	1
PointNum	M	unsigned long	Number of vertices of the detection area. The value must be 4 because currently the detection area can only be a rectangle.	4
Points	C	array	Coordinates of vertices of the detection area. For coordinates of the upper-left corner (X1,Y1) and the lower-right corner (X2,Y2), X1<X2, Y1<Y2.	-
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000

57. FaceDetectionCapInfo

FaceDetectionCapInfo Json Block	{ "IsSupport";, "QualityAnalysisIsSupport";, "FaceQualityAnalysisType";, "AttributeAnalysisIsSupport";, "AttributeAnalysisSkills";, "FeatureIsSupport";, "RecognitionIsSupport";, "FaceRecognitionType";, "FaceLivingAnalysisIsSupport";, "PupilDistanceSizeNum";, "PupilDistanceSizeList": [<PupilDistanceCapInfo>,...] }
--	--

Param	Requirement	Type	Description	Example
IsSupport	M	boolean	Whether smart face detection	0

			is supported: 0: No 1: Yes	
QualityAnalysisIsSupport	C	boolean	Whether the device supports face selection (pushing images). 0: No 1: Yes	1
FaceQualityAnalysisType	C	unsigned long	Face selection type. 0: Selection by the front end 1: Selection by the center	0
AttributeAnalysisIsSupport	C	boolean	Whether the device supports face attribute analysis (push). 0: No 1: Yes	1
AttributeAnalysisSkills	C	array	Configured face attribute analysis capability. Optional when FaceAttributeAnalysisSupported is 0. 0: Age indication bit 1: Gender indication bit 2: Reserved 3: Glasses indication bit 4: Mask indication bit 5: Beard indication bit 6: Hat indication bit 7: Hairstyle indication bit 8: Expression indication bit 9: Facial attractiveness indication bit	[0, 1, 4]
FeatureIsSupport	C	boolean	Whether the device supports binary feature extraction (push). 0: No 1: Yes	1
RecognitionIsSupport	C	boolean	Whether the device supports face recognition. 0: No 1: Yes	1
FaceRecognitionType	C	unsigned long	Face recognition type. 0: Face recognition by the front end. 1: Face recognition by the center.	1
FaceLivingAnalysisIsSupport	C	boolean	Whether the device supports liveness detection. 0: No 1: Yes	1

PupilDistanceSizeNum	C	unsigned long	Number of pupillary distance ranges supported.	1
PupilDistanceSizeList	C	Json Block	Pupillary distance range list. See PupilDistanceCapInfo JsonBlock.	

58. FaceDetectionAreaList

FaceDetectionAreaList Json Block	{ "EntireImageEnabled": , "Num": , "Polygons": [<FaceDetectionPolygonInfo>,...] }
---	---

Param	Requirement	Type	Description	Example
EntireImageEnabled	C	boolean	Enable or disable full screen detection (detect entire image). 0: Full screen detection is disabled (detection of specified areas) 1: Full screen detection is enabled Optional when returned results include capabilities other than smart face detection (IsSupport).	1
Num	M	unsigned long	Maximum number of detection areas. The default is 1.	1
Polygons	C	array	Detection area parameter list. Optional when Num is 0.	--
FaceDetectionPolygonInfo	M	Json Block	Detection area parameter info. See FaceDetectionPolygonInfo Json Block.	--

59. FaceDetectionPolygonInfo

FaceDetectionPolygonInfo Json Block	{ "ID": , "Enabled": , "Sensitivity": , "FaceCalibrationSize": <FaceCalibrationSizeInfo>, "PupilDistanceMin": , "PointNum": , }
--	--

	<pre> "Points": [{ "X": , "Y": },...] } </pre>
--	--

Param	Requirement	Type	Description	Example
ID	C	unsigned long	Index of detection area, starts from 0, and the maximum is the value of Nums in Detections. Optional when the detection area is the entire image (by setting EntireImageEnabled to 1 in FaceDetectionAreaList). ID is optional.	1
Enabled	C	boolean	Enable or disable detection area. 0: Disable 1: Enable Applicable to NVR.	1
Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
FaceCalibrationSize	C	Json Block	Configuration item of detectable face size. See FaceCalibrationSizeInfo Json Block. Optional when returned results include smart face detection (IsSupport) only. This field is only for compatibility use. For new products use the PupilDistanceMin field.	
PupilDistanceMin	C	unsigned long	Minimum pupillary distance, expressed using ‰. The range is determined by PupilDistanceSize in face capability. Required when face capability is supported.	208
PointNum	M	unsigned long	Number of vertices of the detection area.	4
Points	C	array	Coordinates of vertices of the detection area. Optional when the number of vertices is 0.	

X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000

60. FaceDetectionRuleInfo

FaceDetectionRuleInfo Json Block	<pre>{ "FaceFullcheck":<FaceFullcheckInfo>, "FaceCount":<FaceCountInfo>, "FaceSnapshot":<FaceSnapshotInfo>, "FaceQualityAnalysisEx":<FaceQualityAnalysisInfoEx>, "FaceLivingAnalysis":<FaceLivingAnalysisInfo>, "FaceAttributeAnalysis":<FaceAttributeAnalysisInfo>, "FaceRecognition":<FaceRecognitionInfo> }</pre>
---	--

Param	Requirement	Type	Description	Example
FaceFullcheck	C	Json Block	Configuration item of full face detection. See FaceFullcheckInfo. Optional when returned results include capabilities other than smart face detection (IsSupport).	--
FaceCount	C	Json Block	Configuration item of face counting. See FaceCountInfo. Currently not available.	--
FaceSnapshot	C	Json Block	Configuration item of face snapshot. See FaceSnapshotInfo. Optional when returned results include smart face detection (IsSupport) only.	--
FaceQualityAnalysisEx	C	Json Block	Extended field of configuration item of face selection. See FaceQualityAnalysisInfoEx. Optional when returned results include smart face detection (IsSupport) only, or when search results include the face selection (QualityAnalysisIsSupported) field but the capability is not supported.	--
FaceLivingAnalysis	C	Json Block	Configuration item of liveness detection. See FaceAttributeAnalysisInfo. Applicable to PTS.	--
FaceAttributeAnalysis	C	Json Block	Configuration item of attribute analysis. See FaceAttributeAnalysisInfo.	--

			Optional when returned results include smart face detection (IsSupport) only, or when returned results include the face attribute analysis (FaceAttributeAnalysisIsSupported) field but the capability is not supported.	
FaceRecognition	C	Json Block	Configuration item of face recognition. See FaceRecognitionInfo Optional when returned results include smart face detection (IsSupport) only, or when returned results include the face recognition (FaceRecognitionIsSupported) field but the capability is not supported. Not applicable to IPC.	--

61. FaceEnableInfo

FaceEnableInfo Json Block	{ "Enabled": }
-------------------------------------	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable face related smart function (face+body). 0: Disable 1: Enable This parameter is the general switch used to enable/disable various face related smart functions.	1

62. FaceFullcheckInfo

FaceFullcheckInfo Json Block	{ "Enabled": }
--	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable full face detection. 0: Disable 1: Enable	0

63. FaceLivingAnalysisInfo

FaceLivingAnalysisInfo Json Block	{ "Enabled":, "Threshold": }
--	---------------------------------------

Param	Requirement	Type	Description	Example
Enabled	M	unsigned long	Enable or disable liveness detection 0: Disable 1: Enable	0
Threshold	M	unsigned long	Liveness detection threshold. Range: [0,100]	100

64. FacePassRecordInfo

FacePassRecordInfo Json Block	{ "RecordID": , "PassingTime": , "ChannelName": "CompareInfo": <ComapreInfo>, }
--	--

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Pass-through record ID.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
CompareInfo	M	Json Block	Comparison info. See CompareInfo json block.	-

65. FaceQualityAnalysisInfoEx

FaceQualityAnalysisInfoEx Json Block	{ "Enabled":, "Mode":, "PreferenceLastTime":, "QualityNum":, "IntervalTime":, "QualityThreshold":, "ImageQualityThreshold":, }
---	--

	"FaceAngleFilter":<FaceAngleFilterInfo> }
--	--

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable face quality analysis: 0: Disable 1: Enable Face quality analysis is optional for face detection and is required for face recognition.	0
Mode	M	unsigned long	Face quality analysis mode. 0: Effect priority 1: Speed priority 2: Periodic selection	0
PreferenceLastTime	C	unsigned long	Duration of quick mode for face selection (seconds). Range: [1,1800]. Valid when Mode is Speed Priority.	3
QualityNum	C	unsigned long	Number of selected face pictures: [1-3] Valid when Mode is Effect Priority or Speed Priority.	1
IntervalTime	C	unsigned long	Uploading interval (millisecond) Range: [100-1800000] Valid when Mode is Periodic Selection. Applicable to IPC.	1000
QualityThreshold	M	unsigned long	Face quality threshold value. Range: [0, 100]. Applicable to PTS.	64
ImageQualityThreshold	C	unsigned long	Photo quality threshold value. Range: [0, 100]. Applicable to PTS.	80
FaceAngleFilter	C	Json Block	Face filtering angle. Valid on new face platform. See FaceAngleFilterInfo. Applicable to IPC.	

66. FaceRecognitionAlarmLog

FaceRecognitionAlarmLog Json Block	{ "ID": , "Timestamp": , "Type": , "ChannelID": , "AlarmSrc": , "CompareInfo": <ComapreInfo> }
---	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Face recognition alarm record ID.	1
Timestamp	M	unsigned long	Time (UTC format)	
Type	M	unsigned long	Face library type: 0: Blocklist 1: Allowlist	0
ChannelID	C	unsigned long	Channel ID. Required when using the POST method.	1
AlarmSrc	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"Camera"
CompareInfo	M	Json Block	Comparison info.	--

67. FaceRecognitionInfo

FaceRecognitionInfo Json Block	{ "Enabled": }
---	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable face recognition 0: Disable 1: Enable	0

68. FaceSnapShotInfo

FaceSnapShotInfo Json Block	{ "Mode": , "Num": , "MarkFace": }
--	--

Param	Requirement	Type	Description	Example
Mode	M	unsigned	Face snapshot mode	0

		long	(triggered by smart analysis or alarm) 1: Triggered by smart analysis 2: Triggered by an input switch signal 3: Triggered by smart analysis and an input switch signal	
Num	M	unsigned long	Number of captured face snapshots. Range: [1,30].	2
MarkFace	M	boolean	Face bounding box. For the original snapshot image, the face will have a bounding box in the image. 0: Turn off face bounding box. 1: Enable face bounding box.	1

69. FeatureInfo

FeatureInfo Json Block	{ "Age";, "Gender";, "WithGlasses": }
-----------------------------------	---

Param	Requirement	Type	Description	Example
Age	M	unsigned long	Age.	
Gender	M	unsigned long	Gender.	
WithGlasses	M	boolean	Whether glasses is detected.	1

70. FaceSnapshotInfo

FaceSnapInfo Json Block	{ "RecordID";, "ChannelID";, "EventType";, "PassingTime";, "ChannelName";, "CompareInfo": <ComapreInfo> }
------------------------------------	--

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Pass-through record ID.	1
ChannelID	M	unsigned long	Channel ID.	1
EventType	M	unsigned long	Event type. Described by bit order. BIT=1 means the event is this type. BIT=0 means the event is not this type. BIT0: Face snapshot BIT1: Face match alarm BIT2: Face not match alarm	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
CompareInfo	M	Json Block	Comparison info. See CompareInfo json block.	-

71. FileInfo

FileInfo Json Block	{ "Name":, "Size":, "Data":, "URL": }
--------------------------------------	--

Param	Requirement	Type	Description	Example
Name	0	string	File name. Range: [1, 16] Used to push images in real-time or historical records	
Size	0	unsigned long	File size (unit: byte). Range: [0, 1M] Used to push images in real-time or historical records	21783
Data	0	string	Base64 data. Used to push images in real-time or historical records	
URL	0	string	Image URL. Used to push image info when getting historical	photo/20190729/1114/ 20190719181741268_2_56_0_0_unknown.jpg

			records	
--	--	--	---------	--

72. FlowReset

FlowReset Json Block	<pre>{ "FlowReset": { "ResetNowEnabled":, "AutoReset": { "Enabled":, "Time": } } }</pre>
---------------------------------------	--

Param	Requirement	Type	Description	Example
FlowReset	M	Jsonblock	Flow resetting configuration.	----
ResetNowEnabled	M	boolean	Enable or disable immediate counter reset. 0: Disable 1: Enable	1
AutoReset	0	Jsonblock	Required when timed reset is enabled.	----
Enabled	0	boolean	Enable or disable timer reset. Default: Disable 0: Disable 1: Enable	1
Time	0	string	Scheduled reset time. Default: 00:00:00. Reset cycle: 24 hours [00:00:00, 23:59:59]. Length range: [0, 12]	"12:34:56"

73. GeolocationInfo

GeolocationInfo Json Block	<pre>{ "Longitude":, "Latitude": }</pre>
---	--

Param	Requirement	Type	Description	Example
Longitude	M	double	Longitude.	50.1678
Latitude	M	double	Latitude.	50.1678

74. IdentificationInfo

IdentificationInfo Json Block	{ "Type": , "Number": }
--	----------------------------------

Param	Requirement	Type	Description	Example
Type	M	unsigned long	ID type 0: ID card 1: IC card 2: Passport 3: Driver's license 99: Others IC card is applicable to PTS only.	0
Number	M	String	ID number. Range: [1,127].	"33022319890520222X"

75. ImportInfo

ImportInfo Json Block	{ "MaxMemberNum": , "ImportedSum": , "Num":, "DatabaseImportList": [<DatabaseImportInfo>,...] }
--	--

Param	Requirement	Type	Description	Example
MaxMemberNum	M	unsigned long	Maximum number of face library members.	2000
ImportedSum	M	unsigned long	Total number of imported members.	1000
Num	M	unsigned long	Number of face libraries.	2
DatabaseImportList	C	array	Face libraries with imported data Optional when Num is 0.	
DatabaseImportInfo	M	JsonBlock	Face library import info.	See DatabaseImportInfo

76. LightActParamInfo

LightActParamInfo Json Block	{ "Interval":, "Luminance": }
---	--

Param	Requirement	Type	Description	Example
Interval	M	unsigned long	Light alarm interval (unit: second). Range: [1,60]. Default: 10s.	10
Luminance	M	unsigned long	Light brightness. Range: [1,1000]. Default: 50.	50

77. HybridDetectionCapInfo

HybridDetectionCapInfo Json Block	{ "IsSupported": }
--	--------------------------

Param	Requirement	Type	Description	Example
IsSupport	M	boolean	Whether mixed traffic detection is supported: 0: No 1: Yes	0

78. MonitorAlarmInfo

MonitorAlarmInfo Json Block	{ "MonitorReason": , "MonitorAlarmType": , "MemberID": }
--	--

Param	Requirement	Type	Description	Example
MonitorReason	C	unsigned long	Monitoring cause 0: Robbed vehicle 1: Stolen vehicle 2: Suspect vehicle 3: Traffic violation 4: Emergency control 255: All types Valid only for vehicle alarm records and present only when the alarm is a vehicle match alarm.	0
MonitorAlarmType	C	unsigned long	Monitoring alarm type. Expressed using an enumeration value. Required in alarm records only. 0: Match alarm	0

			1: Not match alarm	
MemberID	C	unsignedlong	Vehicle member ID. Valid only for vehicle alarm records and present only when the alarm is a vehicle match alarm.	0

79. MonitorDevResultInfo

MonitorDevResultInfo Json Block	{ "DevID": , "MonitorResult": <MonitorResultInfo> }
--	--

Param	Requirement	Type	Description	Example
DevID	M	unsigned long	Monitoring device ID.	3
MonitorResult	M	Json Block	Info about device monitoring result. See MonitorResultInfo Json Block	-

80. MonitorInfo

MonitorInfo Json Block	{ "ID":, "MointorBaseInfo": <MonitorRuleInfo> , "LinkageStrategyNum", "LinkageStrategy":[{ "Type":, "LinkageActions":<LinkageActionList> }], "WeekPlan": <WeekPlanInfo> }
-----------------------------------	--

Param	Requirement	Type	Description	Example
ID	C	unsigned long	Face monitoring task ID. Required when using the GET method.	-
MointorBaseInfo	M	JsonBlock	Face monitoring task info. See MonitorRuleInfo Json Block.	-
LinkageStrategyNum	M	unsigned long	Number of alarm linkage strategies.	1
LinkageStrategy	C	array	Alarm linkage strategies. Optional when LinkageStrategyNum is 0.	

Type	M	unsigned long	Alarm type 1: Match alarm 2: Not match alarm	1
LinkageActions	C	JsonBlock	Linkage actions of face monitoring task. See LinkageActionListJson Block. Applicable to NVR only.	
WeekPlan	C	JsonBlock	Arming schedule of face monitoring task. See WeekPlanInfo Json Block. Applicable to NVR and IPC.	

81. MonitorList

MonitorList Json Block	{ "Num": , "MonitorList": [<MonitorInfo>, ...] }
-----------------------------------	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of face monitoring tasks.	3
MonitorList	C	array	Face monitoring task list. This field is not returned when Num is 0.	-
MonitorInfo	M	JsonBlock	Info about face monitoring task. See MonitorInfoJson Block.	-

82. MonitorRuleInfo

MonitorInfoRule Json Block	{ "Enabled": , "MonitorType": , "Name": , "Begin": , "End": , "MonitorType": , "Reason": , "DatabaseID", "MemberNum" : , "MemberInfoList": [{ "MemberID": , "MemberName": , },...] "DevNum" : , "MonitorDevIDList": [<DevID>,...], }
---------------------------------------	--

Param	Requirement	Type	Description	Example
Enabled	M	unsigned long	Whether face monitoring task is enabled. 0: Disable 1: Enable	1
MonitorType	C	unsigned long	Monitoring alarm type. Applicable to VMS only. 0: Match alarm 1: Not match alarm	0
Name	M	string	Name of face monitoring task. Range: [1,63].	"Monitoring task"
Begin	C	unsigned long	Start time (unit: second, UTC format). Applicable to VMS only.	1520412142
End	C	unsigned long	End time (unit: second, UTC format). Applicable to VMS only.	1520412142
MonitorType	M	unsigned long	Monitoring alarm type. 0: Match alarm 1: Not match alarm	0
Reason	M	string	Cause of the face monitoring task. Range: [1,63].	"Search target"
DatabaselD	O	unsigned long	Face library ID. Required when getting details about a monitoring task.	1
MemberNum	C	unsigned long	Number of face members. Range: [0,32]. Required when getting details about a monitoring task.	3
MemberInfoList	M	array	Face member info list. Optional when MemberNum is 0.	-
MemberID	M	unsigned long	Face member ID. Required when getting details about a monitoring task.	1
MemberName	M	string	Member name. Range: [1,63].	"Zhangsan"
DevNum	C	unsigned long	Number of monitoring devices used in the face monitoring task. Range: [1-4]. Required when getting details about a monitoring task.	1
MonitorDevIDList	M	array	List of monitoring devices used in the face monitoring task. Absent when DevNum is 0.	-
DevID	M	unsigned long	ID of monitoring device used in the face monitoring task.	1

83. MonitorResultInfo

MonitorResultInfo Json Block	<pre>{ "Num":, "Result":[{ "ID":, "Name":, "ResultCode": }, ...] }</pre>
---	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of returned batch operations.	1
Result	C	array	Result info list. Optional when Num is 0.	-
ID	M	unsigned long	Device ID	1
Name	M	string	Member name. Length range: [1,63].	""
ResultCode	M	unsigned long	Returned error code. See VMS error code.	100

84. MonitorMemberInfo

DatabaselInfo Json Block	<pre>{ "MemberID": , "Name", "DevNum": , "DevMonitorInfoList": [<DevMonitorInfo>,...], "MonitorStatus": }</pre>
---	---

Param	Requirement	Type	Description	Example
MemberID	M	unsigned long	Member ID.	3
Name	M	String	Member name.	-
DevNum	C	unsigned long	Number of monitoring devices. Required when getting monitored face library members.	0
DevMonitorInfoList	M	unsigned long	Monitoring device info list. Optional when DevNum is 0. See DevMonitorInfo Json Block.	3
MonitorStatus	M	unsigned long	Monitoring status: 0: Monitoring failed	0

			1: Monitoring succeeded	
--	--	--	-------------------------	--

85. MonitorVehicleInfo

MonitorVehicleInfo Json Block	{ "PlateNo": , "VehicleImage": }
--	---

Param	Requirement	Type	Description	Example
PlateNo	M	string	Plate number. Range:[1,15].	"A00000"
VehicleImage	0	Json Block	Vehicle image for comparison. See FileInfo Json Block. The size of the encrypted image should not exceed 4M. Absent when the monitoring type is "allowlist".	-

86. NonMotorVehicleRecord

NonMotorVehicleRecord Json Block	{ "RecordID":, "PassingTime":, "ChannelName": , , "NonMotorVehicleInfo":<NonMotorVehicleInfo>, "NoMotorVehicleImage":<FileInfo>, "PanolImage":<FileInfo> }
---	---

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	ID of structured non-monitor vehicle record.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	String	Camera name. Range: [1,63]	"XX road intersection"
NonMotorVehicleInfo	M	Json Block	Non-motor vehicle info. See NonMotorVehicleInfo Json Block.	--
NoMotorVehicleImage	M	Json Block	Snapshot of non-motor vehicle. See FileInfo Json Block. The size of the encrypted image should not exceed 1M.	--
PanolImage	C	Json Block	Panoramic image. See	--

			FileInfo Json Block. Only image URL and size are present. Use /LAPI/V1.0/System/Picture to get image data.	
--	--	--	--	--

87. ObjectCountInfo

ObjectCountInfo Json Block	{ "Enabled": }
---------------------------------------	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable counting in mixed traffic. 0: Disable 1: Enable	0

88. ObjectSize

ObjectSize Json Block	{ "Width":, "Height": }
----------------------------------	----------------------------------

Param	Requirement	Type	Description	Example
Width	M	unsigned long	Object width. Range: [0,9999].	5000
Height	M	unsigned long	Object height. Range: [0,9999].	5000

89. OutputSwitchActParamInfo

OutputSwitchActParamInfo Json Block	{ "Num":, "IDs":[<ID>,<ID>...] }
--	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of linked output switches	M
IDs	C	array	List of linked output switches.	C
ID	C	unsigned long	Alarm output switch channel ID. For NVR, the value is video channel	1

			ID*100+alarm output ID (channel ID of NVR itself is 0). For VMS, the ID is configured on the VMS.	
--	--	--	--	--

90. PersonPassRecordInfo

PersonPassRecordInfo Json Block	{ "RecordID": , "PassingTime": , "ChannelName": , "FaceAttr": <FaceAttr>, "PersonAttr": <PersonAttr>, "FacelImage":<FileInfo>, "PersonImage":<FileInfo>, "PanolImage":<FileInfo> }
--	---

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	ID of structured person record.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
FaceAttr	0	Json Block	Face attribute info. See FaceAttr Json block.	-
PersonAttr	M	Json Block	Person attribute info. See PersonAttr Json Block	
FacelImage	0	Json Block	Captured face image. See FileInfo Json Block. The size of the encrypted image should not exceed 1M.	
PersonImage	M	Json Block	Captured person image. See FileInfo Json Block. The size of the encrypted image should not exceed 1M.	
PanolImage	M	Json Block	Panoramic image. See FileInfo Json Block. Only image URL and size are present. Use /LAPI/V1.0/System/Picture to get image data.	

91. PersonSnapshotInfo

PersonSnapshotInfo Json Block	{ "Enabled": }
--	----------------------

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable capturing person image (snapshot). 0: Disable 1: Enable	1

92. PercentInfo

PercentInfo Json Block	{ "Percent": }
---	----------------------

Param	Ranger	Type	Description	Example
Percent	M	unsigned long	Progress. Range: [0,100]	10

93. PicDataInfo

PicDataInfo Json Block	{ "ID": , "Time":, "Size":, "Name":, "PicData": }
---	---

Param	Ranger	Type	Description	Example
ID	M	unsigned long	Channel ID.	1
Time	M	unsigned long	UTC time (unit: second)	1451577600
Size	M	unsigned long	Image size. Unit: Byte.	61440
Name	M	string	Image name. Length: [1,64]	"aaa"
PicData	M	String	Image data. Base64-encoded.	--

94. PlateAttr

PlateAttr Json Block	{ "PlateNo":, "Color":, "Type": }
---------------------------------------	---

	}
--	---

Param	Requirement	Type	Description	Example
PlateNo	M	string	Plate number. Range:[1,15]	"A00000"
Color	M	unsigned long	Plate color 0: Black 1: White 2: Gray 3: Red 4: Blue 5: Yellow 6: Orange 7: Brown 8: Green 9: Purple 10: Cyan 11: Pink 12: Transparent 99: Others	0
Type	M	unsigned long	Plate type 0: Large vehicle plate 1: Small vehicle plate 2: Embassy vehicle plate 3: Consulate vehicle plate 4: Overseas vehicle plate 5: Foreign vehicle plate 6: Common motorcycle plate 7: Light motorcycle plate 8: Embassy motorcycle plate 9: Consulate motorcycle plate 10: Overseas motorcycle plate 11: Foreign motorcycle plate 12: Low speed vehicle plate 13: Tractor plate 14: Trailer plate 15: Coach vehicle plate 16: Coach motorcycle plate 17: Temporary entry vehicle plate 18: Temporary entry motorcycle plate 19: Temporary plate 20: Police vehicle plate 21: Police motorcycle plate 22: Agricultural vehicle plate 23: Border crossing vehicle plate (Hong Kong) 24: Border crossing vehicle plate (Macau)	1

			25: Armed police plate 26: Military plate 99: Other	
--	--	--	---	--

95. Point

Point Json Block	{ "X":, "Y": }
-----------------------------	-------------------------

Param	Requirement	Type	Description	Example
X	M	unsigned long	x-coordinate. Range: [0,9999]	777
Y	M	unsigned long	y-coordinate. Range: [0,9999]	888

96. PresetActParam

PresetActParam Json Block	{ "Num":, "ChannelPresetList": [<ChannelPreset>, ...] }
--------------------------------------	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of linked PTZ presets	M
ChannelPresetList	C	array	List of linked PTZ presets	C
ChannelPreset	M	Json block	Channel and preset ID	See ChannelPreset Json Block

97. PupilDistanceCapInfo

PupilDistanceCapInfo Json Block	{ "Resolution":< Resolution >, "PupilDistance":<PupilDistanceSize> }
--	---

Param	Requirement	Type	Description	Example
Resolution	M	Json Block	Resolution. See Resolution JsonBlock.	--
PupilDistance	M	Json Block	Pupillary distance range. See PupilDistanceSize JsonBlock.	--

98. PupilDistanceSize

PupilDistanceSize Json Block	{ "Min": , "Max": }
---	------------------------------

Param	Requirement	Type	Description	Example
Min	M	unsigned long	Minimum pupillary distance, expressed using %oo. Range: [0, 9999].	416
Max	M	unsigned long	Maximum pupillary distance, expressed using %oo. Range: [0, 9999].	2604

99. QueryMemberList

QueryMemberList Json Block	{ "Total": "Offset": "Num": "MemberList": [<MemberInfo>, ...] }
---	--

Param	Requirement	Type	Description	Example
Total	M	unsigned long	Total number of members in the face library.	200
Offset	M	unsigned long	Library member ID from where the search starts.	0
Num	M	unsigned long	Total number of face library members returned.	20
MemberList	C	array	List of face library members returned. Optional when Num is 0.	-
MemberInfo	C	JsonBlock	List member info. Optional when Num is 0.	-

100. RegionInfo

RegionInfo Json Block	{ "Nation": , "Province": , "City": , }
--	---

Param	Requirement	Type	Description	Example
Nation	M	string	Nationality. Range: [1,63]	“China”
Province	M	string	Province. Range: [1,63].	“Zhejiang”
City	M	string	City. Range: [1,63].	“Hangzhou”

101. Resolution

Resolution Json Block	<pre>{ "Width": , "Height": }</pre>
----------------------------------	---

Param	Requirement	Type	Description	Example
Width	M	unsigned long	Image width.	1920
Height	M	unsigned long	Image height.	1080

102. RuleInfo

RuleInfo Json Block	<pre>{ "RuleType", "TrigerType", "PointNum": , "PointList": [{ "X": , "Y": }, ...] }</pre>
--------------------------------	--

Param	Requirement	Type	Description	Example
RuleType	M	unsigned long	Rule type: 0: Intrusion detection 1: Cross line detection 2: Leave area 3: Enter area	0
TrigerType	0	unsigned long	Trigger type: 0: None 1: Bi-directional 2: Clockwise 3: Counterclockwise 4: Enter	0

			5: Leave	
PointNum	M	unsigned long	Number of points (vertices)	6
PointList	C	array	Coordinates of vertices of the detection area. Optional when PointNum is 0.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000

103. SmartCompatibleRelationInfo

SmartCompatibleRelationInfo Json Block	{ "Num";, "IDs":["<ID>,..."] }
---	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of other smart functions that can be used simultaneously with a smart function.	1
IDs	M	Json Block	IDs of all the other smart functions that can be used simultaneously with the smart function.	-
ID	M	unsigned long	Smart function ID: 0: Face detection 1: People counting/people flow counting 2: Traffic monitoring/road monitoring 3: Auto tracking 4: Motion object detection 100: Cross line detection (CrossLineDetection) 101: Intrusion detection (IntrusionDetection) 102: Enter area (AccessZone) 103: Leave area (LeaveZone) 104: Loitering detection (HoverZone) 105: Quick moving (FastMove) (100-105 are defined as SMART1) 200: People gathering (Gather) 201: Parking detection (CarStop) 202: Object left behind (RemainArticle) 203: Object removed (RemainArticle) 204: Defocus detection (OutFocus) 205: Scene change (ScenceChange) 206: Heat map/Heatmap (200-206 are defined as SMART2)	1

			300: Helmet detection (SafetyHelmetDetection) 301: Work clothes detection (WorkClothesDetection)	
--	--	--	---	--

104. SmartDetectionMode

SmartDetectionMode Json Block	{ "Mode": }
--	-------------------

Param	Requirement	Type	Description	Example
Mode	M	unsigned long	Service type: 0: Normal mode 1: Filter repetitive movement (objects with repetitive movement will be filtered during smart detection)	0

105. SmartModelInfo

SmartModelInfo Json Block	{ "Mode": }
--------------------------------------	-------------------

Param	Requirement	Type	Description	Example
Mode	M	unsigned long	Smart mode. 0: General 1: Traffic monitoring	0

106. SmartMutexRelationInfo

A. Smart function ID (grouped by mutual exclusion relationship)

Smart Service Group ID	Smart Service ID	Function Description	Remarks
1	0	Face detection	
2	1	People counting (people flow counting)	
3	2	Traffic monitoring (road monitoring)	
4	4	Motion object detection	
5	3	Auto tracking	STRUCT1 (structured)
	6	Mixed traffic detection	
6	5	Box-dome camera linkage	STRUCT2 (structured)

7	3	Auto tracking	SMART1
	100	Cross line detection (CrossLineDetection)	
	101	Intrusion detection (IntrusionDetection)	
	102	Enter area (AccessZone)	
	103	Leave area (LeaveZone)	
	106	Box-dome camera linkage (BoxDomeLink)	
8	200	People gathering (Gather)	SMART2
	201	Parking detection (CarStop)	
	202	Object left behind (RemainArticle)	
	203	Object removed (CareArticle)	
	204	Defocus detection (OutFocus)	
	205	Scene change (ScenceChange)	
	206	Heat map (Heatmap)	
	207	Loitering detection (HoverZone)	
	208	Quick moving (FastMove)	
9	300	Helmet detection (SafetyHelmetDetection)	Production safety
	301	Work clothes detection (WorkClothesDetection)	

NOTE!

1. The smart service groups are mutually exclusive and conflict with each other. Exception: For non-deep learning perimeter protection devices, SMART1 does not conflict with SMART2.
2. Smart services in the same group may co-exist. Exception: In group 7, Auto tracking does not co-exist with box-dome linkage.
3. Auto tracking falls into both Group 5 (STRUCT1) and Group 7 (SMART1). Auto tracking can be enabled simultaneously with other functions in group 5 or group 7; it cannot be enabled simultaneously with other functions in both group 5 and group 7.

B. Compatibility relationship between smart functions/smart functions that can be enabled simultaneously (reference)

Priority level	Smart function	Remarks
0	Face detection	After face detection is enabled, no other smart functions can be enabled.
1	People counting/people flow counting	After people counting is enabled, no other smart functions can be enabled.
2	Traffic monitoring/road monitoring	After traffic monitoring is enabled, no other smart functions can be enabled.
3	Auto tracking	After auto tracking is enabled, no other smart functions can be enabled.
4	Chain calculation phase I (obsolete, no longer supported)	After chain calculation (phase I) is enabled, no other smart functions can be enabled.
5	SMART1+SMART2	SMART1 and SMART2 can be enabled

		simultaneously
--	--	----------------

Remarks:

- 0 is the lowest priority, and 5 is the highest.
- In IPC, high-priority smart functions are configured by default. If conflicting or mutually exclusive smart functions are enabled simultaneously, then high-priority smart functions prevail.

C. Mutually exclusive smart functions/smart functions that can't be enabled simultaneously (for reference)

- Face detection, people counting, traffic monitoring (road monitoring), and auto tracking, these four functions are mutually exclusive, and only one function is available at a time.
- Auto tracking conflicts with face detection, people counting, traffic monitoring, heat map, scene change detection, defocus detection, object left behind/removed, parking detection, and people gathering.
- Face detection conflicts with all the other functions.
- People counting conflicts with face detection, traffic monitoring (road monitoring), and auto tracking because people counting performance will be affected when any of the other three is enabled.
(Remarks: People counting algorithm can be used by the other three; the relationship is defined as "conflicting" because sharing the algorithm will affect people counting performance).
- Traffic monitoring conflicts with all the other functions.
- Heat map conflicts with auto tracking.
- Scene change detection, defocus detection, object left behind/removed, parking detection, people gathering, cross line detection, intrusion detection, enter/leave area, quick moving, loitering detection must not be enabled at the same time with the above conflicting functions.

SmartMutexRelationInfo Json Block	{ "Num":, "CompatibleRelations":[<SmartCompatibleRelationInfo>,...] }
--	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of arrays involving compatibility relationship. Currently 8 groups in total.	8
CompatibleRelations	M	Json Block	Compatibility relations between smart functions. See SmartCompatibleRelationInfo Json block.	-
SmartCompatibleRelationInfo	M	Json Block	Compatibility relationship between smart functions.	-

107. SmartWorkingStatusInfo

SmartWorkingStatusInfo Json Block	{ "EnableNum": "EnableIDs": [<ID>,...], "DisableNum": "DisableIDs": [<ID>,...] }
--	---

Param	Requirement	Type	Description	Example
EnableNum	M	unsigned long	Number of enabled smart functions. Note: EnableNums+DisableNums=Total number of smart functions.	6
EnableIDs	M	array	IDs of enabled smart functions.	[100,101,102,103,104,105]
DisableNum	M	unsigned long	Number of disabled smart functions. Note: EnableNums+DisableNums=Total number of smart functions.	11
DisableIDs	M	array	IDs of disabled smart functions.	[0, 1, 2, 3, 4, 200, 201, 202, 203, 204, 205]
ID	M	unsigned long	Smart function ID: 0: Face detection 1: People counting/people flow counting 2: Traffic monitoring/road monitoring 3: Auto tracking 4: Motion object detection 5: Box-dome camera linkage 6: Mixed traffic detection 100: Cross line detection (CrossLineDetection) 101: Intrusion detection (IntrusionDetection) 102: Enter area (AccessZone) 103: Leave area (LeaveZone) 104: Loitering detection (HoverZone) 105: Quick moving (FastMove) (100-105 are defined as SMART1) 200: People gathering (Gather) 201: Parking detection (CarStop) 202: Object left behind (RemainArticle) 203: Object removed (CareArticle) 204: Defocus detection (OutFocus) 205: Scene change (ScenceChange) 206: Heat map/Heatmap (200-206 are defined as SMART2)	1

			300: Helmet detection (SafetyHelmetDetection) 301: Work clothes detection (WorkClothesDetection)	
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108. SnapshotImage

SnapshotImage Json Block	{ "BigImage":<FileInfo>, "SmallImage": <FileInfo>, "PersonImage":<FileInfo>, "Area": { "left":, "top":, "right":, "bottom": } }
-------------------------------------	---

Param	Requirement	Type	Description	Example
BigImage	M	Json Block	Panoramic image. Absent when searching faces or pushing snapshots. See FileInfo Json Block.	--
SmallImage	C	Json Block	Close-up image of face. See FileInfo Json Block. IPC can be configured not to push close-up images.	--
PersonImage	C	Json Block	Close-up image of body. Present in structured queries only. See FileInfo Json Block.	--
Area	M	Json Block	Coordinates of area.	--
left	M	unsigned long	Coordinates of the left point. Range: [0,9999]	0
top	M	unsigned long	Coordinates of the top point. Range: [0,9999]	0
right	M	unsigned long	Coordinates of the right point. Range: [0,9999]	100
bottom	M	unsigned long	Coordinates of the bottom point. Range: [0,9999]	100

109. StaticObjectDetectionInfo

StaticObjectDetectionInfo Json Block	{ "VehicleEnabled";, "NonVehicleEnabled";, "PersonEnabled";, "FaceEnabled": }
---	--

Param	Requirement	Type	Description	Example
VehicleEnabled	M	boolean	Enable or disable motor vehicle detection. 0: Disable	0

			1: Enable	
NonVehicleEnabled	M	boolean	Enable or disable non-motor vehicle detection. 0: Disable 1: Enable	0
PersonEnabled	M	boolean	Enable or disable pedestrian detection. 0: Disable 1: Enable	0
FaceEnabled	M	boolean	Enable or disable face detection. 0: Disable 1: Enable	0

110. VehicleAttr

VehicleAttr Json Block	{ "Type":, "Color":, "SpeedUnit", "SpeedValue": }
-----------------------------------	--

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Vehicle type 0: 3-wheel vehicle 1: Large bus 999: Others	0
Color	M	unsigned long	Vehicle color 0: Black 1: White 2: Gray 3: Red 4: Blue 5: Yellow 6: Orange 7: Brown 8: Green 9: Purple 10: Cyan 11: Pink 12: Transparent 99: Others	0
SpeedUnit	C	unsigned int	Vehicle speed unit 0: kph 1: mph	0

			Absent when querying vehicle library resources.	
SpeedValue	C	double	Vehicle speed Absent when querying vehicle library resources.	70

111. VehicleInfo

VehicleInfo Json Block	{ "ReqSeq":, "MemberID":, "PlateAttr":<PlateAttr>, "VehicleAttr":<VehicleAttr>, "VehicleImage":<FileInfo>, "IsMonitored":, "DatabaseNum":, "DatabaseIDList":[<DatabaseID>,...] }
-----------------------------------	---

Param	Requirement	Type	Description	Example
ReqSeq	C	unsigned long	Sequence number of request, which will be returned with results. Required in batch adding only. Applicable to VMS only.	
MemberID	C	unsigned long	Face library member ID. Required when using GET or PUT method.	1
PlateAttr	M	Json Block	Plate info. See PlateAttr Json Block.	-
VehicleAttr	M	Json Block	Vehicle info. See VehicleAttr Json Block.	-
VehicleImage	C	Json Block	Vehicle image. See FileInfo Json Block. The size of the encrypted image should not exceed 4M.	-
IsMonitored	M	boolean	Whether monitoring is enabled: 0: No 1: Yes	0
DatabaseNum	M	unsigned long	Number of vehicle libraries containing the vehicle.	3
DatabaseIDList	C	Array	IDs of vehicle libraries containing the vehicle. Absent when DatabaseNum is 0.	-
DatabaseID	M	unsigned long	Vehicle library ID.	3

112. VehicleMonitorInfo

VehicleMonitorInfo Json Block	{ "ID":,
--	-------------

	"Rule": <VehicleMonitorRuleInfo> }
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Param	Requirement	Type	Description	Example
ID	C	unsigned long	Monitoring task ID. Required when using the GET method.	0
Rule	M	Json Block	Monitoring task info. See VehicleMonitorRuleInfo Json Block	-

113. VehicleMonitorRuleInfo

VehicleMonitorInfoRule Json Block	{ "Enabled": , "Name": , "Begin": , "End": , "MonitorAlarmType" : , "MonitorReason": , "Description": , "DatabaseID" , "MemberNum" : , "MemberInfoList": [{ "MemberID": , "MemberName": , },...] }
--	--

Param	Requirement	Type	Description	Example
Enabled	M	Boolean	Whether monitoring task is enabled. 0: Disable 1: Enable	1
Name	M	String	Name of vehicle monitoring task. Range: [1,63]	"Robbed vehicle"
Begin	M	unsigned long	Start time (unit: second, UTC format)	1520412142
End	M	unsigned long	End time (unit: second, UTC format)	1520412142
MonitorAlarmType	M	unsigned long	Monitoring alarm type. 0: Match alarm 1: Not match alarm	0
MonitorReason	C	unsigned long	Monitoring cause 0: Robbed vehicle 1: Stolen vehicle	0

			2: Suspect vehicle 3: Traffic violation 4: Emergency control Absent when MonitorAlarmType is Not Match Alarm.	
Description	M	String	Description of vehicle monitoring task. Range: [1,63]	"Searching robbed vehicle"
DatabaselD	0	unsigned long	Vehicle library ID. Required when getting details about a monitoring task.	1
MemberNum	C	unsigned long	Number of vehicle members. Range: [0,32]. Required when getting details about a monitoring task.	3
MemberInfoList	M	Array	Vehicle member ID list. Not returned when MemberNum is 0.	-
MemberID	M	unsigned long	Vehicle member ID.	1
MemberName	M	string	Vehicle member name. Range: [1,15].	"A66666"

114. VehicleMonitorList

VehicleMonitorList Json Block	{ "Num": , "VehicleMonitorList": [<VehicleMonitorInfo>, ...] }
--	---

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of monitoring tasks.	3
VehicleMonitorList	C	array	Monitoring task list. Absent when Num is 0.	-
VehicleMonitorInfo	M	Json Block	Monitoring task info. See VehicleMonitorInfo Json Block.	-

115. VehicleMonitorRuleInfoIDList

VehicleMonitorRuleInfoIDList Json Block	{ "Num": , "VehicleMonitorRuleInfoIDList": [<VehicleMonitorRuleInfoID>, ...] }
--	---

Param	Requirement	Type	Description	Example
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Num	M	unsigned long	Number of monitoring tasks.	1
VehicleMonitorRuleInfoIDList	C	array	Monitoring task ID list. Optional when Num is 0.	--
VehicleMonitorRuleInfoID	M	Json Block	Monitoring task ID, Optional when Num is 0.	1

116. VehiclePassRecordInfo

VehiclePassRecordInfo Json Block	<pre> { "RecordID":, "PassingTime":, "ChannelName":, "MonitorAlarmInfo": <MonitorAlarmInfo>, "PlateAttr":<PlateAttr>, "VehicleAttr": <VehicleAttr>, "PlateImage": <FileInfo>, "VehicleImage":<FileInfo>, "PanolImage":<FileInfo> } </pre>
---	---

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Pass-through record ID.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	LPR camera name. Range: [1,63].	"XX road intersection"
MonitorAlarmInfo	C	Json Block	Plate monitoring alarm info. See MonitorAlarmInfo Json Block. Required when the record is a vehicle alarm record.	
PlateAttr	M	Json Block	Plate attributes info. See PlateAttr Json Block.	-
VehicleAttr	M	Json Block	Vehicle attributes info. See VehicleAttr Json Block.	-
PlateImage	M	Json Block	Captured vehicle image. See FileInfo Json Block. The size of the encrypted image should not exceed 1M.	-
VehicleImage	C	Json Block	Captured vehicle image. Present in structured queries. See FileInfo Json Block. The size of the encrypted image should not exceed 1M.	
PanolImage	C	Json Block	Panoramic image. Present in structured queries. See	

			FileInfo Json Block. Only image URL and size are present. Use /LAPI/V1.0/System/Picture to get image data.	
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117. VehicleSnapshotInfo

VehicleSnapshotInfo Json Block	<pre>{ "RecordID"; "ChannelID"; "EventType"; "PassingTime"; "ChannelName";, "MonitorReason";, "MemberID";, "PlateAttr":<PlateAttr>, "VehicleAttr": <VehicleAttr>, "PlateImage": <FileInfo> }</pre>
---	--

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Vehicle snapshot record ID.	1
ChannelID	M	unsigned long	Channel ID.	1
EventType	M	unsigned long	Event type. Described by bit order. BIT=1 means the event is this type. BIT=0 means the event is not this type. BIT0: Vehicle capture BIT1: Vehicle match alarm BIT2: Vehicle not match alarm	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	LPR camera name. Range: [1,63].	"XX road intersection"
MonitorReason	C	unsigned long	Monitoring cause 0: Robbed vehicle 1: Stolen vehicle 2: Suspect vehicle 3: Traffic violation 4: Emergency control 255: All types Present when EventType is "vehicle match alarm".	0
MemberID	C	unsigned long	Vehicle member ID. Present when EventType is "vehicle match alarm".	
PlateAttr	M	Json Block	Plate attributes info. See PlateAttr	-

			Json Block.	
VehicleAttr	M	Json Block	Vehicle attributes info. See VehicleAttr Json Block.	-
PlateImage	M	Json Block	Captured vehicle image. See FileInfo Json Block. The size of the encrypted image should not exceed 100K.	-

118. VolumeInfo

VolumeInfo Json Block	{ "IsAbnormal":, "Volume": }
---------------------------------	---------------------------------------

Param	Requirement	Type	Description	Example
IsAbnormal	M	boolean	Whether sound volume is normal. 0: Normal 1: Abnormal	0
Volume	M	unsigned long	Sound volume. Range: [0 400].	0

119. VolumeInfoList

VolumeInfoList Json Block	{ "Nums":, "Volumes":[<VolumeInfo>,...] }
-------------------------------------	--

Param	Requirement	Type	Description	Example
Nums	M	unsigned long	Frame rate. Range: [0 30]	20
Volumes	C	array	Volume info list. Optional when Nums is 0.	-
VolumeInfo	M	Json Block	Volume info. See VolumeInfo Json Block	-

120. WeekPlanInfo

WeekPlanInfo Json Block	{ "Enabled": , "Num": , "Days": [<DayPlanInfo>, ...] }
-----------------------------------	--

Param	Requirement	Type	Description	Example
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Enabled	C	boolean	Enable or disable arming schedule. 0: Disable 1: Enable (default) Applicable to IPC only.	1
Num	M	unsigned long	Number of scheduled days NVR: Max 8 (seven days per week and holiday). IPC: Max 7 (seven days per week).	8
Days	M	array	Arming schedule list of every day in a week.	
DayPlanInfo	M	Json Block	Arming schedule details of every day. See DayPlanInfo Json Block	

121. PassRecordInfo

PassRecordInfo Json Block	{ "RecordID": , "PassingTime": , "ChannelName": , "ChannelID": , "Similarity": , "AttributeInfo": <FaceAttr> , "SnapshotImage": <SnapshotImage> }
--------------------------------------	---

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Pass-through record ID.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
ChannelID	M	unsigned long	Channel ID	1
Similarity	M	unsigned long	Matching degree (similarity)	-
SnapshotImage	M	Json Block	Image info	
AttributeInfo	C	Json Block	Face attribute info. Present if exists. See <u>FaceAttr</u> .	-

122. SnapshotImage

SnapshotImage Json Block	{ "BigImage":<FileInfo>, "SmallImage": <FileInfo>, "Area": { "left":, "top":, "right":, "bottom": } }
-------------------------------------	---

Param	Requirement	Type	Description	Example
BigImage	M	Json Block	Original image. Absent when searching faces or pushing snapshots. See FileInfo Json Block.	
SmallImage	C	Json Block	Close-up image. See FileInfo Json Block. IPC can be configured not to push close-up images.	
Area	M	Json Block	Coordinates of the close-up image in the original image.	
left	M	unsigned long	Coordinates of the left point.	
top	M	unsigned long	Coordinates of the top point.	
right	M	unsigned long	Coordinates of the right point.	
bottom	M	unsigned long	Coordinates of the bottom point.	

123. TimeSectionInfo

TimeSectionInfo Json Block	{ "Begin": , "End": , "ArmingType": , }
---------------------------------------	---

Param	Requirement	Type	Description	Example
Begin	M	string	Start time. Format: hh:mm:ss Length range: [0,31] NVR does not support ss (seconds).	00:00:00
End	M	string	End time. Format: hh:mm:ss Length range: [0,31] NVR does not support	24:00:00

			ss (seconds).	
ArmingType	C	unsigned long	Arming type 0: Schedule 1: Motion detection 2: Alarm 3: Motion detection and alarm 4: Motion detection or alarm 5: No schedule 10: Event Arming schedules only supports "Schedule". Recording/snapshot schedules support all. Optional for IPC, which only supports "Schedule".	

124. ZonePolygonInfo

ZonePolygonInfo Json Block	<pre> { "ID": , "Enabled": , "Sensitivity": , "PointNum": , "PointList": [{ "X": , "Y": }, ...], "Priority":, "Num":, "DetectTargetList":[<DetectTargetInfo>,<DetectTargetInfo>,...] } </pre>
---	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Index of detection area, starts from 1, and the maximum is the value of Nums in Areas.	1
Enabled	M	boolean	Enable or disable detection area 0: Disable 1: Enable	1

Sensitivity	M	unsigned long	Sensitivity. Range: [1,100]	1
PointNum	M	unsigned long	Number of vertices of the polygon detection area.	4
PointList	C	array	Coordinates of vertices of the detection area. Optional when PointNum is 0.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000
Priority	C	unsigned long	Priority level High: 0 Medium: 1 Low: 2	0
Num	C	unsigned long	Number of detection objects.	3
DetectTargetList	C	array	Detection object list. Required when object parameters are present.	
DetectTargetInfo	M	Json Block	Detection object info. See DetectTargetInfo Json Block	

3.9 PeopleLibraries

3.9.1 PeopleLibraries

1. /LAPI/V1.0/PeopleLibraries/BasicInfo

URL	/LAPI/V1.0/PeopleLibraries/BasicInfo
Description	Get info about all the created person libraries on the device.
Method	GET
Input Data	None
Success Return Data	LibraryList
Note	Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/PeopleLibraries/BasicInfo
Description	Change person library info.
Method	PUT
Input Data	LibraryList
Success Return Data	None
Note	1. Currently only one library can be changed at a time (Num is 1). 2. Only the library name can be changed. Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/PeopleLibraries/BasicInfo
------------	--------------------------------------

Description	Create person library info.
Method	POST
Input Data	LibInfo
Success Return Data	{ "ID": }
Note	1. Only one library can be created at a time Applicable to NVR, VMS, and IPC.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Library ID that is generated when a library is created.	3

2. /LAPI/V1.0/PeopleLibraries/<ID>

URL	/LAPI/V1.0/PeopleLibraries/<ID>
Description	Delete a face library from a device.
Method	DELETE
Input Data	{ "IsDeletePerson": }
Success Return Data	None
Note	1. Currently only one library can be deleted at a time. Applicable to NVR, VMS, and IPC.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Person library ID	3
IsDeletePerson	C	unsigned long	Whether to delete persons in the library. 0: Not delete 1: Delete Applicable to VMS.	0

3. /LAPI/V1.0/PeopleLibraries/Capacity

URL	/LAPI/V1.0/PeopleLibraries/Capacity
Description	Get capacity info of all person libraries
Method	GET
Input Data	NONE
Success Return Data	PeopleLibrariesCapacityInfo
Notes	Applicable to NVR and IPC.

3.9.2 People

1. /LAPI/V1.0/PeopleLibraries/<ID>/People

URL	/LAPI/V1.0/PeopleLibraries/<ID>/People
Description	Add persons to a library.
Method	POST
Input Data	PersonInfoList
Success Return Data	PersonResultList
Note	1. If a photo of a person failed to be synced, then all the info of the person failed to be added. 2. The total size of the HTTP packet must not exceed 6M (Content-Length: 6291456). Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/PeopleLibraries/<ID>/People
Description	Edit a person.
Method	PUT
Input Data	PersonInfoList
Success Return Data	PersonResultList
Note	1. Currently only one member can be modified at a time (Num is 1). Applicable to NVR, VMS, and IPC.

2. /LAPI/V1.0/PeopleLibraries/<ID>/People/<ID>

URL	/LAPI/V1.0/PeopleLibraries/<ID>/People/<ID>
Description	Delete a person.
Method	DELETE
Input Data	None
Success Return Data	None
Note	Applicable to NVR, VMS, and IPC.

Param	Requirement	Type	Description	Example
<ID>	M	unsigned long	Person library ID Note: PeopleLibraries/<ID>	3
<ID>	M	unsigned long	Person ID Note: People/<ID>	3

3. /LAPI/V1.0/PeopleLibraries/<ID>/People/Info

URL	/LAPI/V1.0/PeopleLibraries/<ID>/People/Info
Description	Get person info with search criteria.
Method	POST

Input Data	{ "Num":, "QueryInfos": [< QueryInfo >, ...] , "Limit": , "Offset": }
Success Return Data	{ "Total":, "Offset":, "PersonList":< PersonInfoList > }
Note	Search criteria shall be provided by respective products and can specify whether image data need to be returned. Applicable to NVR, VMS, and IPC.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Person library ID	3
Num	M	unsigned long	Number of search criteria.	3
QueryInfos	C	Array	Search criteria list. Optional when Num is 0.	-
QueryInfo	M	Json Block	Detailed info about search criteria. See QueryInfo Json Block	-
Limit	M	unsigned long	Number of members to search. Max 20.	12
Offset	M	unsigned long	Sequence number to start search from. The sequence number starts from 0.	0
Total	M	unsigned long	Total number of results that meet the search criteria.	500
Offset	M	unsigned long	Current sequence number. The sequence number starts from 0.	0
PersonList	C	Json Block	Info list of persons that meet search criteria See PersonInfoList Json Block.	-

3.9.3 MonitorRule

1. /LAPI/V1.0/Smart/Face/Recognition/Monitor?ID=<ID>&Limit=<Limit>&Offset=<Offset>

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor?ID=<ID>&Limit=<Limit >&Offset=<Offset>
Description	Get all face monitoring tasks.
Method	GET
Input Data	None
Success Return Data	MonitorList
Note	Applicable to NVR, VMS, and IPC.

Param	Requirement	Type	Description	Example
ID	C	unsigned long	Channel ID. Required when getting monitoring tasks by channel. Applicable to NVR only.	1
Limit	C	unsigned long	Number of tasks to search each time. Max 25. Optional when getting monitoring tasks by channel. Required when getting all monitoring tasks. Only NVR supports paginated search of monitoring tasks.	12
Offset	C	unsigned long	Sequence number to start search from. The sequence number starts from 0. Optional when getting monitoring tasks by channel. Required when getting all the monitoring tasks. Applicable to NVR only.	0

2. /LAPI/V1.0/Smart/Face/Recognition/Monitor

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor
Description	Add a face monitoring task.
Method	POST
Input Data	MonitorInfo
Success Return Data	MonitorResultInfo
Note	Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor
Description	Delete monitoring tasks in batches.
Method	DELETE
Input Data	{ "Num": "MonitorIDList":[<ID>, <ID>,...] }
Success Return Data	BatchDeleteResultInfo
Note	Applicable to NVR and VMS.

3. /LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>
Description	Get a face monitoring task.
Method	GET
Input Data	None
Success Return Data	MonitorInfo
Note	Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>
Description	Set a face monitoring task.
Method	PUT
Input Data	MonitorInfo
Success Return Data	None
Note	Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>
Description	Delete a face monitoring task.
Method	DELETE
Input Data	None
Success Return Data	None
Note	Applicable to NVR, VMS, and IPC.

4. /LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/RuleInfo

URL	LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/RuleInfo
Description	Get rules of a monitoring task.
Method	GET
Input Data	None
Success Return Data	MonitorRuleInfo
Note	Applicable to NVR, VMS, and IPC.

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/RuleInfo
Description	Set rules of a monitoring task.
Method	PUT
Input Data	MonitorRuleInfo
Success Return Data	None
Note	Applicable to NVR, VMS, and IPC.

5. /LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/LinkageActions

URL	LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/LinkageActions
Description	Get linkage actions of a face monitoring task.
Method	GET
Input Data	None
Success Return Data	<pre> "LinkageStrategyNum", "LinkageStrategy":[{ Type, "LinkageActionList":<LinkageActionList> }, { Type,</pre>

	<pre>"LinkageActionList":<LinkageActionList> }]</pre>
Note	Applicable to NVR and IPC.

Param	Requirement	Type	Description	Example
LinkageStrategyNum	M	unsigned long	Number of alarm linkage strategies	1
LinkageStrategy	C	array	Alarm linkage configurations. Optional when LinkageStrategyNum is 0.	
Type	M	unsigned long	Alarm type 1: Match alarm 2: Not match alarm	1
LinkageActions	C	JsonBlock	Linkage actions of face monitoring task. See LinkageActionList Json Block. Applicable to NVR and IPC.	

URL	LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/LinkageActions
Description	Set linkage actions of a face monitoring task.
Method	PUT
Input Data	<pre>{ "LinkageStrategyNum", "LinkageStrategy":[{ Type, "LinkageActionList":<LinkageActionList> }, { Type, "LinkageActionList":<LinkageActionList> }], }</pre>
Success Return Data	None
Note	Applicable to NVR and IPC.

Param	Requirement	Type	Description	Example
LinkageStrategyNum	M	unsigned long	Number of alarm linkage strategies	1
LinkageStrategy	C	array	Alarm linkage configurations. Optional when LinkageStrategyNum is 0.	
Type	M	unsigned long	Alarm type 1: Match alarm 2: Not match alarm	1
LinkageActions	C	JsonBlock	Linkage actions of face monitoring task. See LinkageActionList Json Block. Applicable to NVR and IPC.	

6. /LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/Plan

URL	LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/Plan
Description	Get the arming schedule of a face monitoring task.
Method	Get
Input Data	None
Success Return Data	{ "WeekPlan": < WeekPlanInfo >, }
Note	Applicable to NVR and IPC.

URL	LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/Plan
Description	Set the arming schedule of a face monitoring task.
Method	PUT
Input Data	{ "WeekPlan": < WeekPlanInfo >, }
Success Return Data	None
Note	Applicable to NVR and IPC.

Param	Requirement	Type	Description	Example
WeekPlans	C	JsonBlock	Arming schedule of face monitoring task. See WeekPlanInfo Json Block. Applicable to NVR and IPC.	

7. /LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/Member?Limit=<Limit>&Offset=<Offset>

URL	/LAPI/V1.0/Smart/FaceRecognition/Monitor/<ID>/Member?Limit=<Limit>&Offset=<Offset>
Description	Get monitored members of a face monitoring task.
Method	GET
Input Data	None
Success Return Data	{ "Total": "Offset": "MemberNum":, "MonitorMemberInfoList":[< MonitorMemberInfo >,...] }
Note	Applicable to IPC.

Param	Requirement	Type	Description	Example
Limit	M	unsigned long	Number of members to search each time. Max 50.	30

Offset	M	unsigned long	Sequence number to start search from. The sequence number starts from 0.	0
Total	M	unsigned long	Total number of monitored face members.	200
Offset	M	unsigned long	Library member ID from where the search starts.	0
MemberNum	M	unsigned long	Total number of monitored members returned.	50
MonitorMemberInfoList	C	array	Monitored member info list. Optional when MemberNum is 0.	3
MonitorMemberInfo	M	Json Block	Info about the monitored member. See MonitorMemberInfo json block.	-

8. /LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/Member

URL	/LAPI/V1.0/Smart/Face/Recognition/Monitor/<ID>/Member
Description	Re-configure monitored face members in batches.
Method	PUT
Input Data	{ "MemberNum":, "MonitorMemberInfoList":[< MonitorMemberInfo >,...] }
Success Return Data	None
Note	Applicable to IPC.

Param	Requirement	Type	Description	Example
MemberNum	M	unsigned long	Total number of monitored members returned.	50
MonitorMemberInfoList	C	array	Monitored member info list. Optional when MemberNum is 0.	3
MonitorMemberInfo	M	Json Block	Info about the monitored member. See MonitorMemberInfo json block.	-

3.9.4 PassRecord

1. /LAPI/V1.0/Smart/Face/PassRecord

URL	/LAPI/V1.0/Smart/Face/PassRecord
Description	Get pass-through records.
Method	POST
Input Data	{ "Num":, "QueryInfos": [< QueryInfo >, ...] , "Limit": , "Offset": , }

	}
Success Return Data	{ "Total"; "Offset": , "Num"; "FacePassRecordInfoList": [< FacePassRecordInfo >, ...] }
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of search criteria.	1
QueryInfos	C	array	Search criteria list. Optional when Num is 0.	-
QueryInfo	M	Json Block	Search criteria. See QueryInfo Json Block.	-
Limit	M	unsigned long	Number of records to search each time. Max 32.	20
Offset	M	String long	Sequence number to start search from. The sequence number starts from 0.	1
Total	M	unsigned long	Total number of pass-through records.	20
Offset	M	unsigned long	Record ID from where the search starts.	1
Num	M	unsigned long	Number of pass-through records returned.	20
FacePassRecordInfoList	C	array	List of pass-through records returned. Optional when Num is 0.	-
FacePassRecordInfo	M	Json Block	Detailed pass-through record info. See FacePassRecordInfo Json Block.	-

3.9.5 Json Block

1. PersonInfoList

PersonInfoList Json Block	{ "Num": , "PersonInfoList": [< PersonInfo >,...] }
--------------------------------------	--

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of members in the person library For PTS, up to 6 members can be added, modified, or deleted at a time.	1
PersonInfoList	M	Array	Person info list	--

PersonInfo	C	JsonBlock	Person info. Optional when PersonNum is 0.	--
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2. PersonInfo

PersonInfo Json Block	<pre> { "ReqSeq":, "PersonID":, "LastChange":, "PersonCode":, "PersonName":, "Gender":, "Birthday": "Nation": "Telephone":, "Region" :<RegionInfo>, "Address" :, "IdentificationNum":, "IdentificationList": [<IdentificationInfo>,...], "ImageNum":, "ImageList":[{ "FaceID":, "ModelStatus":, <FileInfo> },...], "PersonPIN":, "IsMonitored":, "BelongLibNum": , "BelongLibIDList" : [<LibID>,<LibID>,...] "Desc": } </pre>
--	---

Param	Requirement	Type	Description	Example
ReqSeq	C	unsigned long	Sequence number of request, which will be returned with results. Required in batch adding only. Applicable to VMS only.	
PersonID	M	unsigned long	Person ID	1
LastChange	M	unsigned long	Last time person info was modified (Unix timestamp)	1510925018
PersonCode	0	string	Person ID. Can be student ID or employee ID. Range: [1, 15] Applicable to PTS.	"3060911012"
PersonName	M	string	Name of the person. Range: [1, 63].	"Zhangsan"

Gender	M	unsigned long	Gender. 0: Unknown 1: Male 2: Female 9: Unspecified The Unspecified type is applicable to PTS only.	1
Birthday	0	string	Birth date (YYYYMMDD). Range: [1,31]	"20180112"
Telephone	0	string	Phone number Length range: [1, 64].	"123456789"
Region	0	Json Block	Region info. See RegionInfo Json Block.	-
Address	0	string	Address Length range: [1, 64].	"xxx"
IdentificationList	M	array	Member ID info	--
IdentificationInfo	C	Json Block	Optional when IdentificationNum is 0. See IdentificationInfo Json Block.	--
ImageNum	M	unsigned long	Number of face images. Range: [0, 6].	--
ImageList	M	array	Face image info list	--
FaceID	C	unsigned long	Face image ID Optional when ImageNum is 0.	23
IsModeled	C	boolean	Whether modeling is complete (whether face image has been converted into semi-structured data). 0: No 1: Yes Required when getting library members. Applicable to NVR.	
FileInfo	C	Json Block	Image info. See FileInfo Json Block. Optional when ImageNum is 0. At the same level as FaceID, and no need to add extra brackets. For example: {"FaceID", "Name":, "Size":, "Data":} Note: File size range for PTS: [0, 768K]	--
ModelStatus	C	unsigned long	Modeling status (whether face image has been converted into semi-structured data).	1

			0: No 1: Yes 2: Failed Required when getting library members. Applicable to NVR.	
PersonPIN	0	String	Password of the person. Digits only. Range: [8, 32]. Applicable to PTS.	"12345678"
BelongLibNum	C	unsigned long	Number of libraries the person belongs to. Optional when adding a person. Applicable to VMS only.	3
BelongLibIDList	C	array	Person library ID list. Absent when BelongLibNum is 0. Applicable to VMS only.	-
LibID	M	unsigned long	Person library ID. Applicable to VMS only.	1
Desc	0	string	Description of the person. String length range: [1, 128].	"zzz"

3. FacePassRecordInfo

FacePassRecordInfo Json Block	<pre>{ "RecordID": , "Type":, "PassingTime": , "ChannelName":, "ChannelID":, "CompareInfo": <ComapreInfo> }</pre>
--	---

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Pass-through record ID.	1
Type	C	unsigned long	Pass-through record type. 0: Face snapshot 1: Face match alarm 2: Face not match alarm	0
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
CompareInfo	M	Json Block	Comparison info. See ComapreInfo Json Block.	-

4. CompareInfo

CompareInfo Json Block	<pre>{ "Similarity":, "PersonInfo": <PersonInfo> "FaceAttr":<FaceAttr>, "PersonAttr":<PersonAttr>, "SnapshotImage": <SnapshotImage> "FeatureNum": , "FeatureList":[{ "FeatureVersion":, "Feature":, "FeatureManufacturer": },...], "CapSrc": }</pre>
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Param	Requirement	Type	Description	Example
Similarity	C	unsigned long	Matching degree. Required for match alarms Required when Type is “match alarm” or “not match alarm”.	
PersonInfo	C	JsonBlock	Person info. See PersonInfo Json Block. Required for match alarms. Required when Type is “match alarm” or “not match alarm”.	
FaceAttr	C	Json Block	Face attributes. See FaceAttr Json Block. Applicable to VMS when querying structured data.	
PersonAttr	C	Json Block	Person attributes. See PersonAttr Json Block. Applicable to VMS when querying structured data.	
SnapshotImage	M	JsonBlock	Snapshot image. See SnapshotImage Json Block.	
FeatureNum	0	unsigned long	Number of semi-structured features. Optional when there is no semi-structured feature. Required for PTS.	
FeatureList	0	array	List of semi-structured features. Optional when there is no semi-structured feature. Required for PTS.	
FeatureVersion	0	String	Version number of algorithm extracting semi-structured face features, for example, "ISFRFR259.2.0". Range: [0, 20].	

			Required for PTS.	
Feature	0	String	Base64-encoded. Extracted face features. Used to assist the backend server to perform face comparison. Currently 512 bytes before encryption. Required for PTS.	
FeatureManufacturer	0	String	Vendor of the algorithm that extracts semi-structured face features. Range: [0, 20].	"Uniview"
CapSrc	C	unsigned long	Data collection source. 1. Face info collected by face recognition terminal. 2. Access card info collected by card reader. 3. ID card info collected by card reader. 4. Info collected by gate. For FaceInfo, choose 1. Required for PTS.	

5. CorrectImageParam

CorrectImageParam Json Block	<pre>{ "Enabled"; "FaceWidth"; "PupilDistance"; "MinImageSize"; "MaxImageSize"; "BlackAndWhiteThreshold": }</pre>
---	---

Param	Requirement	Type	Description	Example
Enabled	M	boolean	Enable or disable 0: Disable 1: Enable	1
FaceWidth	M	unsigned long	Width of generated face Range: [120,150] Unit: px	150
PupilDistance	M	unsigned long	Pupillary distance Range: [120,150] Unit: px	50
MinImageSize	M	unsigned long	Minimum image size Unit: KB	10
MaxImageSize	M	unsigned long	Maximum image size Unit: KB	2048
BlackAndWhiteThreshold	M	float	Black and white threshold	0.95

			Range: 0-1	
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6. MonitorStatusInfo

MonitorStatusInfo Json Block	{ "DevID": , "ChannelID":, "MonitorStatus": }
---	---

Param	Requirement	Type	Description	Example
DevID	M	unsigned long	Device ID.	3
ChannelID	M	unsigned long	Channel ID.	10
MonitorStatus	M	unsigned long	Monitoring status: 0: Success 1: Failed to create library 2: Failed to import member 3: Monitoring failed	0

7. MonitorMemberInfo

MonitorMemberInfo Json Block	{ "MemberID": , "Name", "MonitorNum": "MonitorStatusList":[< MonitorStatusInfo >, < MonitorStatusInfo >,....], "MonitorResult": }
---	---

Param	Requirement	Type	Description	Example
MemberID	M	unsigned long	Member ID.	3
Name	M	string	Member name.	-
MonitorNum	M	unsigned long	Number of monitored members	1
MonitorStatusList	C	array	Monitoring status info list. Absent when MonitorNum is 0.	-
MonitorStatusInfo	M	Json Block	Monitoring status info. See MonitorStatusInfo Json Block	-
MonitorStatusList	C	array	Monitoring status info list. Absent when MonitorNum is 0.	-
MonitorStatusInfo	M	Json Block	Monitoring status info. See MonitorStatusInfo Json Block.	-
MonitorResult	M	unsigned long	Total monitoring result:	0

			0: Monitoring succeeded 1: Monitoring failed Note: The general monitoring result is “failed” as long as one monitoring result is “failed”.	
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8. MonitorResultInfo

MonitorResultInfo Json Block	<pre>{ "MemberID"; "Name"; "Num"; "ResultList":[<ResultInfo>,....] }</pre>
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Param	Requirement	Type	Description	Example
MemberID	M	unsigned long	Member ID	1
Name	M	string	Member name. Length range: [1,63].	""
Num	M	unsigned long	Number of monitoring locations.	
ResultList	C	array	Result info list. Optional when Num is 0.	
ResultInfo	M	Json Block	Monitoring result info. See ResultInfo Json Block	

9. ResultInfo

ResultInfo Json Block	<pre>{ "DevID"; "ChIID"; "ResultCode" }</pre>
---------------------------------	---

Param	Requirement	Type	Description	Example
DevID	M	unsigned long	Device ID	1
ChIID	C	unsigned long	Channel ID. Required when a channel is used for monitoring.	2
ResultCode	M	unsigned long	Cause of failed monitoring: 1: General execution failure. 1000: Algorithm initialization failed. 1001: Face detection failed. 1002: No face in image. 1003: Failed to decode JPEG image. 1004: Low image quality. 1005: Failed to zoom image.	1000

			1006: Smart function disabled. 1007: Image is too small. 1008: Image is too large. 1009: Image exceeds 1920*1080. 1010: Image does not exist. 1011: Number of face elements reached upper limit. 1012: Smart bar version does not match the device. 1013: Invalid ID number for imported library member. 1014: Invalid image format of imported library member. 1015: The number of monitoring tasks has reached the upper limit. 1016: Face library is being operated by another client 1017: Face library file is being updated. 1018: JSON deserialization failed. 1019: Base64 decode failed. 1020: The size of encoded face image does not match the received. 1021: Each monitoring task can only use monitoring devices that are connected via the same image access protocol. 1022: Face libraries on the device reached the upper limit 1023: Smart bar is not in position 1024: Device is busy 1025: VIID is not connected	
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10. LibraryList

LibraryList Json Block	<pre>{ "Num":, "LibList":[<LibInfo>,...] }</pre>
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Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of libraries in the device Range: [0, 16]	3
LibList	M	Array	Library list info	-
LibInfo	C	JsonBlock	This field is not returned when Num is 0. Detailed library info. See LibInfo Json Block.	--

11. LibInfo

LibInfo Json Block	{ "ID"; "Name"; "PersonNum"; "FaceNum"; "MemberNum"; "LastChange": , "BelongIndex": "IsMonitored"; "IsMonitored": }
-------------------------------------	---

Param	Requirement	Type	Description	Example
ID	C	unsigned long	Library ID. Optional when adding a library. Required when setting a library.	3
Name	M	string	Library name Range: [1,63]	"清沁花园居民"
PersonNum	O	unsigned long	Total number of persons (different PersonIDs) in the library. Returned when using the GET method.	1
FaceNum	C	unsigned long	Total number of face photos in the library. Returned when using the GET method.	10
MemberNum	C	unsigned long	Total number of person members (different FaceIDs) in the library. Note: If any person has more than one image in the library, this field is the total number of images in the library. A person without any image in the library is still counted as one member. Returned when using the GET method.	10
LastChange	M	unsigned long	Last time library info was modified (UNIX timestamp) Returned when using the GET method.	1510811178
BelongIndex	M	string	Unique index that identifies the device that the library belongs to. Identified by MD5-encrypted NVR serial number. Required when involving libraries of a channel. Required when using the GET method.	

IsMonitored	C	Boolean	Whether monitoring is enabled. Required when using the GET method. Applicable to VMS only. 0: No 1: Yes	0
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12. PeopleLibrariesCapacityInfo

PeopleLibrariesCapacityInfo Json Block	{ "MaxPeopleNum":, "FreePeopleNum":, "MaxFaceNum":, "FreeFaceNum":, "MaxCardNum":, "FreeCardNum":, "MaxLibNum":, "FreeLibNum":, "Num":, "CapacityInfoList":[< CapacityInfo >,...] }
---	--

Param	Requirement	Type	Description	Example
MaxPeopleNum	M	unsigned long	Total library capacity. [Unit: person]. The sum of individual libraries must not be greater than the total library capacity. Only for display. Unmodifiable.	100
FreePeopleNum	M	unsigned long	Remaining capacity: [unit: Person]. Not applicable to the PUT method.	80
MaxFaceNum	M	unsigned long	Maximum library capacity: [Unit: Person]. Applicable to PTS	100
FreeFaceNum	0	unsigned long	Remaining face space. [Unit: person]. Applicable to PTS Not applicable to the PUT method.	80
MaxCardNum	M	unsigned long	Maximum card capacity: [Unit: card]. Applicable to PTS	100
FreeCardNum	0	unsigned long	Remaining card capacity. [Unit: card]. Applicable to PTS. Not applicable to the PUT method.	80
MaxLibNum	M	unsigned long	Maximum number of libraries allowed.	5
FreeLibNum	M	unsigned long	Remaining number of libraries allowed. Not applicable to the PUT method.	5
Num	C	unsigned long	Number of existing libraries.	5

			Range: [0,16] Applicable to IPC.	
CapacityInfoList	C	array	Single library capacity info list Optional when Num is 0 Applicable to IPC.	--
CapacityInfo	M	JsonBkck	Library capacity info. Applicable to IPC.	--

13. CapacityInfo

CapacityInfo Json Block	{ "LibID":, "Capacity": }
--	------------------------------------

Param	Requirement	Type	Description	Example
LibID	M	unsigned long	Person library ID	3
Capacity	M	unsigned long	Library capacity info. [Unit: person]	5

14. PersonResultList

PersonResultList Json Block	{ "Num": , "PersonList":[{ "ReqSeq":, "PersonID", "FaceNum", "FaceList":[{ "FaceID", "ResultCode" }], ...] }], ...] }
--	--

Param	Requirement	Type	Description	Example
ReqSeq	C	unsigned long	Sequence number of request, which will be returned with results. Applicable to VMS only.	1
Num	M	unsigned long	Number of persons. Range: [0,6]	3
PersonList	M	Array	Person list	--
PersonID	M	unsigned long	Person ID	2
FaceNum	M	unsigned long	Number of faces	2

			Range: [0, 6].	
FaceList	M	Array	Face list	--
FaceID	C	unsigned long	Face ID Optional when FaceNum is 0.	22
ResultCode	C	unsigned long	Status code for processing result. 0: Success. 1: General execution failure. 1000: Algorithm initialization failed. 1001: Face detection failed. 1002: No face in image. 1003: Failed to decode JPEG image. 1004: Low image quality. 1005: Failed to zoom image. 1006: Smart function disabled. 1007: Image is too small. 1008: Image is too large. 1009: Image exceeds 1920*1080. 1010: Image does not exist. 1011: Number of face elements reached upper limit. 1012: Smart bar version does not match the device. 1013: Invalid ID number for imported library member. 1014: Invalid image format of imported library member. Optional when FaceNum is 0.	0

15. BatchDeleteResultInfo

BatchDeleteResultInfo Block	Json	{ "Num": "Result":[{"ID":, "ResultCode":},] }
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Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of batch operation results returned	1
Result	C	array	Result info list. Optional when Num is 0.	-
ID	M	unsigned long	Resource ID	1
ResultCode	M	unsigned long	Returned error code	100

3.10 Event

3.10.1 Event

1. /LAPI/V1.0/System/Event/Subscription

URL	/LAPI/V1.0/System/Event/Subscription
Description	Alarm subscription.
Method	POST
Input Data	{ "AddressType": , "IPAddress": , "Port": , "Duration": , "SubscribePersonCondition":< SubscribePersonCondition >, }
Success Return Data	{ "ID":, "Reference":, "CurrentTime":, "TerminationTime": }
Note	Applicable to IPC, NVR, and VMS.

Param	Requirement	Type	Description	Example
AddressType	M	unsigned long	IP address type: 0: IPv4 1: IPv6 2: Domain name 3: IPv4 and IPv6 Currently only IPv4 is supported.	0
IPAddress	M	string	Length range: [0,256].	
Port	M	unsigned long	Port. Range: [1, 65535].	
Duration	M	unsigned long	Subscription period. Unit: s. Range: [30, 3600].	
SubscribePersonCondition	C	Json Block	Conditions for person data subscription. See SubscribePersonCondition Json Block.	
ID	M	unsigned long	Same as the last segment of the URL of the Reference parameter and is used to identify a subscription. Required when updating or deleting a subscription.	0
Reference	M	string	Descriptive information of the subscriber, in the form of a URL.	"192.168.0.13:80 /Subscription/Su

				bscribers/0"
CurrentTime	M	unsigned long	UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1477104900
TerminationTime	M	unsigned long	UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1477104949

2. /LAPI/V1.0/System/Event/Subscription/<ID>

URL	/LAPI/V1.0/System/Event/Subscription/<ID>
Description	Update subscription.
Method	PUT
Input Data	{ "Duration": }
Success Return Data	{ "Reference";, "CurrrntTime";, "TerminationTime": }
Note	
Note	Applicable to IPC, NVR, and VMS.

Param	Requireme nt	Type	Description	Example
ID	M	unsigned long	Subscription ID, assigned by the device. If subscription is successful, the ID will be returned to the subscriber as a return value. Required when updating or cancelling alarm subscription.	0
Duration	M	unsigned long	The seconds from subscriptionstart(POST succeeded) to finish. The subscription server will be terminated when timeout. IPC will nerver push message to the subscriber unless subscribe again. If termination time is not reached, subscriber can renew(PUT) subscrtpction. Subscription period. Unit: s. Range: [30, 3600].	300
Reference	C	string	Descriptive information of the subscriber, in the form of a URL.	"192.168.0.13:80/Su bscription/Subscriber s/1"
CurrrntTime	M	unsigned long	Subscription start time UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1477104900
TerminationTime	M	unsigned long	Subscription finish time UTC time, number of seconds since 0'o	1477104949

			clock, Jan. 1, 1970.	
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3. /LAPI/V1.0/System/Event/Subscription/<ID>

URL	/LAPI/V1.0/System/Event/Subscription/<ID>
Description	Update subscription.
Description	Cancel subscription
Method	DELETE
Input Data	None
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Subscription ID, assigned by the device. If subscription is successful, the ID will be returned to the subscriber as a return value. Required when updating or cancelling alarm subscription.	0

4. /LAPI/V1.0/System/Event/Notification/Alarm

URL	/LAPI/V1.0/System/Event/Notification/Alarm
Description	Push alarms.
Method	POST
Input Data	{ "Reference":, "AlarmInfo":< AlarmInfo >, "RelatedObjects":< RelatedObjectInfo > }
Success Return Data	None
Note	Applicable to IPC, NVR, VMS, and VM.

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
AlarmInfo	M	Json Block	Alarm push message. See AlarmInfo Json Block.	-
RelatedObjects	C	Json Block	Alarm-related object info. Required when there are alarm-related object data to push. See RelatedObjectInfo	-

5. /LAPI/V1.0/System/Event/Notification/Structure

URL	/LAPI/V1.0/System/Event/Notification/Structure
Description	Push structured events.
Method	POST
Input Data	<pre>{ "Reference":, "TimeStamp":, "Seq":, "SrcID":, "SrcName":, "RelatedID":, "StructureInfo":<StructureDataInfo> }</pre>
Success Return Data	None
Note	

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subsc ription/Subscribers/1"
TimeStamp	M	unsigned long	Alarm time. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Alarm sequence number Applicable to VMS and IPC.	327
SrcID	M	unsigned long	Alarm source ID.	1
SrcName		String	Alarm source name. Length: [1,63]	Channel1
RelatedID		string	ID that relates alarm with alarm data or data of objects from other channels. Unique in each camera. 15 characters. Required when there is related alarm or related object from other channels.	"5ED9FE4C0000001"
StructureDataInfo	M	Json Block	Structured data info. See StructureDataInfo Json Block	-

6. /LAPI/V1.0/System/Event/Notification/PersonInfo

URL	/LAPI/V1.0/System/Event/Notification/PersonInfo
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Description	Push person related events.
Method	POST
Input Data	<pre>{ "Reference":, "PersonEventInfo":<PersonEventInfo> }</pre>
Success Return Data	None
Note	Applicable to IPC, NVR, and VMS. Use this interface to push person related events when the subscription type in /LAPI/V1.0/System/Event/Subscription is "face recognition".

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
PersonEventInfo	C	Json Block	Person event info. See PersonEventInfo Json Block	-

7. /LAPI/V1.0/System/Event/Notification/PeopleCount/LineRuleData

URL	/LAPI/V1.0/System/Event/Notification/PeopleCount/LineRuleData
Description	Push rules data of tripwire people counting.
Method	POST
Input Data	<pre>{ "Reference": , "TimeStamp": , "Seq": , "ChannelID": , "LineNum": , "LineRuleDataList": [<LineRuleDataInfo>,...] }</pre>
Success Return Data	None
Note	Applicable to IPC.

Param	Requirement	Type	Description	Example
Reference	C	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
TimeStamp	M	unsigned long	Alarm notification time. UTC time, number of seconds since 0'o	1489040894

			clock, Jan. 1, 1970.	
Seq	C	unsigned long	Sequence number of data to push. Applicable to IPC.	327
ChannelID	M	unsigned long	Channel ID, starts from 0.	1
LineNum	M	unsigned long	Number of tripwires, starts from 0. 0 means no tripwire.	1
LineRuleDataList	C	Json Block	People counting statistics collected based on a tripwire. Required when LineNum is not 0. See LineRuleDataInfo Json Block	-

8. /LAPI/V1.0/System/Event/Notification/PeopleCount/AreaRuleData

URL	/LAPI/V1.0/System/Event/Notification/PeopleCount/AreaRuleData
Description	Push rules data of area people counting.
Method	POST
Input Data	{ "Reference": , "TimeStamp": , "Seq": , "ChannelID": , "AreaNum": , "AreaRuleDataList": [AreaRuleDataInfo],...] }
Success Return Data	None
Note	Applicable to IPC.

Param	Requirement	Type	Description	Example
Reference	C	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
TimeStamp	M	unsigned long	Alarm notification time. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Sequence number of data to push. Applicable to IPC.	327
ChannelID	M	unsigned long	Channel ID, starts from 0.	1
AreaNum	M	unsigned long	Number of areas, starts from 0. 0 means no area.	1
AreaRuleDataList	C	Json Block	People counting statistics collected based on a detection area. Required when AreaNum is not 0. See AreaRuleDataInfo Json Block	

9. /LAPI/V1.0/System/Event/Notification/FastMovingDetection

URL	/LAPI/V1.0/System/Event/Notification/FastMovingDetection
Description	Push quick moving detection data.
Method	POST
Input Data	<pre>{ "Reference":, "AlarmType":, "TimeStamp":, "Seq":, "SourceID":, "SourceName":, "AlarmPicture":<AlarmPicture> }</pre>
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type	"FastMoving"
TimeStamp	M	unsigned long	Time when the pushed data are generated. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Message ID. Applicable to VMS and IPC.	327
SourceID	C	unsigned long	Alarm source ID. Applicable to VMS and IPC.	1
SourceName	C	string	Alarm source name. Length: [1,63]	Channel1
AlarmPicture	M	Json Block	Image data carried in the alarm. See AlarmPicture JsonBlock	-

10. /LAPI/V1.0/System/Event/Notification/HatDetection

URL	/LAPI/V1.0/System/Event/Notification/HatDetection
Description	Push helmet and chef hat detection data.
Method	POST

Input Data	<pre>{ "Reference":, "AlarmType":, "TimeStamp":, "Seq":, "SourceID":, "SourceName":, "AlarmPicture":<AlarmPicture> }</pre>
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"SafetyHelmetAlarm" "ChefHatAlarm"
TimeStamp	M	unsigned long	Time when the pushed data are generated. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Message ID. Applicable to VMS and IPC.	327
SourceID	C	unsigned long	Alarm source ID. Applicable to VMS and IPC.	1
SourceName	C	string	Alarm source name. Length: [1,63]	Channel1
AlarmPicture	M	Json Block	Image data carried in the alarm. See AlarmPicture JsonBlock	-

11. /LAPI/V1.0/System/Event/Notification/WorkClothesDetection

URL	/LAPI/V1.0/System/Event/Notification/WorkClothesDetection
Description	Push work clothes detection data.
Method	POST
Input Data	<pre>{ "Reference":, "AlarmType":, "TimeStamp":, "Seq":, "SourceID":, "SourceName":,</pre>

	"AlarmPicture":< AlarmPicture > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"WorkClothesAlarm"
TimeStamp	M	unsigned long	Time when the pushed data are generated. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Message ID. Applicable to VMS and IPC.	327
SourceID	C	unsigned long	Alarm source ID. Applicable to VMS and IPC.	1
SourceName	C	string	Alarm source name. Length: [1,63]	Channel1
AlarmPicture	M	Json Block	Image data carried in the alarm. See AlarmPicture Json Block	-

12. /LAPI/V1.0/System/Event/Notification/TelephoningDetection

URL	/LAPI/V1.0/System/Event/Notification/TelephoningDetection
Description	Push calling detection data.
Method	POST
Input Data	{ "Reference"; "AlarmType"; "TimeStamp"; "Seq"; "SourceID"; "SourceName"; "AlarmPicture":< AlarmPicture > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
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Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"TelephoningAlarm"
TimeStamp	M	unsigned long	Alarm notification time. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Message ID. Applicable to VMS and IPC.	327
SourceID	C	unsigned long	Alarm source ID. Applicable to VMS and IPC.	1
SourceName	C	string	Alarm source name. Length: [1,63]	Channel1
AlarmPicture	M	Json Block	Image data carried in the alarm. See AlarmPicture Json Block	-

13. /LAPI/V1.0/System/Event/Notification/SmokingDetection

URL	/LAPI/V1.0/System/Event/Notification/SmokingDetection
Description	Push smoking detection data.
Method	POST
Input Data	{ "Reference";, "AlarmType";, "TimeStamp";, "Seq";, "SourceID";, "SourceName";, "AlarmPicture":< AlarmPicture > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Reference	M	string	URL used by the client to locate the alarm notification. <SubscribersID>: Used to differentiate different subscribing clients that a user is logged in to with the same IP and port.	"192.168.0.13:80/Subscription/Subscribers/1"
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"SmokingAlarm"

TimeStamp	M	unsigned long	Time when the pushed data are generated. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	1489040894
Seq	C	unsigned long	Message ID. Applicable to VMS and IPC.	327
SourceID	C	unsigned long	Alarm source ID. Applicable to VMS and IPC.	1
SourceName	C	string	Alarm source name. Length: [1,63]	Channel1
AlarmPicture	M	Json Block	Image data carried in the alarm. See AlarmPicture Json Block	-

3.10.2 Json Block

1. AlarmInfo

AlarmInfo Json Block	{ "AlarmType";, "AlarmLevel";, "TimeStamp";, "AlarmSeq";, "AlarmSrcID";, "AlarmSrcName";, "RelatedID": }
-------------------------	--

Param	Requirement	Type	Description	Example
AlarmType	M	string	Alarm type. See Alarm&Operate Log Type .	"MotionAlarmOn"
AlarmLevel	C	unsigned long	Alarm level 0: Level 1 1: Level 2 2: Level 3 3: Level 4 4: Level 5 Level 1 is the most serious, and the severity decreases from level 1 to level 5. Applicable to VM, NVR, and VMS.	0
TimeStamp	M	unsigned long	Alarm time	1489040894
AlarmSeq	C	unsigned long	Alarm sequence number Applicable to VM, NVR, VMS, and IPC.	327
AlarmSrcID	C	unsigned long	Resource ID Applicable to VM, NVR, and VMS.	860
AlarmSrcName	C	string	Alarm source name	"206.9.252.7_V_6"

			String length range: [0, 128]. Applicable to VMS.	
RelatedID	C	string	ID that relates an alarm with alarm data. Unique in each camera. 15 characters. Required when there are alarm-related data.	"5ED9FE4C0000001"

2. AlarmPicture

AlarmPicture Json Block	{ "ImageNum", "ImageList": [< ImageInfo >, < ImageInfo >...] }
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Param	Requirement	Type	Description	Example
ImageNum	M	unsigned long	Number of images.	2
ImageList	C	array	Image data list. See ImageInfo Json Block. Required when ImageNum is not 0.	-

3. AreaRuleDataInfo

AreaRuleDataInfo Json Block	{ "AreaID": , "ObjectNum": }
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Param	Requirement	Type	Description	Example
AreaID	M	unsigned long	Detection area ID, starts from 0.	3
ObjectNum	M	unsigned long	Number of people present in the area	20

4. FaceAttr

Faceattr Json Block	{ "Gender": , "AgeRange": , "GlassFlag": , "Temperature": }
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Param	Requirement	Type	Description
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Gender	C	unsigned long	Gender. Optional when detection is disabled. 0: Unknown 1: Male 2: Female
AgeRange	C	unsigned long	Age range. Optional when detection is disabled. 0: Unknown 1: Child 2: Juvenile 3: Youth 4: 4: Middle-aged 5: Senior
GlassFlag	C	unsigned long	Whether glasses are detected. Optional when detection is disabled. 0: Unknown 1: No 2: Yes
MaskFlag	C	unsigned long	Whether mask is detected. Optional when detection is disabled. 0: Unknown 1: No 2: Yes
MaskColor	C	unsigned long	Mask color. Optional when detection is disabled. See definitions of person info elements.
Temperature	C	float	Body temperature. Optional when detection is disabled. Unit: Celsius degree Precision: 2 decimal places

5. FaceInfo

FaceInfo Json Block	<pre>{ "FaceID";, "FaceDoforPersonID";, "FaceDoforNonMotorVehicleID";, "FaceDoforVehicleID";, "Position";, "AppearTime";, "DisAppearTime";, "Confidence";, "SmallPicAttachIndex";, "LargePicAttachIndex";, "FeaturVersion";, "Feature";, "AttributeInfo":<FaceAttr>, "RuleInfo":<RuleInfo> }</pre>
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Param	Requirement	Type	Description	Example
FaceID	M	unsigned long	Face ID, starts from 1.	1
FaceDoForPersonID	C	unsigned long	Corresponding person ID, which starts from 1. 0 is invalid. Optional when the face is not linked to any person.	1
FaceDoForNonMotorVehicleID	C	unsigned long	Corresponding non-motor vehicle ID, which starts from 1. 0 is invalid. Required when the face is linked to a non-motor vehicle.	1
FaceDoForVehicleID	C	unsigned long	Corresponding motor vehicle ID, which starts from 1. 0 is invalid. Required when the face is linked to a motor vehicle.	1
Position	M	string	Position of the face area. Coordinates: 0-10000. The top left and bottom right points of the rectangle: "138,315;282,684" Comma (,) and semicolon (;) are half-width characters. Range: [0,32]	"138,315;282,684"
AppearTime	C	string	Time when face appears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM. The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18.	"20190422083030789"
DisAppearTime	C	string	Time when face disappears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM. The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	"20190422083030789"
Confidence	C	unsigned long	Confidence of face detection. Optional when detection is	70

			disabled. Value range: [0-100]	
SmallPicAttachIndex	C	unsigned long	Index of the close-up image of face in the image list, which starts from 1. 0 is invalid. Optional when no close-up image is pushed.	0
LargePicAttachIndex	C	unsigned long	Index of the original image of the face in the image list, which starts from 1. 0 is invalid. Optional when no original image is pushed.	1
FeatureVersion	C	string	Version number of algorithm extracting semi-structured features. Range: [0,32]	"ISFRFR259.2.0"
Feature	C	string	Base64-encoded. Extracted face features. The maximum length before encoding is 4096 bytes.	
AttributeInfo	C	Json Block	Face attribute info. Optional when attribute detection is disabled. See FaceAttr Json Block.	-
RuleInfo	0	Json Block	Rule info. See RuleInfo Json Block	-

6. FacePassRecordInfo

FacePassRecordInfo Json Block	{ "RecordID": , "PassingTime": , "ChannelName": "CompareInfo": < CompareInfo >, }
----------------------------------	--

Param	Requirement	Type	Description	Example
RecordID	M	unsigned long	Pass-through record ID.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
CompareInfo	M	Json Block	Comparison info. See CompareInfo json block.	-

7. FileInfo

FileInfo Json Block	{ "Name":, "Size":, "Data":, "URL": }
------------------------	--

Param	Requirement	Type	Description	Example
Name	0	String	File name. Range: [1, 16]. Used to push images in real-time or historical records	
Size	0	unsigned long	File size (unit: byte). Range: [0, 1M] Used to push images in real-time or historical records	21783
Data	0	String	Base64-encoded file data. Used to push images in real-time or historical records	
URL	0	String	Image URL. Used to push image info when getting historical records	photo/20190729/1114/ 20190719181741268_2_56 _0_0_unknown.jpg

8. ImageInfo

ImageInfoJson Block	{ "Index":, "Type":, "Format":, "Width":, "Height":, "CaptureTime":, "Size":, "Data": }
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Param	Requirement	Type	Description	Example
Index	M	unsigned long	Image index, starts from 0. Index used to link original (large) image and close-up (small) images of objects in ObjectInfo when pushing structured data.	0
Type	M	unsigned long	Image type. See the ImageType of the extended	

			members.	
Format	M	unsigned long	Image format. 0: JPEG 1: BMP 2: PNG 3: GIF 4: TIFF	1
Width	M	unsigned long	Image width.	1920
Height	M	unsigned long	Image height.	1080
CaptureTime	M	unsigned long	Time when image was captured. UTC time, number of seconds since 0'o clock, Jan. 1, 1970.	21090103121759999
Size	C	unsigned long	Length of Base64-encoded image. Unit: Byte. Max. 3M. Required when the pushed is Base64-encoded data.	128000
Data	C	String	Base64-encoded image data. Required when the pushed is Base64-encoded data.	-
URL	C	String	Image URL. Use the URL to get the image. Length range: [0,256]. Required when the pushed is an image URL.	/LAPI/V1.0/System/Picture?Type=1&Index=ubs_5_0_0_0_0_5278_41_8719_1545200097&Size=80564

9. LineRuleDataInfo

LineRuleDataInfo Json Block	{ "LineID": , "BeginPassTime": , "EndPassTime": , "ObjectIn": , "ObjectOut": }
--------------------------------	--

Param	Requirement	Type	Description	Example
LineID	M	unsigned long	Tripwire ID, starts from 0.	3
BeginPassTime	M	string	Detection start time: YYYYMMDDHHMMSS, in 24H format. Length range: [0,18].	"20190919121212"
EndPassTime	M	string	Detection end time: YYYYMMDDHHMMSS, in 24H format.	"20190919121212"
ObjectIn	M	unsigned long	Number of people that have entered within the set time.	10

ObjectOut	M	unsigned long	Number of people that have left within the set time.	15
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10. NonMotorVehicleInfo

NonMotorVehicleInfo Json Block	<pre>{ "ID "; "Position"; "SmallPicAttachIndex"; "LargePicAttachIndex"; "PlatePicAttachIndex"; "AppearTime": , "DisAppearTime": , "Confidence": , "Speed": , "DirectionType": , "AttributeInfo":<NonMotorVehicleAttr> "PlateAttributeInfo":<PlateAttr>, "PersonOnNoVehicleNum"; "PersonOnNoVehicleInfoList":[<PersonAttr> ,...], "RuleInfo":<RuleInfo> }</pre>
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Param	Requirement	Type	Description	Example
ID	M	unsigned long	Non-motor vehicle ID, starts from 1.	1
Position	M	string	Position of the non-motor vehicle Coordinates: 0-10000. The top left and bottom right points of the rectangle: "138,315;282,684" Comma (,) and semicolon (;) are half-width characters. Range: [0,32].	"138,315;282,684"
SmallPicAttachIndex	C	unsigned long	Index of the close-up image of non-motor vehicle in the image list, which starts from 1. 0 is invalid. Optional when no close-up image is pushed.	0
LargePicAttachIndex	C	unsigned long	Index of the original image of the non-motor vehicle in the image list, which starts from 1. 0 is invalid. Optional when no original image is pushed.	1
PlatePicAttachIndex	C	unsigned long	Index of the close-up image of plate in the image list, which	2

			starts from 1. Optional when no close-up image of plate is pushed.	
AppearTime	C	string	Time when the detection object appears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM, The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	"20190422083030789"
DisAppearTime	C	string	Time when the detection object disappears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM, The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	"20190422083030789"
Confidence	C	unsigned long	Confidence of detection. Optional when detection is disabled. Value range: [0-100]	70
Speed	C	unsigned long	Speed. Optional when detection is disabled. Speed: km/h.	70
DirectionType	C	string	Driving direction. Optional when detection is disabled.	See Direction
FeatureVersion	C	string	Version number of algorithm extracting semi-structured features. Range: [0,32]	"ISFRFR259.2.0"
Feature	C	string	Base64-encoded. Extracted feature info of non-motor vehicles. The maximum length before encoding is 4096 bytes.	
AttributeInfo	C	Json Block	Non-motor vehicle attribute info. Optional when detection is disabled. See NonMotorVehicleAttr	0
PlateAttributeInfo	C	Json Block	Plate attribute info. Optional when detection is disabled. See PlateAttr	-
PersonOnNoVehicle Num	M	unsigned long	Total number of persons in vehicle	0

PersonOnNoVehicleInfoList	C	Json Block	Vehicle occupant info list. Optional when PersonOnNoVehicleNum is 0. See PersonAttr	-
RuleInfo	O	Json Block	Rule info. See RuleInfo Json Block	-

11. NonMotorVehicleAttr

NonMotorVehicleAttrJson Block	{ "SpeedType"; "ImageDirection", "NonVehicleType"; }
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Param	Requirement	Type	Description	Example
SpeedType	C	unsigned long	Speed type of non-motor vehicles in structured scenes. Optional when detection is disabled. 0: Unknown 1: Static 2: Low speed 3: Medium speed 4: Fast	0
ImageDirection	C	unsigned long	Movement direction (relative to the picture) of non-motor vehicles in structured scenes. Optional when detection is disabled. 0: Unknown 1: Static 2: Up 3: Down 4: Left 5: Right 6: Up left 7: Down left 8: Up right 9: Down right	0
NonVehicleType	C	unsigned long	Type of non-motor vehicle. Optional when detection is disabled. 0: Unknown 1: Bicycle 2: 3-wheel 3: Motorcycle	0

			4: Electric bicycle 5: 2-wheel (motorcycle, bicycle, electric bicycle)	
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12. PersonEventInfo

PersonEventInfo Json Block	<pre>{ "ID";, "Timestamp";, "NotificationType";, "FaceInfoNum";, "FaceInfoList":[<FacePassRecordInfo> ,...], }</pre>
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Param	Requirement	Type	Description	Example
ID	M	unsigned long	Notification record ID	
Timestamp	M	unsigned long	Notification time (unit: second, UTC format).	
NotificationType	M	unsigned long	Notification type 0: Real-time notification 1: Historical notification	
FaceInfoNum	0	unsigned long	Face info number Range: [0, 1]. FaceInfo fields are optional when no face info is collected	
FaceInfoList	0	array	Face info list FaceInfo fields are optional when no face info is collected	

13. PersonInfo

PersonInfo Json Block	<pre>{ "PersonID";, "Position";, "AppearTime";, "DisAppearTime": , "Confidence";, "SmallPicAttachIndex";, "LargePicAttachIndex";, "FeaturVersion";, "Feature";, "AttributeInfo":<PersonAttr>, "RuleInfo":<RuleInfo> }</pre>
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Param	Requirement	Type	Description	Example
PersonID	M	unsigned long	Person ID. Starts from 1. Optional when detection is disabled.	1
Position	M	string	Position of the person Coordinates: 0-10000. The top left and bottom right points of the rectangle: "138,315;282,684" Comma (,) and semicolon (;) are half-width characters. Range: [0,32].	"138,315;282,684"
AppearTime	C	string	Time when the detection object appears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM, The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	"20190422083030789"
DisAppearTime	C	string	Time when the detection object disappears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM, The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	"20190422083030789"
Confidence	C	unsigned long	Confidence of face detection. Optional when detection is disabled. Value range: [0-100]	70
SmallPicAttachIndex	C	unsigned long	Index of the close-up image of person in the image list, which starts from 1. 0 is invalid. Optional when no close-up image is pushed.	0
LargePicAttachIndex	C	unsigned long	Index of the original image of the person in the image list, which starts from 1. 0 is invalid. Optional when no original image is pushed.	1
FeatureVersion	C	string	Version number of algorithm extracting semi-structured features. Range: [0,32]	"ISFRFR259.2.0"

Feature	C	string	Base64-encoded. Extracted body feature info. The maximum length before encoding is 4096 bytes.	
AttributeInfo	C	Json Block	Human body attribute info. Optional when attribute detection is disabled. See PersonAttr	-
RuleInfo	O	Json Block	Rule info. See RuleInfo Json Block	-

14. PersonAttr

PersonAttrJson Block	<pre>{ "Gender":, "AgeRange":, "SleevesLength":, "CoatColor":, "TrousersLength":, "TrousersColor":, "BodyToward":, "ShoesTubeLength":, "HairLength":, "BagFlag": }</pre>
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Param	Requirement	Type	Description	Example
Gender	C	unsigned long	Gender. Optional when detection is disabled. 0: Female 1: Male 98: Unknown	0
AgeRange	C	unsigned long	Age range. Optional when detection is disabled. 0: Child 1: Juvenile 2: Youth 3: Middle-aged 4: Senior 98: Unknown	0
SleevesLength	C	unsigned long	Sleeve length. Optional when detection is disabled. 0: Unknown/not sure 1: Short sleeves 2: Long sleeves	0
CoatColor	C	unsigned long	Coat color. Optional when detection is disabled.	"A"

			See extended data type CommonColorType	
TrousersLength	C	unsigned long	Trousers length. Optional when detection is disabled. 0: Unknown 1: Short 2: Long	0
TrousersColor	C	string	Trousers color. Optional when detection is disabled. See extended data type CommonColorType	"A"
BodyToward	C	unsigned long	Body direction. Optional when detection is disabled. 0: Not sure/Unknown 1: Front 2: Side 3: Back	0
ShoesTubeLength	C	unsigned long	Shoes tube length. Optional when detection is disabled. 0: Unknown 1: Long tubes 2: Short tubes/ordinary shoes	0
HairLength	C	unsigned long	Hair length. Optional when detection is disabled. 0: Unknown/not sure 1: Long hair 2: Short hair	0
BagFlag	C	unsigned long	Whether bag is detected. Optional when detection is disabled. 0: No bag 1: Handbag 2: Backpack 98: Unknown	0

15. PlateAttr

PlateAttr Json Block	{ "PlateNo";, "Color";, "Type";, }
----------------------	--

Param	Requirement	Type	Description	Example
PlateNo	C	string	Plate number. Optional when detection is disabled.	"A00000"

Color	C	unsigned long	Plate color. Optional when detection is disabled. See extended data type: plate color	1
Type	C	unsigned long	Plate type. Optional when detection is disabled. See extended data type: plate type	0

16. RegionInfo

RegionInfo Json Block	{ "Nation": , "Province": , "City": , }
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Param	Requirement	Type	Description	Example
Nation	M	string	Nationality. Range: [1,63]	"China"
Province	M	string	Province. Range: [1,63].	"Zhejiang"
City	M	string	City. Range: [1,63].	"Hangzhou"

17. RelatedObjectInfo

RelatedObjectInfo Json Block	{ "ObjectNum", "ObjectList": [< ObjectIDInfo >, < ObjectIDInfo >...] }
---------------------------------	---

Param	Requirement	Type	Description	Example
ObjectNum	M	unsigned long	Number of objects.	1
ObjectList	C	array	Object list. Required when ObjectNum is not 0. See ObjectIDInfo	-

18. RuleInfo

RuleInfo Json Block	{ "RuleType", "TrigerType", "PointNum": , "PointList": [{ "X": , "Y": },]
------------------------	--

	<pre> ...] }</pre>
--	---------------------------------

Param	Requirement	Type	Description	Example
RuleType	M	unsigned long	Rule type: 0: Intrusion detection 1: Cross line detection 2: Leave area 3: Enter area	0
TrigerType	0	unsigned long	Type of rule trigger: 0: None 1: Bi-directional 2: Clockwise 3: Counterclockwise 4: Enter 5: Leave	0
PointNum	M	unsigned long	Number of points (vertices)	6
PointList	C	array	Coordinates of vertices of the detection area. Optional when PointNum is 0.	
X	M	unsigned long	x-coordinate. Range: [0,10000]	5000
Y	M	unsigned long	y-coordinate. Range: [0,10000]	5000

19. ObjectInfo

ObjectInfo Json Block	<pre> { "FaceNum"; "FaceInfoList":[<FaceInfo> ,...], "PersonNum"; "PersonInfoList":[<PersonInfo> ,...], "NonMotorVehicleNum"; "NonMotorVehicleInfoList":[:<NonMotorVehicleInfo> ,...] "VehicleNum"; "VehicleInfoList":[<VehicleInfo> ,...] }</pre>
-----------------------	--

Param	Requirement	Type	Description	Example
FaceNum	M	unsigned long	Number of faces.	1
FaceInfoList	C	Json Block	Face info list. See FaceInfo Json Block; Optional if FaceNum is 0.	-
PersonNum	M	unsigned long	Number of persons.	1
PersonInfoList	C	Json Block	Person info list. See PersonInfo	-

			Json Block. Optional when PersonNum is 0.	
NonMotorVehicleNum	C	unsigned long	Number of non-motor vehicles. Optional when the subscribed service type is face or body.	1
NonMotorVehicleInfoList	C	Json Block	Non-motor vehicle info list. Optional when NonMotorVehicleNum is 0 or when the subscribed service type is face or body. See NonMotorVehicleInfo Json Block	-
VehicleNum	C	unsigned long	Number of motor vehicles. Optional when the subscribed service type is face or body.	1
VehicleInfoList	C	Json Block	Motor vehicle info list. Optional when VehicleNum is 0 or when the subscribed service type is face or body. See VehicleInfo Json Block	-

20. ObjectIDInfo

ObjectIDInfo Json Block	{ "ObjectType", "ObjectID" }
----------------------------	---------------------------------------

Param	Requirement	Type	Description	Example
ObjectType	M	unsigned long	Object type: 0: Unknown/unrecognized object type 1: Face 2: Human body 3: Motor vehicle 4: Non-motor vehicle	1
ObjectID	M	unsigned long	Object ID	2

21. SubscribePersonCondition

SubscribePersonCondition Json Block	{ "LibIDNum";, "LibIDList":[{"LibID";},{LibID";},...] }
--	--

Param	Requirement	Type	Description	Example
LibIDNum	M	unsigned long	Number of IDs of subscribed libraries. LibIDNum and LibIDList are optional when subscription type is 0. When the subscription type is 1, it is required to specify the number and value of library IDs to subscribe from. When LibIDNum is 0xFFFF, it means subscription from all libraries.	0
LibIDList	M	Array	List of IDs of subscribed libraries.	--
LibID	C	unsigned long	Optional when LibIDNum is 0.	3

22. StructureDataInfo

StructureInfo Json Block	<pre>{ "ObjInfo":<ObjectInfo>, "ImageNum":, "ImageInfoList":,[<ImageInfo> ,...], "FinishFaceNum": , "FinishFaceList" [<FaceID>,<FaceID>...] }</pre>
--------------------------	---

Param	Requirement	Type	Description	Example
ObjInfo	C	Json Block	Object info list. Optional when ObjNum is 0. See ObjectInfo	-
ImageNum	M	unsigned long	Number of images	1
ImageInfoList	C	Json Block	Image info list. Optional when ImageNum is 0. See ImageInfo	-
FinishFaceNum	0	unsigned long	Number of faces that have finished detection Range: [0,40] Absent when no face has finished detection.	
FinishFaceList	0	array	List of faces that have finished detection. Absent when FinishFaceNum is 0.	

23. VehicleAttr

VehicleAttr Json Block	{ "Type":, "Color":, "SpeedType", "VehicleBrand":, "AimStatus": , "DriverSunVisorStatus": , "CodriverSunVisorStatus": , "DriverSeatBeltStatus": , "CodriverSeatBeltStatus": , "DriverMobileStatus": , "PendantStatus": , "ImageDirection":, }
---------------------------	--

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Vehicle type. See extended data type: vehicle type	0
Color	M	unsigned long	Vehicle body color. Optional when detection is disabled. See extended data type CommonColorType	0
SpeedType	C	unsigned long	Speed type of motor vehicles in structured scenes. Optional when detection is disabled. 0: Unknown 1: Static 2: Slow 3: Medium speed 4: Fast	
VehicleBrand	C	string	Vehicle brand ID (custom). Optional when detection is disabled. See extended data type: vehicle brand	
AimStatus	C	String	Status of inspection sticker. Optional when detection is disabled. 0: Not in use 1: In use 2: Unknown Range: [0-2]	"1"
DriverSunVisorStatus	C	String	Status of driver-side sun visor. Optional when detection is disabled.	"0"

			0-Not in use 1-In use 2-Unknown Range: [0-2]	
CodriverSunVisorStatus	C	String	Status of passenger-side sun visor. Optional when detection is disabled. 0-Not in use 1-In use 2-Unknown Range: [0-2]	"0"
DriverSeatBeltStatus	C	String	Status of driver-side seat belt. Optional when detection is disabled. 0: Not in use 1: In use 2: Unknown Range: [0-2]	"0"
CodriverSeatBeltStatus	C	String	Status of passenger-side seat belt. Optional when detection is disabled. 0: Not in use 1: In use 2: Unknown Range: [0-2]	"0"
DriverMobileStatus	C	String	Whether driver is calling while driving. Optional when detection is disabled. 0: No 1: Yes 2: Unknown Range: [0-2]	"0"
PendantStatus	C	String	Pendant. Optional when detection is disabled. 0: Not detected 1: Detected 2: Unknown Range: [0-2]	"0"
ImageDirection	C	unsigned long	Driving direction (relative to the image) of motor vehicle in structured scenes. Optional when detection is disabled. Same value as ImageDirection of non-motor vehicle	

24. VehicleInfo

VehicleInfo Json Block	{ "ID "; "Position"; "SmallPicAttachIndex"; "LargePicAttachIndex"; "PlatePicAttachIndex"; "AppearTime": ,
---------------------------	---

	"DisAppearTime": , "Confidence": , "FeaturVersion":, "Feature":, "VehicleAttributeInfo":< VehicleAttr > "PlateAttributeInfo":< PlateAttr >, }
--	---

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Vehicle ID, starts from 1.	1
Position	M	string	Position of the vehicle Coordinates: 0-10000. The top left and bottom right points of the rectangle: "138,315;282,684" Comma (,) and semicolon (;) are half-width characters. Range: [0,32].	138,315;282,684
SmallPicAttachIndex	C	unsigned long	Index of the close-up image of vehicle in the image list, which starts from 1. 0 is invalid. Optional when no close-up image is pushed.	0
LargePicAttachIndex	C	unsigned long	Index of the original image of the vehicle in the image list, which starts from 1. 0 is invalid. Optional when no original image is pushed.	1
PlatePicAttachindex	C	unsigned long	Index of the close-up image of plate in the image list, which starts from 1. 0 is invalid. Optional when no close-up image is pushed.	1
AppearTime	C	string	Time when the detection object appears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM, The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	AppearTime
DisAppearTime	C	string	Time when the detection object disappears. Optional when detection is disabled. Format: YYYYMMDDHHMMSSMMM, The first MM represents month, the second MM represents minute, and the third MMM represents millisecond. Fixed length: 18	DisAppearTime

Confidence	C	unsigned long	Confidence of detection Value range: [0-100]	Confidence
FeatureVersion	C	string	Version number of algorithm extracting semi-structured features. Range: [0,32].	"ISFRFR259.2.0"
Feature	C	string	Base64-encoded. Extracted motor vehicle feature info. The maximum length before encoding is 4096 bytes.	
VehicleAttributeInfo	C	Json Block	Vehicle attribute info. Optional when attribute detection is disabled. See VehicleAttr	-
PlateAttributeInfo	C	Json Block	Plate attribute info. Optional when face attribute detection is disabled. See PlateAttr	-

3.11 SearchByImage

3.11.1 SearchByImage

1. /LAPI/V1.0/Smart/SearchByImage/Task

URL	/LAPI/V1.0/Smart/SearchByImage/Task
Description	Create a search-by-image task
Method	POST
Input Data	None
Success Return Data	{ "TaskID": }
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
TaskID	M	unsigned long	Task ID	854

2. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/Control

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/Control
Description	Control a search-by-image task
Method	PUT
Input Data	Control
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Control	M	Json Block	Operation action	
ID	M	unsigned long	Task ID	

3. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/Target

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/Target			
Description	Add search parameters			
Method	POST			
Input Data	<pre>{ "SearchTarget":<SearchTarget>, "CompareTarget":<CompareTarget> }</pre>			
Success Return Data	None			
Note	Applicable to NVR.			

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	
SearchTarget	M	Json Block	Search target. See SearchTarget Json Block.	
CompareTarget	M	Json Block	Comparison target. See CompareTarget Json Block	

4. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/SearchTarget

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/SearchTarget			
Description	Add search target			
Method	POST			
Input Data	<pre>{ "SearchTarget":<SearchTarget> }</pre>			
Success Return Data	None			
Note	Applicable to NVR.			

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	
SearchTarget	M	Json Block	Search target. See SearchTarget Json Block.	

5. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/CompareTarget

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/CompareTarget			
Description	Add object search range.			
Method	POST			

Input Data	{ "CompareTarget":< CompareTarget > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	
CompareTarget	M	Json Block	Comparison target. See CompareTarget Json Block	

6. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/SearchTarget

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/SearchTarget
Description	Add search target
Method	POST
Input Data	{ "SearchTarget":< SearchTarget > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	
SearchTarget	M	Json Block	Search target. See SearchTarget Json Block.	

7. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/CompareTarget

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/CompareTarget
Description	Update target search range.
Method	PUT
Input Data	{ "CompareTarget":< CompareTarget > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	
CompareTarget	M	Json Block	Search target. See CompareTarget Json Block	

8. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>/SearchTarget

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>/SearchTarget
Description	Update search target
Method	POST
Input Data	{ "SearchTarget":< SearchTarget > }
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	
SearchTarget	M	Json Block	Search target. See SearchTarget Json Block.	

9. /LAPI/V1.0/Smart/SearchByImage/Progress?TaskID=<ID>

URL	/LAPI/V1.0/Smart/SearchByImage/Progress?TaskID=<ID>
Description	Get the search progress of a task.
Method	GET
Input Data	None
Success Return Data	SearchProgress
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	1
SearchProgress	M	Json Block	Search progress	

10. /LAPI/V1.0/Smart/SearchByImage/Query?Task=<ID>&Limit=<Limit>&Offset=<Offset>

URL	/LAPI/V1.0/Smart/SearchByImage/Query?Task=<ID>&Limit=<Limit>&Offset=<Offset>
Description	Get search results
Method	GET
Input Data	None
Success Return Data	SearchResult
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID	1
Limit	M	unsigned long	Number of tasks to search each time. Max 20.	20
Offset	M	unsigned long	Sequence number to start search from. The sequence number starts from 0.	10
SearchResult	M	Json Block	Search results	

11. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>
Description	Delete a task.
Method	DELETE
Input Data	None
Success Return Data	None
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
ID	M	unsigned long	Task ID.	

12. /LAPI/V1.0/Smart/SearchByImage/Task

URL	/LAPI/V1.0/Smart/SearchByImage/Task
Description	Get the search-by-image task list
Method	GET
Input Data	None
Success Return Data	{ "Num":, "TaskList":[< SearchTaskBaseInfo >,...] }
Note	Applicable to NVR.

Param	Requirement	Type	Description	Example
Num	M	unsigned long	Number of tasks.	1
TaskList	C	array	Task list. Invalid when Num is 0.	-
SearchTaskBaseInfo	M	Json Block	Basic info about search-by-image task. See SearchTaskBaseInfo Json Block	-

13. /LAPI/V1.0/Smart/SearchByImage/Task/<ID>

URL	/LAPI/V1.0/Smart/SearchByImage/Task/<ID>
Description	Get task info
Method	GET
Input Data	None
Success Return Data	{ "TaskInfo": < TaskInfo > }
Note	Applicable to NVR.

3.11.2 Json Block

1. SearchPassRecordResult

SearchPassRecordResult Json Block	<pre>{ "Total";, "Offset";, "Num";, "FacePassRecordList": [{ "TargetFileName";, "PassRecordInfo":<PassRecordInfo> },...] }</pre>
--	--

Param	Requirement	Type	Description	Example
Total	M	unsigned long	Total number of search results.	5000
Offset	M	unsigned long	Offset from the start of the data	100
Num	M	unsigned long	Number of returned records	10
FacePassRecordList	C	array	Pass-through records. Absent when there are no data.	
TargetFileName	M	string	Filename of object image. Specified in SearchTarget . Length range: [0,770]	
PassRecordInfo	M	Json Block	Pass-through record info	

2. PassRecordInfo

PassRecordInfo Json Block	<pre>{ "RecordID";, "PassingTime";, "ChannelName";, "ChannelID";, "Similarity";, "AttributeInfo":<FaceAttr> "SnapshotImage": <SnapshotImage> }</pre>
--	--

Param	Requirement	Type	Description	Example
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RecordID	M	unsigned long	Pass-through record ID.	1
PassingTime	M	unsigned long	Pass-through time (unit: second, UTC format)	1520412142
ChannelName	M	string	Name of the channel capturing the snapshot. Range: [1, 63].	"XX road intersection"
ChannelID	M	unsigned long	Channel ID	1
Similarity	M	unsigned long	Matching degree (similarity)	-
SnapshotImage	M	Json Block	Image info See SnapshotImage	
AttributeInfo	C	Json Block	Face attribute info. Present when there is face attribute info. See FaceAttr .	-

3. SnapshotImage

SnapshotImage Json Block	{ "BigImage":< FileInfo >, "SmallImage": < FileInfo >, "Area": { "left":, "top":, "right":, "bottom": } }
------------------------------------	---

Param	Requirement	Type	Description	Example
BigImage	M	Json Block	Original image. Absent when searching faces or pushing snapshots. See FileInfo Json Block.	
SmallImage	C	Json Block	Close-up image. See FileInfo Json Block. IPC can be configured not to push close-up images.	
Area	M	Json Block	Coordinates of the close-up image in the original image.	
left	M	unsigned long	Coordinates of the left point.	
top	M	unsigned long	Coordinates of the top point.	
right	M	unsigned long	Coordinates of the right point.	
bottom	M	unsigned long	Coordinates of the bottom point.	

4. SearchPeopleLibrarieResult

SearchPeopleLibrarieResult Json Block	{ "Total":, "Offset": , "Num":,
---	--

	<pre> "PersonList": [{ "TargetFileName": "TargetInfo":<PersonInfo> },...] </pre>
--	--

Param	Requirement	Type	Description	Example
Total	M	unsigned long	Total number of search results.	5000
Offset	M	unsigned long	Offset from the start of the data	100
Num	M	unsigned long	Returned records.	10
PersionList	C	array	Person list. Absent when there are no data.	
TargetFileName	M	string	Filename of object image. Length range: [0,770].	
TargetInfo	M	Json Block	Person info. See PersonInfo	

5. Control

Control Json Block	<pre> { "Control":, } </pre>
---------------------------	----------------------------------

Param	Requirement	Type	Description	Example
Control	M	unsigned long	Action type 0: Start 1: Pause 2: Stop (currently not supported)	0

6. SearchTarget

SearchTarget Json Block	<pre> { "TargetInfo":<TargetInfo> } </pre>
--------------------------------	--

Param	Requirement	Type	Description	Example
TargetInfo	M	Json Block	Object info	

7. TargetInfo

TargetInfo Json Block	{ "SimilarityThreshold":, "TargetNum":, "TargetInfoList":[< FileInfo >,...] }
------------------------------	---

Param	Requirement	Type	Description	Example
SimilarityThreshold	M	unsigned long	Matching threshold (result below the threshold will not be returned). Max: 100. Min: 0.	
TargetNum	M	unsigned long	Number of objects. Currently only one object is supported.	
TargetInfoList	M	array	Object list info	
FileInfo	M	json block	Object image info. JPEG. Max resolution: 1080*1960. Max. file size: 300KB.	

8. CompareTarget

CompareTarget Json Block	{ "Type":, "CompareLib":< CompareLib >, "CompareRecord":< CompareRecord >, }
---------------------------------	--

Param	Requirement	Type	Description	Example
Type	M	unsigned long	Object type: 1: Face 2: Motor vehicle (currently not supported) 3: Human body (currently not supported)	
CompareLib	C	Json Block	If the object type is "Face", the object is compared with a face library. If the object type is "Motor vehicle", the object is compared with a vehicle library. Required.	
CompareRecord	C	Json Block	If the object type is "Face", the object is compared with a person pass-through record. If the object type is	

			"Motor vehicle", the object is compared with a vehicle pass-through record. Required.	
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9. CompareLib

CompareLib Json Block	<pre>{ "LibType"; "LibNum"; "LibList":["<ID>,<ID>...] }</pre>
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Param	Requirement	Type	Description	Example
LibType	M	unsigned long	Type of compared library. 0: Face library 1: Vehicle library	0
LibNum	M	unsigned long	Number of compared libraries.	
LibList	M	array	Library list	

10. CompareRecord

CompareRecordJson Block	<pre>{ "BeginTime"; "EndTime"; "Num"; "ChannelList":["<ID>,<ID>...] }</pre>
--------------------------------	---

Param	Requirement	Type	Description	Example
BeginTime	M	unsigned long	Start time (UTC format)	
EndTime	M	unsigned long	End time (UTC format)	
Num	0	unsigned long	Number of channels. If not specified, all channels will be searched by default.	
ChannelList	C	array	Channel list. Required when Num is not 0.	
ID	M	unsigned long	Channel ID	

11. SearchProgress

SearchProgress Json Block	<pre>{ "Percent"; }</pre>
----------------------------------	-----------------------------

	<pre> "BeginTime"; "EndTime"; , "MatchNum"; "TotalNum"; "CompleteNum"; "RunStatus"; } </pre>
--	--

Param	Requirement	Type	Description	Example
Percent	M	unsigned long	Search progress. Range: [0,100].100 means the search task has completed.	
BeginTime	M	unsigned long	Start time of the searched data (UTC format)	
EndTime	M	unsigned long	End time of the searched data (UTC format)	
MatchNum	M	unsigned long	Number of matched records. NVR can buffer 1000 matching results for each task.	
TotalNum	0	unsigned long	Total number of records to compare. Numerical. Applicable to VMS only.	
CompleteNum	0	unsigned long	Number of compared records. Numerical. Applicable to VMS only.	
RunStatus	M	unsigned long	Task status: 0: Normal 1: Paused 2: Ended 3: Failed (common error) 4: Operation failed (modeling failed)	

12. SearchResult

Comparison object type and result structure

Object Type	Comparison Object	SearchResult Structure
Face	Pass-through record	SearchPassRecordResult
Face	Face library	SearchPeopleLibrarieResult

13. SearchTaskBaseInfo

SearchTaskBaseInfo Block	Json	<pre> { "TaskID"; "Speed"; "Priority"; "SimilarityThreshold"; "BeginTime"; "EndTime"; "Status"; } </pre>
---------------------------------	-------------	--

Param	Requirement	Type	Description	Example
TaskID	C	unsigned long	Task ID. Optional when creating a task.	1
Speed	0	unsigned long	Search speed level (0-5). Currently not in use.	0
Priority	0	unsigned long	Execution priority level (0-3). High priority tasks are executed first. Currently not in use.	0
SimilarityThreshold	M	unsigned long	Matching threshold (results below the threshold will not be returned). Max: 100. Min: 0. Default: 80.	80
BeginTime	M	unsigned long	Start time (UTC format)	
EndTime	M	unsigned long	End time (UTC format)	
Status	M	unsigned long	0: Not started 1: Searching 2: Paused 4: Stopped	

14. TaskInfo

PassRecordSearchTaskInfo Json Block	<pre>{ "TaskBaseInfo":<SearchTaskBaseInfo>, "SearchTarget":<SearchTarget>, "CompareTarget":<CompareTarget> }</pre>
--	--

Param	Requirement	Type	Description	Example
SearchTaskBaseInfo	M	Json Block	Basic info about search-by-image task. See SearchTaskBaseInfo Json Block	-
SearchTarget	M	Json Block	Search target. See SearchTarget Json Block.	
CompareTarget	M	Json Block	Comparison target. See CompareTarget Json Block	

4 Extended Data Types

4.1 Common Color (CommonColorType)

Vehicle body colors are defined as common colors, and other things use this definition, such as hat color, shoe color, pedestrian clothes color.

Code	Description	Remarks
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0	Black	
1	White	
2	Gray	
3	Red	
4	Blue	
5	Yellow	
6	Orange	
7	Brown	
8	Green	
9	Purple	
10	Cyan	
11	Pink	
12	Transparent	
13	Silver	
14	Dark	
15	Light	
16	No color	
17	Yellow & green	
18	Gradient green	
99	Others	
100	Unknown	

4.2 Structured Attributes (AttributeType)

Code	Description	Remarks
0	Gender	
1	Age	
2	/	
3	Emotion	
4	/	
5	Glasses	
6	Smile	
7	Mask	
8	Mustache	
9	Facial attractiveness	
10	Upper half body color	
11	Lower half body color	
12	Coat style	
13	Coat textures	
14	Trousers style	
15	Hat	
16	Bag	
17	Trolley case	
18	Posture	
19	Umbrella	
20	Hairstyle	

21	Shoes	
22	Temperature	
Reserved	/	
100	Movement speed	
101	Movement direction	
Reserved	/	
200	Non-motor vehicle type	
201	Non-motor vehicle plate	
Reserved	/	
300	Plate number	
301	Plate type	
302	Plate color	
303	Vehicle type detection	
304	Vehicle color	
305	Vehicle brand detection	
306	Sun visor detection	
307	Seat belt detection	
308	Calling detection	
309	Vehicle model detection	
310	Yellow label vehicle detection	
311	Close-up face image of driver	
312	Close-up face image of front-row passenger	
313	Front-row passenger seat belt detection	
314	Pendant detection	
315	Inspection sticker detection	
316	Dangerous goods detection	